

Chapter 15

Relations and Functions

Solutions

SECTION - A

Objective Type Questions (One option is correct)

1. If $A = \{4, 8\}$ and $B = \{p, q, r\}$, then $A \times B$ is equal to

(1) $\{(4, p), (8, q), (4, q)\}$

(2) $\{(4, p), (4, r), (8, p), (8, r)\}$

(3) $\{(4p, 8q), (4r, 8q)\}$

(4) $\{(4, p), (4, q), (4, r), (8, p), (8, q), (8, r)\}$

Sol. Answer (4)

2. If $A = \{2, 3, 7, 9\}$ and $n(A \times B) = 12$, then the number of elements in set B is

(1) 4

(2) 3

(3) 2

(4) 8

Sol. Answer (2)

3. If $A = \{5, 7, 8\}$ and $B = \{9, 11\}$, then

(1) $(9, 5) \in A \times B$

(2) $(7, 8) \in A \times B$

(3) $(5, 11) \in A \times B$

(4) $(9, 11) \in A \times B$

Sol. Answer (3)

4. If A and B are any two sets, then

(1) $A \times B = B \times A$

(2) $A \times B \subset B \times A$

(3) $B \times A \subset A \times B$

(4) $n(A \times B) = n(B \times A)$

Sol. Answer (4)

5. The relation R from A to B is given as $R = \left\{ (1, 2), (2, 5), \left(\frac{1}{2}, 7 \right) \right\}$. The domain of R is

(1) $\{1, 2\}$

(2) $\{2, 5, 7\}$

(3) $\left\{ 1, 2, \frac{1}{2} \right\}$

(4) $\left\{ 1, 2, 5, \frac{1}{2}, 7 \right\}$

Sol. Answer (3)

6. The relation R from A to B is given as $R = \{(5, 3), (2, 7), (8, 5)\}$. The range of R is

(1) $\{5, 2, 8\}$

(2) $\{3, 7, 5\}$

(3) $\{2, 3, 5, 7, 8\}$

(4) $\{2, 3, 5, 7\}$

Sol. Answer (2)

7. If R is a relation from a set A to set B , then

- (1) $R \subset B \times A$ (2) $R \subset A \times B$ (3) $R = A \times B$ (4) $A \times B \subset R$

Sol. Answer (2)

8. If $A = \{2, 4, 6\}$, then domain of the relation $R = \{(a, b) : a, b \in A, |a| - |b| \text{ is even number}\}$ defined on A is

- (1) $\{2, 4\}$ (2) $\{4, 6\}$ (3) $\{(2, 6)\}$ (4) $\{2, 4, 6\}$

Sol. Answer (4)

9. Number of relations that can be defined on the set $A = \{a, b, c\}$ is

- (1) 2^3 (2) 6 (3) 3^2 (4) 2^9

Sol. Answer (4)

10. If A and B are two sets such that $A \times B = \phi$, then which of the following must be true?

- (1) $A = \phi$ and $B \neq \phi$ (2) $A = \phi$ and/or $B = \phi$ (3) $A \neq \phi$ and $B = \phi$ (4) $A = \phi$ and $B = \phi$

Sol. Answer (2)

11. If $A = \{-1, 1\}$, then $A \times A \times A$ is equal to

- (1) $\{(-1, 1), (1, -1)\}$
 (2) $\{(-1, -1), (1, 1)\}$
 (3) $\{(-1, -1), (-1, 1), (1, -1), (1, 1)\}$
 (4) $\{(-1, -1, -1), (-1, -1, 1), (-1, 1, -1), (-1, 1, 1), (1, -1, -1), (1, -1, 1), (1, 1, -1), (1, 1, 1)\}$

Sol. Answer (4)

12. If $A = \{x_1, y_1, z_1\}$ and $B = \{x_2, y_2\}$, then the number of relations from A to B is

- (1) 8 (2) 16 (3) 32 (4) 64

Sol. Answer (4)

13. Let X be any non-empty set containing n elements, then the number of relations on X is

- (1) 2^{n^2} (2) 2^n (3) 2^{2n} (4) n^2

Sol. Answer (1)

14. If $A = \{1, 2, 3, 4, 5\}$, then the number of relations that can be defined on A is

- (1) 2^5 (2) 5^2 (3) 2^{25} (4) 4^5

Sol. Answer (3)

$$\begin{aligned} \text{Number of relations that can be defined from } A \text{ to } A &= 2^{n^2}, \text{ where } n = 5 \\ &= 2^{25} \end{aligned}$$

15. If $A = \{1, 2, 3, 4, 5\}$ and $B = \{a, b, c, d\}$, then the number of relations that can be defined from A to B is

- (1) 20 (2) 2^{20} (3) 2^9 (4) 9

Sol. Answer (2)

$$\begin{aligned} \text{Number of relations that can be defined from } A \text{ to } B &= 2^{mn}, \text{ where } m = 5 \text{ and } n = 4 \\ &= 2^{20} \end{aligned}$$

16. Let $A = \{a, b, c, d\}$. Which of the following relations is a function from A to A ?

- (1) $\{(a, b), (b, d), (c, a), (d, b)\}$ (2) $\{(a, b), (a, c), (b, a)\}$
 (3) $\{(a, b), (b, c), (c, d)\}$ (4) $\{(a, a), (a, b), (b, c), (b, d)\}$

Sol. Answer (1)

A relation will be a function from A to B if every element of A has one image in B .

All the elements of A are connected to B only in option (1).

17. Let R_1 and R_2 be equivalence relations on a set A , then $R_1 \cup R_2$ may or may not be

- (1) Reflexive (2) Symmetric (3) Transitive (4) None of these

Sol. Answer (3)

If R_1 and R_2 are transitive on a set A , then $R_1 \cup R_2$ may or may not be transitive. As an example, consider the set $A = \{1, 2, 3\}$.

Let $R_1 = \{(1,1), (2,2), (3,3), (1,2), (2,1)\}$ and $R_2 = \{(1,1), (2,2), (3,3), (1,3), (3,1)\}$ then both R_1 and R_2 are equivalence relations on A . However, $R_1 \cup R_2 = \{(1,1), (2,2), (3,3), (1,2), (2,1), (3,1), (1,3)\}$ is not transitive as $(2, 1)$ and $(1, 3) \in R_1 \cup R_2$ but $(2, 3) \notin R_1 \cup R_2$.

18. Let R be the relation defined on the set W of whole numbers by the rule xRy iff $4x + 5y = 20$, then domain of R is

- (1) $\{4, 5\}$ (2) $\{0, 5\}$ (3) $\{0, 4\}$ (4) $\{0, 2\}$

Sol. Answer (2)

$$\text{Here, } y = \frac{20 - 4x}{5}$$

$$\Rightarrow y = 4 - \frac{4}{5}x$$

Values of x for which x and y both are whole numbers is 0, 5.

$\therefore$ Domain of R is $\{0, 5\}$.

19. A relation $R = \{(x, y) : |x - y| \text{ is divisible by } 3; x, y \in R; (\text{set of real numbers})\}$, then R is

- (1) Symmetric (2) Reflexive (3) Transitive (4) All of these

Sol. Answer (4)

Reflexivity : Let x be an arbitrary element of R and $x \in R$.

$$\Rightarrow |x - x| = 0 \text{ which is divisible by } 3$$

$$\Rightarrow xR_1x$$

Hence, R_1 is reflexive.

Symmetric : Let $(x, y) \in R_1$

If $|x - y|$ is divisible by 3

$$\Rightarrow |y - x| \text{ is also divisible by } 3$$

$$\Rightarrow (y, x) \in R_1$$

Hence, R_1 is symmetric.

Transitive : Let xRy and yRz

$$\Rightarrow |x - y| = 3m \text{ and } |y - z| = 3n, \text{ where } m \text{ and } n \text{ are integers.}$$

According to sign of modulus, after subtracting or adding both equations, we can observe that $|x - z|$ is also divisible by 3.

$$\Rightarrow xRz$$

Hence, R is transitive also.

20. Let $A = \{a, b, c, d\}$ and $R = \{(a, a), (b, b), (a, c), (c, a)\}$, then R is

- (1) Reflexive (2) Symmetric (3) Transitive (4) Equivalence

Sol. Answer (2)

Reflexivity : R is not reflexive because for $x \in A$, $(x, x) \notin R$ ($\because$ each element of A is not related)

Symmetric : For $(x, y) \in R$

$$\Rightarrow (y, x) \text{ also } \in R$$

Hence, R is symmetric.

Transitive : We observe that $(c, a) \in R$ and $(a, c) \in R$, but $(c, c) \notin R$.

Hence, R is not transitive.

21. Let $A = \{1, 2, 3, 4\}$. Which of the following is an equivalence relation of A ?

- (1) $\{(1, 1), (2, 2), (1, 2), (2, 1)\}$ (2) $\{(1, 2), (2, 1), (3, 1), (1, 3)\}$
 (3) $\{(1, 2), (4, 1), (1, 4), (3, 1)\}$ (4) $\{(1, 1), (2, 2), (3, 3), (4, 4), (1, 2), (2, 1)\}$

Sol. Answer (4)

Options (1), (2), (3) are not reflexive.

Only option (4) is equivalence relation on A .

22. Let R be a relation defined on set $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$ such that $R = \{(2, 3), (4, 5), (7, 8)\}$. If the domain of R is set B and range is set C , then $B \cap C$ is

- (1) ϕ (2) $\{2, 4, 7\}$ (3) $\{3, 5, 8\}$ (4) $\{3\}$

Sol. Answer (1)

$$B = \{2, 4, 7\}, C = \{3, 5, 8\} \Rightarrow B \cap C = \phi$$

23. Consider two sets $A = \{a, b\}$, $B = \{e, f\}$. If maximum numbers of relations from A to B , A to A , B to B are l, m, n respectively then the value of $2l - m - n$ is

- (1) 8 (2) 0 (3) 16 (4) 32

Sol. Answer (2)

$$l = 2^4 = 16, m = 2^4 = 16, n = 2^4 = 16,$$

$$\Rightarrow 2l - m - n = 0$$

24. Consider three sets $A = \{1, 2, 3\}$, $B = \{3, 4, 5, 6\}$, $C = \{6, 7, 8, 9\}$. R_1 is defined from A to B such that $R_1 = \{(x, y) : 4x < y, x \in A, y \in B\}$. Similarly R_2 is defined from B to C such that $R_2 = \{(x, y) : 2x \leq y, x \in B \text{ and } y \in C\}$, then $R_2^{-1} \circ R_1$ is

- (1) $\{(3, 1)\}$ (2) $\{(1, 1)\}$ (3) $\{(1, 3)\}$ (4) ϕ

Sol. Answer (3)

$$R_1 = \{(1, 5), (1, 6)\}, R_2 = \{(3, 6), (3, 7), (3, 8), (3, 9), (4, 8), (4, 9)\}$$

$$R_2^{-1} = \{(6, 3), (7, 3), (8, 3), (9, 3), (8, 4), (9, 4)\}$$

$$R_2^{-1} \circ R_1 = \{(1, 3)\}$$

25. Consider the set $A = \{3, 4, 5\}$ and the numbers of null relations, identity relation, universal relations, reflexive relations on A are respectively n_1, n_2, n_3 and n_4 then the value of $n_1 + n_2 + n_3 + n_4$ is equal to

- (1) 8 (2) 7 (3) 73 (4) 67

Sol. Answer (4)

$$n_1 = 1, n_2 = 1, n_3 = 1, n_4 = 2^{9-3} = 2^6 = 64$$

$$\Rightarrow n_1 + n_2 + n_3 + n_4 = 1 + 1 + 1 + 64 = 67$$

26. If $n(A) = 3, n(B) = 2, n(A \cap B) = 2$, then total number of relations from A to B is

- (1) 64 (2) 32 (3) 8 (4) 2

Sol. Answer (1)

$$\text{Relations} = 2^6 = 64$$

27. Let L be the set of all straight lines in a plane. l_1 and l_2 are two lines in the set. R_1, R_2 and R_3 are defined relations.

(i) $l_1 R_1 l_2 : l_1$ is parallel to l_2

(ii) $l_1 R_2 l_2 : l_1$ is perpendicular to l_2

(iii) $l_1 R_3 l_2 : l_1$ intersects l_2 .

Then which of the following is true?

- (1) R_1, R_2 and R_3 are equivalence (2) R_1 is equivalence
(3) R_2 and R_3 are reflexive (4) R_1, R_2 and R_3 are not symmetric

Sol. Answer (2)

R_1 is reflexive, symmetric as well as transitive

R_2 is not reflexive and transitive but it is symmetric

R_3 is reflexive, symmetric as well as transitive hence R_1 is equivalence

28. Let R be a relation on $A = \{a, b, c\}$ such that

$$R = \{(a, a), (b, b), (c, c)\},$$

then R is

- (1) Reflexive only (2) Symmetric only (3) Non-transitive (4) Equivalence

Sol. Answer (4)

Clearly R is equivalence

29. Consider the following statements on a set

$$A = \{1, 2, 3\}$$

$S_1 : R = \{(1, 1), (2, 2)\}$ is a reflexive relation on A

$S_2 : R = \{(3, 3)\}$ is symmetric and transitive but not reflexive on A

Which of the following statement on set A is true?

- (1) S_1 only (2) S_2 only (3) Both S_1 and S_2 (4) Neither S_1 nor S_2

Sol. Answer (2)

S_1 : R is not reflexive due to the absence of $(3, 3)$

S_2 : R is symmetric and transitive

30. Let R be the relation on the set of all real numbers defined by xRy iff $|x - y| \leq \frac{1}{2}$. Then R is

(1) Reflexive only

(2) Symmetric only

(3) Transitive only

(4) Reflexive and symmetric both

Sol. Answer (4)

$x R x$: $|x - x| \leq \frac{1}{2}$, Which is true hence R is reflexive

$$x R y \Rightarrow y R x$$

Because $|x - y| \leq \frac{1}{2} \Rightarrow |y - x| \leq \frac{1}{2}$

Hence, R is symmetric, but R is not transitive.

31. Let $R = \{(1, 2), (2, 3)\}$ be a relation defined on set $\{1, 2, 3\}$. The minimum number of ordered pairs required to be added in R , such that enlarged relation becomes an equivalence relation is

(1) 3

(2) 5

(3) 7

(4) 9

Sol. Answer (3)

$$R = \{(1, 2), (2, 3)\}$$

If R is equivalence then R will as

$$R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1), (2, 3), (3, 2), (1, 3), (3, 1)\}$$

Added order pairs = 7

32. Let n be a fixed positive integer. Let a relation R be defined on Z , as aRb iff, n is divisible by $a - b$, then relation R is

(1) Reflexive

(2) Symmetric

(3) Transitive

(4) All of these

Sol. Answer (2)

R is not reflexive because if $a = b$ then $a - b = 0$

or $a - a = 0$, in this case $\frac{n}{0}$ is not possible

but R is clearly symmetric

33. Let S be the set of all real numbers. Then, the relation $R = \{(a, b) : 1 + ab > 0\}$ on S is

(1) Reflexive and symmetric but not transitive

(2) Reflexive and transitive but not symmetric

(3) Symmetric and transitive but not reflexive

(4) Reflexive, symmetric and transitive

Sol. Answer (1)

R is reflexive as $1 + a \times a = 1 + a^2 > 0$

R is symmetric because, if $1 + ab > 0 \Rightarrow 1 + ba > 0$

But R is not transitive in all case

For example

$$\left(\frac{1}{2}\right) R \left(-\frac{1}{3}\right) \text{ as } 1 + \frac{1}{2} \left(-\frac{1}{3}\right) > 0$$

$$\text{and } \left(-\frac{1}{3}\right) R (-4) \text{ as } 1 + \left(-\frac{1}{3}\right) (-4) > 0$$

$$\text{But } \left(\frac{1}{2}\right) R (-4) \text{ is not possible as } 1 + \frac{1}{2} (-4) = -1 < 0$$

Hence, R is not transitive.

34. Let $A = \{1, 2, 3, 4, 5\}$ and let $B = A \times A$. Define the relation R on A as follows $(a, b) R (c, d)$ if and only if $ad = cb$. Then R is

- (1) Reflexive only (2) Symmetric only (3) Transitive only (4) Equivalence

Sol. Answer (4)

$$(1) (a, b) R (a, b) : \frac{a}{b} = \frac{a}{b}, \text{ hence } R \text{ is reflexive}$$

$$(2) (a, b) R (c, d) : \frac{a}{b} = \frac{c}{d}$$

$$(c, d) R (a, b) : \frac{c}{d} = \frac{a}{b}$$

Hence, R is symmetric.

$$(3) (a, b) R (c, d) : \frac{a}{b} = \frac{c}{d} \quad \dots (i)$$

$$(c, d) R (e, f) : \frac{c}{d} = \frac{e}{f} \quad \dots (ii)$$

$$\text{By (i), (ii)} \Rightarrow \frac{a}{b} = \frac{e}{f}$$

$\Rightarrow (a, b) R (e, f)$, hence R is transitive.

Hence R is equivalence.

35. For real numbers x and y define a relation R , xRy iff $|x| = |y|$, the relation R is

- (1) Reflexive only (2) Symmetric only (3) Transitive only (4) Equivalence only

Sol. Answer (4)

Reflexivity : Let x be an arbitrary element of R and $x \in R$.

$$\Rightarrow |x| = |x|$$

$$\Rightarrow xR_1x$$

Hence, R_1 is reflexive.

Symmetric : Let $(x, y) \in R_1$.

$$\text{If } |x| = |y|$$

$$\Rightarrow |y| = |x|$$

$$\Rightarrow (y, x) \in R_1$$

Hence, R_1 is symmetric.

Transitive : Let (x, y) and $(y, z) \in R$.

$$\Rightarrow |x| = |y| \text{ and } |y| = |z|$$

$$\Rightarrow |x| = |z|$$

$$\Rightarrow (x, z) \in R_1$$

Hence, R_1 is transitive.

Thus, R is an equivalence relation.

36. If R_1 and R_2 are two non-empty relations in a set A . Which of the following is not true?

- (1) If R_1 and R_2 are transitive, then $R_1 \cup R_2$ is transitive
- (2) If R_1 and R_2 are transitive, then $R_1 \cap R_2$ is transitive
- (3) If R_1 and R_2 are symmetric, then $R_1 \cup R_2$ is symmetric
- (4) If R_1 and R_2 are reflexive, then $R_1 \cap R_2$ is reflexive

Sol. Answer (2)

Clearly (1),(3),(4) are true

37. For real numbers x and y defined a relation R , xRy iff $x < y$, the relation R is

- (1) Reflexive only
- (2) Symmetric only
- (3) Transitive only
- (4) Equivalence

Sol. Answer (3)

Reflexivity : Let x be an arbitrary element of R and $x \in R$.

$$\Rightarrow x < x \text{ which is not true}$$

Hence, R is not reflexive.

Symmetric : Let $(x, y) \in R$.

$$\text{If } x < y \Rightarrow y < x \text{ which is not true.}$$

Hence, R is not symmetric.

Transitive : Let xRy and yRz

$$\Rightarrow x < y \text{ and } y < z$$

$$\Rightarrow x < z$$

So, R is transitive.

38. Let R be the real line. Consider the following subset of the plane $R \times R$. $S = \{(x, y) : y = x + 1 \text{ and } 0 < x < 2\}$, $T = \{(x, y) : x - y \text{ is an integer}\}$. Which one of the following is true?

- (1) Neither S nor T is an equivalence relation on R
- (2) Both S and T are equivalence relations on R
- (3) S is an equivalence relation on R but T is not
- (4) T is an equivalence relation on R but S is not

Sol. Answer (4)

Clearly, T is an equivalence relation i.e., reflexive, symmetric and transitive.

$$\text{But for } S = \{(x, y) : y = x + 1\}$$

$$\because x \neq x + 1 \Rightarrow (x, x) \notin S \Rightarrow S \text{ is not reflexive.}$$

$\therefore S$ is not equivalence relation.

39. Let A be a finite set containing n distinct elements. The number of functions that can be defined from A to A is
- (1) 2^n (2) n^2 (3) $n!$ (4) n^n

Sol. Answer (4)

Since each element of A can be associated to any one of the n elements of A , therefore, number of functions that can be defined from A to A is n^n .

40. Which of the following relation is a function?

- (1) $\{(1, 4), (2, 6), (1, 5), (3, 9)\}$ (2) $\{(3, 3), (2, 1), (1, 2), (2, 3)\}$
(3) $\{(1, 2), (2, 2), (3, 2), (4, 2)\}$ (4) $\{(3, 1), (3, 2), (3, 3), (3, 4)\}$

Sol. Answer (3)

From definition, option (3) represents a function.

41. Let $A = \{2, 3, 5\}$, $B = \{10, 12, 15\}$, then which of the following is a function from A to B ?

- (1) $\{(2, 10), (2, 12), (2, 15)\}$ (2) $\{(2, 10), (3, 12), (3, 15)\}$
(3) $\{(2, 12), (3, 10), (5, 10)\}$ (4) $\{(2, 10), (3, 12), (3, 15), (5, 15)\}$

Sol. Answer (3)

42. Which of the following is a function?

- (1) $\{(x, y) : y = |x|; x, y \in \mathbb{R}\}$ (2) $\{(x, y) : y^2 = x; x, y \in \mathbb{R}\}$
(3) $\{(x, y) : x^2 + y^2 = 1; x, y \in \mathbb{R}\}$ (4) $\{(x, y) : x^2 - y^2 = 1; x, y \in \mathbb{R}\}$

Sol. Answer (1)

43. Let $n(A) = m$ and $n(B) = n$. The total number of non-empty relations that can be defined from A to B is

- (1) m^n (2) $n^m - 1$ (3) $mn - 1$ (4) $2^{mn} - 1$

Sol. Answer (4)

44. If $f(x) = ax + b$, where a and b are integers, $f(-1) = -5$ and $f(3) = 3$, then a and b are respectively

- (1) $a = -3, b = -1$ (2) $a = 2, b = 3$ (3) $a = 2, b = -3$ (4) $a = 0, b = 2$

Sol. Answer (3)

45. Domain of the function defined by $f(x) = \frac{x^2 + 2x + 1}{x^2 - x - 6}$ is given by

- (1) $\mathbb{R} - \{3, -2\}$ (2) $\mathbb{R} - \{-3, 2\}$ (3) $\mathbb{R} - [3, -2]$ (4) $\mathbb{R} - (3, 2)$

Sol. Answer (1)

46. Domain of the function f given by $f(x) = 2 - |x - 5|$ is

- (1) $\mathbb{R}^+$ (2) $\mathbb{R} - \{5\}$ (3) $\mathbb{R} - \{-5\}$ (4) $\mathbb{R}$

Sol. Answer (4)

47. The domain of $f(x) = \frac{\log_2(x+3)}{x^2 + 3x + 2}$ is

- (1) $\mathbb{R} - \{-1, -2\}$ (2) $(-2, +\infty)$ (3) $\mathbb{R} - \{-1, 2, -3\}$ (4) $(-3, +\infty) - \{-1, -2\}$

Sol. Answer (4)

$$x + 3 > 0 \Rightarrow x > -3$$

$$x^2 + 3x + 2 \neq 0 \Rightarrow (x+1)(x+2) \neq 0 \Rightarrow x \neq -1, -2$$

Hence,

$$D_f \in (-3, \infty) - \{-2, -1\}$$

48. Let $B = \{1, 2, 3, 4, 5, \dots, 30\}$, $A = \{2, 3, 4, 5, 6, 7, 8, 9, 10\}$. A relation is defined from A to B defined by $R = \{(a, b) : b = 4a, a \in A, b \in B\}$, then which of the following is false?

- (1) Domain of $R = \{2, 3, 4, 5, 6, 7\}$ (2) Range of $R = \{8, 12, 16, 20, 24, 28\}$
 (3) Codomain of $R = B$ (4) Pre-image of 27 is 3

Sol. Answer (4)

$$R = \{(2, 8), (3, 12), (4, 16), (5, 20), (6, 24), (7, 28)\}$$

It is clear that option 4 is false. Pre-image of 27 does not exist

49. Let A and B are two sets such that $A \times B$ consists of 6 elements. If three elements of $A \times B = (1, 2), (2, 3), (4, 3)$ then the remaining order pairs of $A \times B$ are

- (1) $(1, 3), (2, 2), (4, 2)$ (2) $(3, 1), (2, 2), (4, 2)$
 (3) $(3, 1), (2, 4), (2, 2)$ (4) $(1, 3), (2, 2), (2, 4)$

Sol. Answer (1)

$$A = \{1, 2, 4\}$$

$$B = \{2, 3\}$$

$$A \times B = \{(1, 2), (1, 3), (2, 2), (2, 3), (4, 2), (4, 3)\}$$

50. Let $A = \{4, 5, 7\}$ and $B = \{2, 4, 6\}$ be two sets and let a relation R be a relation from A to B is defined by $R : \{(x, y) : x < y, x \in A, y \in B\}$ then the difference between the sum of elements of domain and range of R is

- (1) 2 (2) 3 (3) 4 (4) 5

Sol. Answer (2)

$$R = \{(4, 6), (5, 6)\}$$

$$\text{Dom}(R) = \{4, 5\}$$

$$\text{Range}(R) = \{6\}$$

51. Let $A = \{2, 3, 4\}$, $B = \{3, 4, 5\}$ be two sets and a relation R is defined from A to B is such that $R = \{(x, y) : x \text{ divides } y, x \in A, y \in B\}$ then domain of R is

- (1) $\{2, 3, 4\}$ (2) $\{2, 3\}$ only (3) $\{2, 4\}$ only (4) $\{4, 3\}$ only

Sol. Answer (1)

$$R = \{(2, 4), (3, 3), (4, 4)\}$$

$$\text{Dom}(R) = \{2, 3, 4\}$$

52. Range of $f(x) = \frac{3}{2-x^2}$ is

- (1) $\left(-\infty, \frac{3}{2}\right]$ (2) $(-\infty, 0) \cup \left[\frac{3}{2}, \infty\right)$ (3) $(-\infty, 0] \cup \left[\frac{3}{2}, \infty\right)$ (4) $\left(-\infty, \frac{2}{3}\right]$

Sol. Answer (2)

$$-x^2 \in (-\infty, 0]$$

$$2 - x^2 \in (-\infty, 2]$$

$$\Rightarrow \frac{1}{2-x^2} \in (-\infty, 0) \cup \left[\frac{1}{2}, 0\right) \Rightarrow \frac{3}{2-x^2} \in (-\infty, 0) \cup \left(\frac{3}{2}, \infty\right)$$

53. Range of $f(x) = |x-2|$ is

- (1) $(-\infty, \infty)$ (2) $(-\infty, 2]$ (3) $[0, \infty)$ (4) $(-2, 2]$

Sol. Answer (3)

54. Range of $f(x) = -|x-3|$ is

- (1) $(-\infty, \infty)$ (2) $(-\infty, 3]$ (3) $(-3, 3)$ (4) $(-\infty, 0]$

Sol. Answer (4)

55. Range of $f(x) = \frac{1}{2x-1}$ is

- (1) $(-\infty, \infty)$ (2) $(-\infty, 0]$ (3) $[0, \infty)$ (4) $R - \{0\}$

Sol. Answer (4)

$$2x-1 \in (-\infty, \infty)$$

$$\Rightarrow \frac{1}{2x-1} \in R - \{0\}$$

56. Range of $f(x) = \frac{|x-5|}{x-5}$ is

- (1) $[-5, 5]$ (2) $\{-5, 5\}$ (3) $\{-1, 1\}$ (4) $\{1, 5\}$

Sol. Answer (3)

For $x > 5$

$$f(x) = \frac{x-5}{x-5} = 1$$

for $x < 5$

$$f(x) = \frac{-(x-5)}{(x-5)} = -1$$

Hence $R_f = \{-1, 1\}$

57. If $f(x)$ is an identity function, then $f(5)$ is equal to

- (1) 0 (2) $\frac{1}{5}$ (3) 1 (4) 5

Sol. Answer (4)

58. For a constant function $f(x)$, given that $f\left(\frac{1}{2}\right) = 4$. The value of $f(2)$ is

- (1) $\frac{1}{2}$ (2) 2 (3) 4 (4) 8

Sol. Answer (3)

59. If f be a function defined as $f(x) = x$ for each $x \in R$, then f is called

- (1) Constant function (2) Identity function (3) Modulus function (4) Signum function

Sol. Answer (2)

60. If f be a function defined as $f(x) = p$ for each $x \in R$, where p is a real number, then f is called

- (1) Constant function (2) Identity function (3) Modulus function (4) Signum function

Sol. Answer (1)

61. If $f(x)$ and $g(x)$ are polynomial functions of x , then domain of $\frac{f(x)}{g(x)}$ is

(1) $(-\infty, 0]$ (2) $[0, \infty)$ (3) R (4) $R - \{x : g(x) = 0, x \in R\}$

Sol. Answer (4)

62. The function $f(x) = x^2 - 3x + 7$ is an example of

- (1) Identity function (2) Constant function
(3) Polynomial function (4) Modulus function

Sol. Answer (3)

63. If $f(x)$ is a signum function, then $f(10)$ is equal to

- (1) -1 (2) 0 (3) 1 (4) 10

Sol. Answer (3)

64. For a signum function $f(x)$, the value of $f(x)$ at $x = -4$ is

- (1) -1 (2) 0 (3) -4 (4) 4

Sol. Answer (1)

65. If $f(x)$ is a greatest integer function, then $f(-2.5)$ is equal to

- (1) -3 (2) -2 (3) 2 (4) 3

Sol. Answer (1)

66. The value of $\frac{f(1.5) - f(1)}{0.25}$, where $f(x) = x^2$, is

- (1) 1 (2) 1.5 (3) 2.5 (4) 5

Sol. Answer (4)

67. Which of the following can represent a linear function for each $x \in R$?

- (1) $f(x) = x^2 + 5$ (2) $f(x) = \frac{1}{x^2}$
 (3) $f(x) = mx + c, m \neq 0$ (4) $f(x) = x^3 + 3x^2 + 2$

Sol. Answer (3)

68. The maximum number of values of x if

$$|x - 2| + |x - 4| = 2 \text{ is}$$

- (1) 1 (2) 2 (3) 3 (4) Infinitely many

Sol. Answer (4)

If $2 \leq x \leq 4$ then

$$|x - 2| = x - 2$$

$$|x - 4| = -(x - 4)$$

$$\text{Then } |x - 2| + |x - 4| = 2$$

Therefore infinite values of x $|x - 2| + |x - 4| = 2$

69. The value of x if $0 \leq |2x + 3| \leq 3$ belongs to

- (1) $[-3, 0]$ (2) $[1, 2]$ (3) $[2, 3]$ (4) $[3, 4]$

Sol. Answer (1)

$$0 \leq |2x + 3| \leq 3$$

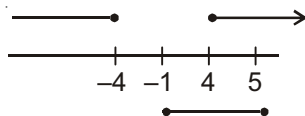
$$\Rightarrow 0 \leq 2x + 3 \leq 3 \text{ or } -3 \leq 2x + 3 \leq 0$$

On solving we get $x \in [-3, 0]$

70. If $|x + 4| + |x - 4| = 2|x|$ and $|x + 1| + |5 - x| = 6$, then x belongs to

- (1) $[4, 6]$ (2) $[-1, 4]$ (3) $[4, 5]$ (4) $[-2, -1]$

Sol. Answer (3)



We know that

$$|x + y| = |x| + |y| \text{ if } xy \geq 0$$

$$\text{Therefore } (x + 4)(x - 4) \geq 0$$

$$x \in (-\infty, -4] \cup [4, \infty) \quad \dots (i)$$

$$\text{And } (x + 1)(5 - x) \geq 0$$

$$x \in [-1, 5] \quad \dots (ii)$$

From (i) and (ii) $x \in [4, 5]$

71. If $[]$ and $\{ \}$ represents the greatest integer function and fractional function then the graph of $y = [x]$ and $y = \{x\}$ respectively are parallel to
- (1) x-axis, y-axis (2) y-axis, x-axis (3) x-axis, $y = x$ (4) $y = 2x$, $y = x + 1$

Sol. Answer (3)

It is clear by the graph of $[x]$ and $\{x\}$

72. If $-1 \leq [2x^2 - 3] < 2$, then x belongs to

- (1) $-\sqrt{\frac{5}{2}} < x \leq -1$ only (2) $1 \leq x < \sqrt{\frac{5}{2}}$ only
- (3) $-\sqrt{\frac{5}{2}} < x \leq -1$ or $1 \leq x < \sqrt{\frac{5}{2}}$ (4) $-1 \leq x \leq 1$

Sol. Answer (3)

$$-1 \leq [2x^2 - 3] < 2$$

$$-1 \leq 2x^2 - 3 < 2 \text{ [because } n_1 \leq [x] < n_2, n_1 \leq x < n_2 \text{]}$$

$$2 \leq 2x^2 < 5$$

$$1 \leq x^2 < \frac{5}{2}$$

$$\text{On solving } x \in \left(-\sqrt{\frac{5}{2}}, -1\right] \cup \left[1, \sqrt{\frac{5}{2}}\right)$$

73. If $f(x) = \log\left(\frac{1+x}{1-x}\right)$, then $f\left(\frac{3x+x^3}{1+3x^2}\right)$ is equal to

- (1) $(f(x))^3$ (2) $(f(x))^2$ (3) $3f(x)$ (4) $2f(x)$

Sol. Answer (3)

$$f\left(\frac{3x+x^3}{1+3x^2}\right) = \log\left(\frac{1+\frac{3x+x^3}{1+3x^2}}{1-\frac{3x+x^3}{1+3x^2}}\right) = \log\left(\frac{x^3+3x^2+3x+1}{1-x^3+3x^2-3x}\right) = \log\frac{(x+1)^3}{(1-x)^3} = \log\left(\frac{1+x}{1-x}\right)^3 = 3f(x)$$

74. The domain of $f(x) = \log|x| + \frac{1}{\sqrt{|x|}} + \frac{1}{\log|x|}$ is $R - A$ where A is the set

- (1) $\{-1, 0, 1\}$ (2) $\{-1, 1\}$ (3) $\{2, 3, 4\}$ (4) $\{0, 1, 2\}$

Sol. Answer (1)

$$f(x) = \log|x| + \frac{1}{\sqrt{|x|}} + \frac{1}{\log|x|}$$

Clearly $x \neq 0, 1$ and -1

75. The largest set of real values of x for which

$$f(x) = \sqrt{(x+2)(5-x)} - \frac{1}{\sqrt{x^2-4}} \text{ is a real function is}$$

(1) $[1, 2) \cup (2, 5]$

(2) $(2, 5]$

(3) $[3, 4]$

(4) $\left[\frac{1}{\sqrt{2}}, 1\right]$

Sol. Answer (2)

$$f(x) = \sqrt{(x+2)(5-x)} - \frac{1}{\sqrt{x^2-4}}$$

For $f(x)$ to be real, we have

$$(x+2)(5-x) \geq 0 \quad \text{and} \quad x^2 - 4 > 0$$

$$\text{i.e. } (x+2)(x-5) \leq 0 \quad \& \quad (x+2)(x-2) > 0$$

Required Domain $(2, 5]$

76. The domain of the function $f(x) = \frac{x^2+1}{\ln(x^2+1)}$ is

(1) $[0, \infty)$

(2) $(-\infty, 0]$

(3) $(-\infty, \infty)$

(4) $(-\infty, \infty) - \{0\}$

Sol. Answer (4)

$$f(x) = \frac{x^2+1}{\ln(x^2+1)}$$

for $f(x)$ to be real $\frac{x^2+1}{\ln(x^2+1)}$ should be real which is real for all $x \in \mathbb{R} - \{0\}$

77. The domain of definition of the function

$$f(x) = \log_{3/2} \log_{1/2} \log_{\pi} \log_{\pi/4} x \text{ is}$$

(1) $(0, \infty)$

(2) $\left(0, \left(\frac{\pi}{4}\right)^{\pi}\right)$

(3) $\left(\left(\frac{\pi}{4}\right)^{\pi}, \frac{\pi}{4}\right)$

(4) $\left(\left(\frac{\pi}{2}\right)^{\pi}, \infty\right)$

Sol. Answer (3)

$$f(x) = \log_{\frac{3}{2}} \log_{\frac{1}{2}} \log_{\pi} \log_{\frac{\pi}{4}} x$$

For $f(x)$ to be real

$$\log_{\frac{1}{2}} \log_{\pi} \log_{\frac{\pi}{4}} x > 0$$

$$\Rightarrow 0 < \log_{\pi} \log_{\frac{\pi}{4}} x < \left(\frac{1}{2}\right)^0 = 1 \Rightarrow \left(\frac{\pi}{4}\right)^{\pi} < x < \frac{\pi}{4}$$

$$\Rightarrow \pi^0 < \log_{\frac{\pi}{4}} x < \pi \Rightarrow x \in \left(\left(\frac{\pi}{4}\right)^{\pi}, \frac{\pi}{4}\right)$$

78. The domain of $\log_{[x+1]}(x^2 - 1)$, where $[\]$ represents greatest integer function, is

- (1) $(-\infty, 0)$ (2) $(-\infty, \infty)$ (3) $(1, \infty)$ (4) $(0, \infty)$

Sol. Answer (3)

$$f(x) = \log_{[x+1]}(x^2 - 1)$$

For 'f' to be defined

$$x^2 - 1 > 0 \Rightarrow x \in (-\infty, -1) \cup (1, \infty) \quad \dots(i)$$

$$[x + 1] > 0, [x + 1] \neq 1$$

$$x + 1 \geq 1 \cup x + 1 \notin [1, 2)$$

$$x \geq 0, x \notin [0, 1) \quad \dots(ii)$$

From (i) and (ii), $x \in (1, \infty)$

79. The domain of the function $f(x) = \sqrt{x^2 - [x]^2}$, where $[\cdot]$ denotes greatest integer function less than or equal to x

- (1) R (2) $[0, \infty) \cup Z$ (3) $(-\infty, 0]$ (4) R^+

Sol. Answer (2)

$$f(x) = \sqrt{x^2 - [x]^2}, \text{ where } [\cdot] \text{ denotes the G.I.F.}$$

for $f(x)$ to be real

$$x^2 - [x]^2 \geq 0$$

$$\Rightarrow (x + [x])(x - [x]) \geq 0$$

$$\Rightarrow x + [x] \geq 0, \text{ as } x - [x] \geq 0$$

$$\Rightarrow x \in [0, \infty) \cup Z$$

Hence domain $(f) = [0, \infty) \cup Z$

80. The domain of the function $f(x) = \sqrt{x - \sqrt{1 - x^2}}$ is

(1) $\left[-1, \frac{-1}{\sqrt{2}}\right] \cup \left[\frac{1}{\sqrt{2}}, 1\right]$

(2) $[-1, 1]$

(3) $\left(-\infty, -\frac{1}{2}\right] \cup \left[\frac{1}{\sqrt{2}}, +\infty\right)$

(4) $\left[\frac{1}{\sqrt{2}}, 1\right]$

Sol. Answer (4)

$$f(x) = \sqrt{x - \sqrt{1 - x^2}}$$

for $f(x)$ to be real

$$x - \sqrt{1 - x^2} \geq 0 \quad \text{and} \quad \sqrt{1 - x^2} \geq 0$$

$$x^2 \geq 1 - x^2 \quad \text{and} \quad 1 - x^2 \geq 0$$

$$\Rightarrow 2x^2 - 1 \geq 0 \quad \text{and} \quad x^2 - 1 \leq 0$$

$$\Rightarrow \left(x + \frac{1}{\sqrt{2}}\right) \left(x - \frac{1}{\sqrt{2}}\right) \geq 0 \quad \text{and } (x+1)(x-1) \leq 0$$

$$\Rightarrow x \leq -\frac{1}{\sqrt{2}} \text{ or } x \geq \frac{1}{\sqrt{2}} \quad \text{and } x \in [-1, 1]$$

But when $x \in \left(-1, \frac{1}{\sqrt{2}}\right)$

$x - \sqrt{1-x^2}$ is negative

Hence required domain is $\left[\frac{1}{\sqrt{2}}, 1\right]$

81. The range of $f(x) = 2 + 2x - x^2$ is

- (1) $(-\infty, \infty)$ (2) $(-\infty, 1)$ (3) $(-\infty, 3]$ (4) $(0, \infty)$

Sol. Answer (3)

$$f(x) = -[x^2 - 2x - 2]$$

Making perfect square

$$f(x) = -[x^2 - 2x + 1 - 1 - 2]$$

$$f(x) = -(x-1)^2 + 3$$

Range of function is $(-\infty, 3]$

82. The range of the expression $2^x + 2^{-x} + 3^x + 3^{-x}$ for $x \in R$, is

- (1) $(-\infty, \infty)$ (2) $[4, \infty)$ (3) $(6, \infty)$ (4) $(8, \infty)$

Sol. Answer (2)

$$f(x) = 2^x + 2^{-x} + 3^x + 3^{-x}$$

Applying A.M. – G.M. inequality

$$\frac{2^x + 2^{-x} + 3^x + 3^{-x}}{4} \geq (2^x \cdot 2^{-x} \cdot 3^x \cdot 3^{-x})^{\frac{1}{4}}$$

$$\Rightarrow 2^x + 2^{-x} + 3^x + 3^{-x} \geq 4 \quad (\text{equality holds at } x = 1)$$

83. The range of the function $y = f(x)$ if $(34)^x + (34)^y = 34$ equals

- (1) $(-\infty, -1)$ (2) $(-\infty, 1)$ (3) $(-\infty, 1]$ (4) $(1, \infty)$

Sol. Answer (2)

$$(34)^x + (34)^y = 34$$

$$\Rightarrow (34)^x = 34 - 34^y$$

$$\Rightarrow x = \log_{34}(34 - 34^y)$$

$$\Rightarrow 34 - 34^y > 0 \Rightarrow y < 1$$

$$\Rightarrow y \in (-\infty, 1)$$

84. Which of the following is a function?

(1) $\{(x, y) : y^2 = x, x, y \in \mathbb{R}\}$

(2) $\{(x, y) : y = |x|, x, y \in \mathbb{R}\}$

(3) $\{(x, y) : x^2 + y^2 = 1, x, y \in \mathbb{R}\}$

(4) $\{(x, y) : x^2 - y^2 = 1, x, y \in \mathbb{R}\}$

Sol. Answer (2)

$y = |x|$ is a function because any line parallel to y -axis intersects the given graph at one point only. Also for any values of x is $y^2 = x$, $x^2 + y^2 = 1$ and $x^2 - y^2 = 1$ gives more than one values of y .

$\Rightarrow$ (1), (3), (4) are not functions.

85. If $f(x)$ is a polynomial function of the second degree such that $f(-3) = 6$, $f(0) = 6$ and $f(2) = 11$, then the graph of the function $f(x)$ cuts the ordinate at $x = 1$ at the point

(1) (1, 8)

(2) (1, 4)

(3) (-2, 1)

(4) (1, 9)

Sol. Answer (1)

Let

$$f(x) = ax^2 + bx + c$$

$$f(0) = 6 \Rightarrow c = 6$$

$$f(-3) = 6 \Rightarrow 9a - 3b + 6 = 6 \Rightarrow b = 3a$$

$$f(2) = 11 \Rightarrow 4a + 2b + 6 = 11 \Rightarrow 10a = 5 \Rightarrow a = \frac{1}{2}$$

$$\Rightarrow b = \frac{3}{2} \text{ and } c = 6$$

$$f(x) = \frac{1}{2}x^2 + \frac{3}{2}x + 6$$

At $x = 1$

$$f(1) = \frac{1}{2} \times 1 + \frac{3}{2} \times 1 + 6 = 8$$

86. Let $f(x) = \frac{1}{x - [x]}$

$$g(x) = \sqrt{\ln\{x\} - \ln[x]}$$

$$h(x) = \log_{(0.5+x)} \left(\frac{x^2 + 2x - 3}{4x^2 - 4x - 3} \right), \text{ where } \{x\}, [x] \text{ represent fractional function and greatest integer function then}$$

(1) The domain of $f(x)$ is $\mathbb{R} - \mathbb{Z}^+$

(2) The domain of $h(x)$ is $\left(-\frac{1}{2}, \frac{1}{2}\right) \cup \left(\frac{1}{2}, 1\right) \cup \left(\frac{3}{2}, \infty\right)$

(3) The domain of $g(x)$ is $\mathbb{R}^+$

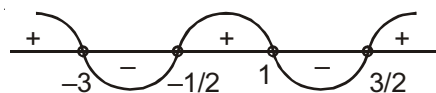
(4) The domain of $g(x)$ is $\mathbb{R}$

Sol. Answer (2)

$\{x\} = 0$ if $x \in \mathbb{I}$ hence $\{x\}^{\{x\}}$ will not be defined at any integer $[x] = 0$ if $x \in [0, 1)$ hence $[x]^{[x]}$ will not be defined in $[0, 1)$, hence domain of $f(x) = \{x\}^{\{x\}} + [x]^{[x]}$ will be $\mathbb{R} - \{\mathbb{I} \cup (0, 1)\}$.

$$h(x) = (x + 0.5)^{\log_{(x+0.5)} \left(\frac{x^2 + 2x - 3}{4x^2 - 4x - 3} \right)}$$

For $\log_{(0.5+x)} \left(\frac{x^2 + 2x - 3}{4x^2 - 4x - 3} \right)$ to be defined.



$$\frac{x^2 + 2x - 3}{4x^2 - 4x - 3} > 0$$

$$\frac{(x+3)(x-1)}{\left(x+\frac{1}{2}\right)\left(x-\frac{3}{2}\right)} > 0$$

And $(0.5 + x) > 0$ and $x + 0.5 \neq 1$

$$\boxed{x > -\frac{1}{2}}$$

$$x \neq 0.5$$

Hence domain $\left(-\frac{1}{2}, \frac{1}{2}\right) \cup \left(\frac{1}{2}, 1\right) \cup \left(\frac{3}{2}, +\infty\right)$

$$g(x) = \sqrt{\ln\{x\} - \ln[x]}$$

$$\ln\{x\} - \ln[x] \geq 0$$

$$\ln\{x\} \geq \ln[x]$$

$$\{x\} \geq [x]$$

$$\Rightarrow x \in (-\infty, 1)$$

For logarithm to be defined $\{x\} > 0 \mid [x] > 0$
 $\Rightarrow x \in \mathbb{R} \mid x \in [1, \infty)$

Hence no common domain, hence domain is null set.

87. The solution set for $[x]\{x\} = 1$ [where $\{x\}$ and $[x]$ are respectively fractional part and greatest integer function.] is

(1) $\mathbb{R}^+ - (0, 1)$

(2) $\mathbb{R}^+ - \{1\}$

(3) $\left\{m + \frac{1}{m} \mid m \in \mathbb{I} - \{0\}\right\}$

(4) $\left\{m + \frac{1}{m} \mid m \in \mathbb{N} - \{1\}\right\}$

Sol. Answer (4)

$$[x] = \frac{1}{\{x\}}$$

$x \notin \mathbb{I}$ therefore if $x \in \mathbb{I}, \{x\} = 0$ hence A and B options are wrong.

$$\begin{array}{l} \therefore \{x\} \in [0, 1) \quad \left| \quad \{x\} = \frac{1}{[x]} \quad \Rightarrow \quad \text{If } \{x\} \text{ gives } \frac{1}{m} \right. \\ \therefore \frac{1}{\{x\}} \in (1, \infty) \quad \left| \quad \therefore \{x\} = \frac{1}{m}, m \in \mathbb{I} \quad \text{[x] gives } m. \right. \end{array}$$

If $x = m + \frac{1}{m}, m \in \mathbb{I}$

$$\Rightarrow [x] = \left[m + \frac{1}{m} \right] = m$$

$$\{x\} = \left\{ m + \frac{1}{m} \right\} = \frac{1}{m}$$

$$\therefore \left[m + \frac{1}{m} \right] = \frac{1}{\left\{ m + \frac{1}{m} \right\}} \text{ but } m \neq 0 \Rightarrow x = \left\{ m + \frac{1}{m} \mid m \in \mathbb{N} - \{1\} \right\}$$

88. Domain of the function

$$f(x) = \sqrt{(5x-6-x^2)[\ln\{x\}^4]} + \sqrt{7x-5-2x^2} + \left(\ln\left(\frac{7}{2}-x\right) \right)^{-1}$$

$\{ \cdot \}$ represents fractional part function and $[\cdot]$ represents G.I.F.

- (1) $\mathbb{R}$ (2) $\left[1, \frac{5}{2} \right]$ (3) $(1, 2)$ (4) $(1, 2) \cup \left(2, \frac{5}{2} \right]$

Sol. Answer (4)

$$f(x) = \sqrt{(5x-6-x^2)[\ln\{x\}^4]} + \sqrt{7x-5-2x^2} + \left(\ln\left(\frac{7}{2}-x\right) \right)^{-1}$$

For log to be defined $\{x\} \neq 0$

$$\Rightarrow x \notin \mathbb{I}$$

$$7x-5-2x^2 \geq 0$$

$$(x-1)\left(x-\frac{5}{2}\right) \leq 0$$



$$x \in \left[1, \frac{5}{2} \right]$$

$$\ln\left(\frac{7}{2}-x\right) \neq 0 \text{ being in denominator,}$$

$$\frac{7}{2}-x \neq 1$$

$$x \neq \frac{5}{2}$$

$$\text{Also } \frac{7}{2}-x > 0$$

$$x < \frac{7}{2}$$

$$\text{Domain } (1, 2) \cup \left(2, \frac{5}{2} \right) - \left\{ p^{1/4} \right\} \text{ where } p \in \{2, 3, \dots, 38, 39\}$$

89. If $f(x)$ is defined on $(0, 1)$, the domain of definition of $f(e^x) + f(\ln|x|)$ is

- (1) $(0, 1)$ (2) R (3) $(-e, -1)$ (4) $(-e, e) - [-1, 1]$

Sol. Answer (3)

$\therefore f(x)$ is defined from $(0, 1) \therefore x \in (0, 1)$ for $f(e^x)$ to be defined $0 < e^x < 1 \Rightarrow -\infty < x < 0$

$f(\ln|x|)$ to define $0 < \ln|x| < 1$

$$\Rightarrow e^0 < |x| < e^1$$

$$\Rightarrow 1 < |x| < e$$

$$\Rightarrow x \in (-e, -1) \cup (1, e)$$

Hence domain of given function will be $(-e, -1)$

90. The domain of function $f(x) = \log_3(-(\log_3 x)^2 + 5(\log_3 x) - 6)$ is

- (1) $(9, 27)$ (2) $[9, 27]$ (3) $(0, 9) \cup (27, \infty)$ (4) $(0, 9] \cup [27, \infty)$

Sol. Answer (1)

$$f(x) = \log_3(-(\log_3 x)^2 + 5(\log_3 x) - 6)$$

$$-(\log_3 x)^2 + 5(\log_3 x) - 6 > 0$$

$$(\log_3 x)^2 - 5(\log_3 x) + 6 < 0$$

$$(\log_3 x - 2)(\log_3 x - 3) < 0$$

$$2 < \log_3 x < 3$$

$$9 < x < 27$$

91. If domain of $y = f(x)$ is $[-4, 3]$, then domain of $g(x) = f([x])$ is, where $[.]$ denotes greatest integer function

- (1) $(-3, 4)$ (2) $[-3, 4)$ (3) $[-5, 3]$ (4) $[-4, 3]$

Sol. Answer (2)

By given condition, $-4 \leq [x] \leq 3$

$$\Rightarrow 0 \leq [x] \leq 3$$

$$\Rightarrow -3 \leq [x] \leq 3$$

$$\Rightarrow -3 \leq x < 4$$

$\therefore$ Domain of $g(x)$ is $[-3, 4)$.

92. Domain of definition of the function

$$f(x) = \frac{9}{9-x^2} + \log_{10}(x^3 - x), \text{ is}$$

- (1) $(-1, 0) \cup (1, \infty)$ (2) $(1, 3) \cup (3, \infty)$
(3) $(-1, 0) \cup (1, 3) \cup (3, \infty)$ (4) $(-1, 0) \cup (3, \infty)$

Sol. Answer (3)

$$f(x) = \frac{9}{9-x^2} + \log_{10}(x^3 - x)$$

$f(x)$ is real if $9 - x^2 \neq 0$ and $x^3 - x > 0$

$$\Rightarrow x \neq \pm 3 \text{ and } x(x-1)(x+1) > 0$$

$$\Rightarrow x \neq \pm 3 \text{ and } x \in (-1, 0) \cup (1, \infty)$$

$$\therefore \text{Domain of } f(x) \text{ is } (-1, 0) \cup (1, 3) \cup (3, \infty)$$

93. Range of function $f(x) = \cos(k \sin x)$ is $[-1, 1]$, then the least positive integral value of k will be

(1) 1

(2) 2

(3) 3

(4) 4

Sol. Answer (4)

$$\therefore -1 \leq \sin x \leq +1$$

$$-k \leq k \sin x \leq +k$$

For $\cos \theta$ to show complete range its domain should be $[0, \pi]$ or $[-\pi, 0]$

Hence $k = \pi$

Hence minimum integral value is $k = 4$

94. If $f(x) = 4^x - 2^{x+1} + 5$, then range of $f(x)$ is

(1) R (2) $[5, \infty)$ (3) $[4, \infty)$ (4) $[0, \infty)$ **Sol.** Answer (3)

$$\begin{aligned} f(x) &= 4^x - 2^{x+1} + 5 = (2^x)^2 - 2 \cdot 2^x + 5 \\ &= (2^x - 1)^2 + 4 \end{aligned}$$

$$\therefore (2^x - 1)^2 \geq 0$$

$$f(x) = (2^x - 1)^2 + 4 \geq 4. \text{ Hence range } [4, \infty).$$

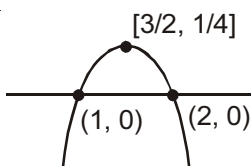
95. If $f(x) = \ln(x^2 + 7|x| + 10)$ is a single valued real function then the range of $f(x)$ in its natural domain will be

(1) $[0, +\infty)$ (2) $[\ln 10, +\infty)$ (3) $[0, 10]$ (4) R **Sol.** Answer (2)

$$f(x) = \ln(x^2 + 7|x| + 10)$$

$$\therefore 10 \leq x^2 + 7|x| + 10 < \infty \text{ hence range is } [\ln 10, \infty)$$

96. Range of the function $f(x) = \log_{0.5}(3x - x^2 - 2)$

(1) $(-\infty, 2]$ (2) $[2, +\infty)$ (3) R (4) $R - (2)$ **Sol.** Answer (2)

$$f(x) = \log_{0.5}(3x - x^2 - 2)$$

Domain is $1 < x < 2$

$\therefore$ If $1 < x < 2$

$$0 < 3x - x^2 - 2 \leq \frac{1}{4}$$

$$\infty > \ln_{1/2}(3x - x^2 - 2) \geq 2$$

$\therefore$ Range $[2, \infty)$

97. If $f(x)$ is a real valued function defined as $f(x) = \begin{cases} \min\left\{|x|, \frac{1}{x^2}, \frac{1}{x^3}\right\} & , x \neq 0 \\ 1 & , x = 0 \end{cases}$ then range of $f(x)$ is

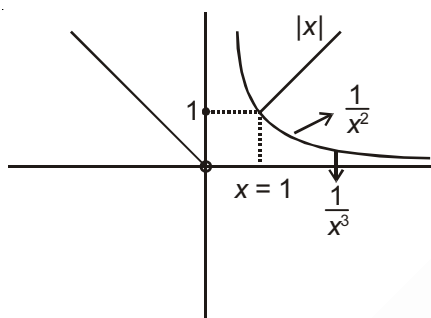
(1) $(-\infty, 1]$

(2) $(-\infty, 1] - \{0\}$

(3) $[1, \infty)$

(4) R

Sol. Answer (2)



$$f(x) = \begin{cases} \min\left\{|x|, \frac{1}{x^2}, \frac{1}{x^3}\right\} & , x \neq 0 \\ 1 & , x = 0 \end{cases}$$

$$\text{Redefine the function} = \begin{cases} \frac{1}{x^3} & \text{if } x < 0 \\ 1 & x = 0 \\ |x| & 0 < x < 1 \\ \frac{1}{x^3} & 1 < x \end{cases}$$

Hence range is $= (-\infty, 1] - \{0\}$

98. Let $f(x) = [x]^2 + [x + 1] - 3$, where $[.]$ denotes the greatest integer function. Then

(1) $f(x) \neq 0$ for all real values of x

(2) $f(x) = 0$ for only two real value of x

(3) $f(x) = 0$ for infinite values of x

(4) $f(x) = 0$ for no real value of x .

Sol. Answer (3)

$$f(x) = [x]^2 + [x + 1] - 3 = 0$$

$$[x]^2 + [x] + 1 - 3 = 0$$

$$[x]^2 + [x] - 2 = 0$$

$$\{[x] + 2\} \{[x] - 1\} = 0$$

$$\Rightarrow [x] = -2, [x] = 1$$

$$\Rightarrow x \in [-2, -1) \cup [1, 2) \text{ Hence } f(x) = 0 \text{ for infinite values of } x.$$

99. The range of the function $f(x) = \cos^2 x - 5\cos x - 9$ is

- (1) $[-13, 3]$ (2) $[0, 3]$ (3) $[-13, -3]$ (4) $[-13, -9]$

Sol. Answer (3)

$$f(x) = \cos^2 x - 5\cos x - 9$$

$$= \left(\cos^2 x - 5\cos x + \frac{25}{4} \right) - 9 - \frac{25}{4}$$

$$= \left(\cos x - \frac{5}{2} \right)^2 - \frac{61}{4} \quad \dots(i)$$

$$\text{where } \frac{9}{4} \leq \left(\cos x - \frac{5}{2} \right)^2 \leq \frac{49}{4} \quad \dots(ii)$$

$$\therefore \text{ from (i) and (ii) } -13 \leq f(x) \leq -3$$

$$\Rightarrow \text{ Range of } f(x) \in [-13, -3]$$

100. Is the function $f: N \rightarrow N$ (N is set of the natural numbers) defined by $f(n) = 2n + 3$ for all $n \in N$ Surjective?

- (1) Yes (2) No
(3) Can't say (4) Information is not sufficient

Sol. Answer (2)

No, this is not a surjective function. As Range of function is $[5, \infty]$, which is not equal to its co-domain.

101. If ' f ' is a function from a set A to A , then f is invertible iff ' f ' is

- (1) One-one (2) Onto
(3) Both one-one and onto (4) Many one

Sol. Answer (3)

A function is invertible iff it is both one-one and onto.

102. Set A has 3 elements and set B has 4 elements. The number of injections that can be defined from A to B is

- (1) 144 (2) 12 (3) 24 (4) 64

Sol. Answer (3)

$$\text{The number of injective mappings} = {}^4P_3 = 24$$

103. The number of Surjections from $A = \{1, 2, \dots, n\}$, $n \geq 2$, onto $B = \{a, b\}$ is

- (1) nP_2 (2) $2^n - 2$ (3) $2^n - 1$ (4) 2^n

Sol. Answer (2)

$$\text{The number of Surjections} = 2^n - {}^nC_1 \cdot 1^n = 2^n - 2$$

104. The total number of injective mappings from a set with m elements to a set with n elements, $m \leq n$, is

- (1) m^n (2) n^m (3) $\frac{n!}{(n-m)!}$ (4) $n!$

Sol. Answer (3)

$$\text{The total number of injective mappings} = {}^nP_m = \frac{n!}{(n-m)!}$$

105. Let $E = \{1, 2, 3, 4\}$ and $F = \{1, 2\}$. Then the number of onto functions from E to F is

- (1) 14 (2) 16 (3) 6 (4) 4

Sol. Answer (1)

$$\text{The number of onto functions} = 2^4 - 2 = 14$$

106. If $f: R \rightarrow R, f(x) = x^3 + 1$, then f is

- (1) One - one but not onto (2) Onto but not one-one
(3) One-one onto (4) Many one

Sol. Answer (3)

$$\text{Given } f: R \rightarrow R, f(x) = x^3 + 1,$$

$$\text{let } x_1, x_2 \in R$$

$$\text{let } f(x_1) = f(x_2) \Rightarrow x_1^3 + 1 = x_2^3 + 1 \Rightarrow x_1 = x_2.$$

So, function is one-one

Now, range of $f(x)$ is equal to its co-domain

So, function is onto

107. Function $f: R \rightarrow R, f(x) = x|x|$ is

- (1) One-one but not onto (2) Onto but not one-one
(3) One-one onto (4) Neither one-one nor onto

Sol. Answer (3)

Obviously function $f: R \rightarrow R, f(x) = x|x|$ is one-one onto function.

108. Function $f(x) = \frac{x^2}{1+x^2}$, is

- (1) Many-one function (2) Odd function (3) One-one function (4) Onto function

Sol. Answer (1)

$$\text{Given } f(x) = \frac{x^2}{1+x^2}$$

$$\text{As, } f(1) = f(-1) = \frac{1}{2}.$$

So $f(x)$ is Many-one function

Also, as $f(x)$ is always positive, so not onto function

109. Let $f: N \rightarrow N$ be defined as $f(x) = 2x$ for all $x \in N$, then f is

- (1) Onto (2) Invertible (3) One-one (4) Many one

Sol. Answer (3)

Obviously $f: N \rightarrow N$ be defined as $f(x) = 2x$ is one-one but not onto.

110. Let $A = \{1, 2, 3\}$. Which of the following functions on A is invertible ?

(1) $f = \{(1,1), (2,1), (3,1)\}$

(2) $f = \{(1,2), (2,3), (3,1)\}$

(3) $f = \{(1,2), (2,3), (3,2)\}$

(4) $f = \{(1,1), (2,2), (3,1)\}$

Sol. Answer (2)

As, only option (2) is one-one onto function. so this is invertible.

111. Function $f : R \rightarrow R, f(x) = x + |x|$, is

(1) One-one

(2) Onto

(3) One-one onto

(4) Many one

Sol. Answer (4)

As, $f(-1) = f(-2) = 0$. So, $f(x)$ is Many-one into function.

112. Function $f : \left[\frac{\pi}{2}, \frac{3\pi}{2}\right] \rightarrow [-1, 1], f(x) = \sin x$ is

(1) Many-one onto

(2) Onto

(3) One-one onto

(4) Many-one into

Sol. Answer (3)

Function $f : \left[\frac{\pi}{2}, \frac{3\pi}{2}\right] \rightarrow [-1, 1], f(x) = \sin x$

Obviously it is One-one onto function.

113. Function $f : \left[\frac{1}{2}\pi, \frac{3}{2}\pi\right] \rightarrow [-1, 1], f(x) = \cos x$ is

(1) Many-one onto

(2) Onto

(3) One-one onto

(4) Many-one into

Sol. Answer (4)

Function $f : \left[\frac{1}{2}\pi, \frac{3}{2}\pi\right] \rightarrow [-1, 1], f(x) = \cos x$

Obviously it is Many-one into.

114. If $f : R \rightarrow R, f(x) = \sin^2 x + \cos^2 x$, then f is

(1) One-one but not onto

(2) Onto but not one-one

(3) Neither one-one nor onto

(4) Both one-one and onto

Sol. Answer (3)

Obviously Many-one into.

115. If function $f(x) = (1 + 2x)$ has the domain $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ and co-domain $(-\infty, \infty)$ then function is

(1) One-one onto

(2) One-one but not onto

(3) Many-one but not onto

(4) Many-one onto

Sol. Answer (2)

Obviously One-one into.

116. The function $f : (0, \infty) \rightarrow [0, \infty)$, $f(x) = \frac{x}{1+x}$ is

- (1) One-one and onto (2) One-one but not onto
(3) Onto but not one-one (4) Neither one-one nor onto

Sol. Answer (2)

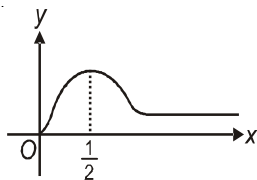
Obviously One-one into.

117. If $f : [0, \infty) \rightarrow [0, \infty)$ and $f(x) = \frac{x^2}{1+x^4}$, then f is

- (1) One-one and onto (2) One-one but not onto
(3) Onto but not one-one (4) Neither one-one nor onto.

Sol. Answer (4)

Clearly, function is neither one-one nor onto.



118. If $f_1(x) = 2x + 3$, $f_2(x) = 3x^3 + 5$, $f_3(x) = x + \cos x$ are defined from $R \rightarrow R$, then f_1 , f_2 and f_3 are

- (1) One-one-onto (2) Many one into (3) One-one-into (4) Many one onto

Sol. Answer (1)

$$f_3(x) = x + \cos x$$

$$f_3'(x) = 1 - \sin x \geq 0$$

$\therefore f_3'(x) = 0$ hold for only point $x = (2n+1)\frac{\pi}{2}$, $n \in I$ i.e. at discrete points not in interval hence function is strictly increasing. Hence function is one-one and onto.

Similarly we can prove for f_1 , f_2 .

119. Which of the functions defined below is one-one function?

- (1) $f(x) = \frac{(x^2+1)}{x^4+x^2+1}$; $(x \in R)$ (2) $g(x) = x + \frac{1}{x}$, $(x \in R^+)$
(3) $h(x) = x^2 + 4x - 5$ ($x \in R$) (4) $f(x) = e^{-x}$, $(x \in R, x \geq 0)$

Sol. Answer (4)

(1) $f = \left(\frac{x^2+1}{x^4+x^2+1} \right)$, $x \in R$ cannot be one-one since even.

(2) $g(x) = x + \frac{1}{x}$, $x \in R^+$ cannot be one-one since $f(a) = f\left(\frac{1}{a}\right)$

(3) $h(x) = x^2 + 4x - 5$, $x \in R$. Graph is upward opening parabola hence many-one.

(4) $f(x) = e^{-x}$, $x \in R, x \geq 0$. Strictly decreasing function hence one-one.

120. Which of the following is an onto function?

- (1) $f(x) = x^6 + x^4 + x^2 + 1, f: R \rightarrow R$
 (2) $f(x) = x^2|x|$, on $[-1, 1]$ $[-1, 1] \rightarrow [-1, 1]$
 (3) $y = x^{11} - x^8 + x^6 + 5$ on $(-\infty, \infty)$ $R \rightarrow R$
 (4) $y = x^{2006} + x^{-2006} + 5$ on $(-\infty, \infty)$ $R \rightarrow R$

Sol. Answer (3)

- (1) $y = x^6 + x^4 + x^2 + 1 = (1 + x^2)(1 + x^4)$
 $\Rightarrow y \geq 1$ hence it is not onto
 (2) $f(x) = x^2|x|$, range = $[0, 1]$
 Hence not onto.
 (3) $y = x^{11} - x^8 + x^6 + 5$
 $y \in R$ hence it is onto
 (4) $y = x^{2006} + x^{-2006} + 5$
 $y \geq 7$ hence it is not onto

121. If a function $f: R \rightarrow R$ is defined by $f(x) = x^3 + 4x^2 + cx + d$, where $c, d \in R$, if $f(x)$ is one-one onto, then c may be

- (1) 6 (2) 4 (3) 0 (4) -1

Sol. Answer (1)

$f(x) = x^3 + 4x^2 + cx + d$ is one-one.

$\therefore f'(x)$ will be either > 0 or < 0

$$f'(x) = 3x^2 + 8x + c$$

Hence, its discriminant $D < 0$

$$8^2 - 4(3)(c) < 0$$

$$\Rightarrow c > \frac{16}{3}$$

Hence, c can take value 6.

122. Let $A = \{1, 2, 3\}$, $B = \{a, b, c\}$, $C = \{a_1, b_1, c_1, d_1, e_1\}$ and consider the following statements

- S_1 : The number of one-one functions from A to C is 60
 S_2 : The number of onto functions from C to A is 150
 S_3 : The number of onto functions from B to C is zero
 S_4 : The number of bijective functions from A to B is 6

Which of the following combinations is true?

- (1) S_1 and S_2 only (2) S_1 and S_3 only (3) S_2 and S_4 only (4) S_1, S_2, S_3 and S_4

Sol. Answer (4)

All statements are true by using the standard result.

123. Which of the followings is not true?

- (1) $f: R \rightarrow R, f(x) = x^3 + x^2 + 3x + \sin x$ is bijective
 (2) $f(x) = x^3 + (a + 2)x^2 + 3ax + 5$ is one-one then $a \in (1, 4)$
 (3) $f(x) = ax + 3\sin x + 4\cos x$ is injective then $a \in (-\infty, -5] \cup [5, \infty)$
 (4) $f(x) = 2x^3 + 6x^2 + 12x + a\sin x + b\cos x$ is injective then the maximum value of $a^2 + b^2$ is 5

Sol. Answer (4)

$$f(x) = x^3 + x^2 + 3x + \sin x$$

$$f'(x) = 3x^2 + 2x + 3 + \cos x = 3\left(x + \frac{1}{3}\right)^2 + \frac{8}{3} + \cos x > 0$$

$$\Rightarrow f'(x) > 0, \forall x \in R$$

Hence $f(x)$ is one-one and onto.

Similarly (2), (3) are true but (4) is false. The maximum value of $a^2 + b^2$ is 36.

124. If $f: R \rightarrow R$ and $f(x) = x \left(\frac{2^x - 2^{-x}}{2^x + 2^{-x}} \right)$ is

(1) One-one and into function

(2) Many one and into function

(3) One-one and onto function

(4) Many one and onto function

Sol. Answer (2)

$$f(x) = x \left(\frac{2^x - 2^{-x}}{2^x + 2^{-x}} \right)$$

$$\text{and } f(-x) = (-x) \left[\frac{2^{-x} - 2^x}{2^{-x} + 2^x} \right] = x \left[\frac{2^x - 2^{-x}}{2^x + 2^{-x}} \right]$$

$\therefore f(-x) = f(x)$ hence function is even. Hence not one-one. Also function is continuous.

Any even-continuous function cannot have range ' R '. Hence function is many one into.

125. Let S be the set of all triangles and R^+ be the set of positive real numbers, then the function

$f: S \rightarrow R^+$, $f(\Delta) = \text{Perimeter of } \Delta \in S$, then ' f ' is

(1) Injective but not surjective

(2) Surjective but not injective

(3) Injective as well as surjective

(4) Neither injective nor surjective

Sol. Answer (2)

Since many triangles may have same perimeter hence function is not injective but many one.

Since there exist a triangle for every positive value of perimeter hence surjective.

126. Let $A = \{0, 1\}$ and the set of all natural numbers. Then the mapping $f: N \rightarrow A$ defined by $f(2n-1) = 0, f(2n) = 1, \forall n \in N$, is

(1) One-one onto

(2) Many one onto

(3) One-one into

(4) Many one into

Sol. Answer (2)

$$f(2n) = 1 \Rightarrow f(2) = f(4) = f(6) = 1. \text{ It is many one and range} = \text{co-domain}$$

$\therefore$ It is many-one onto.

127. Let $f: \left(-1, \frac{-1}{\sqrt{3}}\right) \rightarrow B$, be a function defined by $f(x) = \tan^{-1} \frac{3x - x^3}{1 - 3x^2}$, then f is both one-one and onto when

B is the interval

(1) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

(2) $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

(3) $\left(0, \frac{3\pi}{4}\right)$

(4) $\left(-\frac{3\pi}{4}, 0\right)$

Sol. Answer (2)

In question $f(x)$ must be $3 \tan^{-1}x$

Now, $-1 < x < 1$

$$\Rightarrow -\frac{\pi}{4} < \theta < \frac{\pi}{4} \Rightarrow \frac{-3\pi}{4} < 3\theta < \frac{3\pi}{4}$$

$$\therefore f(x) = 3\theta = 3\tan^{-1}x \text{ and } -\frac{3\pi}{4} < f(x) < \frac{3\pi}{4}$$

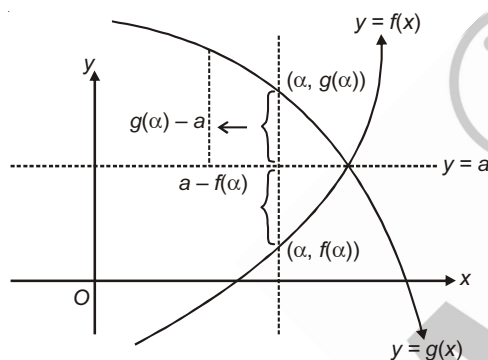
For which $f(x)$ is one-one as well as onto.

$$\text{So, } B = \left(-\frac{3\pi}{4}, \frac{3\pi}{4}\right)$$

128. Let $f: R \rightarrow R$ and $g: R \rightarrow R$ be two one-one and onto function such that they are the mirror images of each other about the line $y = a$ if, $h(x) = f(x) + g(x)$, then $h(x)$ is

- | | |
|----------------------|------------------------------|
| (1) One-one and onto | (2) One-one only |
| (3) Onto only | (4) Neither one-one nor onto |

Sol. Answer (4)



$$g(\alpha) - a = a - f(\alpha)$$

$$f(\alpha) + g(\alpha) = 2a$$

$$\text{In general } f(x) + g(x) = 2a$$

$$\therefore h(x) = 2a = \text{constant}$$

$h(x)$ is neither one-one and onto.

129. Let $f(x) = [x]$ and $g(x) = x - [x]$, then which of the following functions is the zero function ?

- | | | | |
|------------------|---------------|------------------|----------------|
| (1) $(f + g)(x)$ | (2) $(fg)(x)$ | (3) $(f - g)(x)$ | (4) $(fog)(x)$ |
|------------------|---------------|------------------|----------------|

Sol. Answer (4)

Here $f(g(x)) = f(x - [x]) = f(\{x\}) = [\{x\}] = 0$, where $\{.\}$ represents fractional part function.

130. $g \circ f$ exists, when

- | | |
|--------------------------------------|---|
| (1) Domain of f = domain of g | (2) Co-domain of f = domain of g |
| (3) Co-domain of g = domain of f | (4) Co-domain of g = co-domain of f |

Sol. Answer (2)

Obviously $g \circ f$ exists, when co-domain of f = domain of g

131. If $f: R \rightarrow R, f(x) = x^2 + 2x - 3$ and $g: R \rightarrow R, g(x) = 3x - 4$ then the value of $fg(x)$ is

- (1) $3x^2 + 6x - 13$ (2) $9x^2 - 18x + 5$ (3) $(3x - 4)^2 + 2x - 3$ (4) $x^2 + 1$

Sol. Answer (2)

$$\text{We have, } f(g(x)) = f(3x - 4) = (3x - 4)^2 + 2(3x - 4) - 3 = 9x^2 - 18x + 5$$

132. If $f: R \rightarrow R, f(x) = x^2 - 5x + 4$ and $g: R^+ \rightarrow R, g(x) = \log x$, then the value of $(g \circ f)(2)$ is

- (1) 0 (2) ∞ (3) $-\infty$ (4) Undefined

Sol. Answer (4)

$$\text{We have, } g(f(x)) = g(x^2 - 5x + 4) = \log(x^2 - 5x + 4)$$

$$\text{So, } g(f(2)) = \log(2^2 - 5 \cdot 2 + 4) = \log(-2) \text{ i.e. undefined}$$

133. If $f(x) = 2 \sin x, g(x) = \cos^2 x$, then the value of $(f + g)\left(\frac{\pi}{3}\right)$

- (1) 1 (2) $\frac{2\sqrt{3}+1}{4}$ (3) $\sqrt{3} + \frac{1}{4}$ (4) 0

Sol. Answer (3)

$$\text{Here, } f\left(\frac{\pi}{3}\right) + g\left(\frac{\pi}{3}\right) = 2 \sin \frac{\pi}{3} + \cos^2 \frac{\pi}{3} = \sqrt{3} + \frac{1}{4}$$

134. Given $f(x) = \frac{1}{(1-x)}$, $g(x) = f\{f(x)\}$ and $h(x) = f\{f\{f(x)\}\}$, then the value of $f(x) \cdot g(x) \cdot h(x)$ is

- (1) 0 (2) -1 (3) 1 (4) 2

Sol. Answer (2)

$$\text{Here, } f(x) = \frac{1}{(1-x)}, g(x) = f\{f(x)\} = \frac{1}{1 - \frac{1}{1-x}} = \frac{x-1}{x}$$

$$\text{and } h(x) = f\{f\{f(x)\}\} = \frac{1}{1 - \left(\frac{x-1}{x}\right)} = x,$$

$$\text{So, } f(x) \cdot g(x) \cdot h(x) = -1.$$

135. If $f(x) = \frac{x}{1-x}$; $x \neq 1$, then $f^{100}(x)$ is [where $(f \circ f)(x) = f^2(x)$]

- (1) $\frac{x}{1-100x}$ (2) $\frac{100x}{1-100x}$ (3) $\frac{x}{1-x}$ (4) $\frac{x}{100-x}$

Sol. Answer (1)

We have,

$$f(x) = \frac{x}{1-x}$$

$$\Rightarrow f^2(x) = f(f(x)) = \frac{\frac{x}{1-x}}{1 - \frac{x}{1-x}} = \frac{x}{1-2x}$$

$$\text{Also, } f^3(x) = \frac{\frac{x}{1-x}}{1 - 2\left(\frac{x}{1-x}\right)} = \frac{x}{1-3x}$$

$$\text{Similarly, } f^{100}(x) = \frac{x}{1-100x}$$

136. If $g(f(x)) = |\sin x|$ and $f(g(x)) = (\sin \sqrt{x})^2$, then:

(1) $f(x) = \sin^2 x, g(x) = \sqrt{x}$

(2) $f(x) = \sin x, g(x) = |x|$

(3) $f(x) = x^2, g(x) = \sin \sqrt{x}$

(4) f and g cannot be determined

Sol. Answer (1)

Obviously $f(x) = \sin^2 x, g(x) = \sqrt{x}$

137. Let $g(x) = 1 + x - [x]$ and $f(x) = \begin{cases} -1, & x < 0 \\ 0, & x = 0 \\ 1, & x > 0 \end{cases}$. Then for all x , $f[g(x)]$ is equal to

(1) x

(2) 1

(3) $f(x)$

(4) $g(x)$

Sol. Answer (2)

$$\text{As, } g(x) > 0 \text{ and } f(x) = \begin{cases} -1, & x < 0 \\ 0, & x = 0 \\ 1, & x > 0 \end{cases}$$

So, $f[g(x)] = 1$.

138. Let $f(x) = x^2$ and $g(x) = 2^x$ then the solution set of $(fog)(x) = (gof)(x)$ is

(1) R

(2) $\{0\}$

(3) $\{0, 2\}$

(4) $\{2, 3\}$

Sol. Answer (3)

$f(x) = x^2$

$g(x) = 2^x$

$\Rightarrow$ Domain of $f(x) = R$

Domain of $g(x) = R$

and range of $f(x) = R^+ \cup \{0\}$

Range of $g(x) = R^+$

$\Rightarrow fog$ is defined for R^+ whereas gof is defined $R^+ \cup \{0\}$

If $(fog)(x) = (gof)(x)$

$\Rightarrow 2^{2x} = 2^{x^2} \Rightarrow x^2 - 2x = 0 ; x = 0, 2$

$\Rightarrow$ Thus the required solution set is $\{0, 2\}$

139. Let $f(x) = \tan x$, $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ and $g(x) = \sqrt{1-x^2}$ then $g(f(x))$ is

- (1) $\frac{(\sqrt{\cos 2x})}{\cos x}$ (2) $-\frac{(\sqrt{\cos 2x})}{\cos x}$ (3) $\frac{(\sqrt{\cos 2x})}{|\cos x|}$ (4) Not defined

Sol. Answer (4)

Range of $f(x) = R$

Domain of $f(x) = \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Range of $g(x) = [0, 1]$

Domain of $g(x) = [-1, 1]$

For $g[f(x)]$ to be defined Range of $f(x) \subseteq$ domain of $g(x)$

But range of $f(x) \not\subseteq$ domain of $g(x)$

Hence $g[f(x)]$ is not defined.

140. Let f , g and h be real valued functions defined from R to R by

$$f(x) = x^2 - 1, \forall x \in R$$

$$g(x) = \sqrt{1+x^2} \quad \forall x \in R$$

$$h(x) = \begin{cases} 0, & x < 0 \\ x, & x \geq 0 \end{cases}$$

then the composite function $(h \circ f \circ g)(x)$ is given by

- (1) x (2) x^2 (3) $x^2 + 1$ (4) $x^2 - 1$

Sol. Answer (2)

$$f(x) = x^2 - 1 \quad \forall x \in R$$

$$g(x) = \sqrt{1+x^2} \quad \forall x \in R, \quad h(x) = \begin{cases} 0, & x < 0 \\ x, & x \geq 0 \end{cases}$$

$$f \circ g(x) = 1 + x^2 - 1 = x^2$$

$$h(f \circ g(x)) = x^2 \quad \text{as } x^2 > 0$$

141. Let g be a real valued function defined on the interval $(-1, 1)$ such that $e^{-x}(g(x) - 2e^x) = \int_0^x \sqrt{y^4 + 1} dy$ for

all $x \in (-1, 1)$ and f be an another function such $f(g(x)) = g(f(x)) = x$. Then the value of $f'(2)$ is

- (1) $\frac{1}{2}$ (2) $\frac{1}{4}$ (3) $\frac{1}{5}$ (4) $\frac{1}{3}$

Sol. Answer (4)

The given equation can be written as

$$g'(x) = 2e^x + e^x \int_0^x \sqrt{1+y^4} dy$$

$$\Rightarrow g'(x) = 2 + e^x \sqrt{1+x^4} + \left(\int_0^x \sqrt{1+y^4} dy \right) e^x$$

$$\text{But } g(f(x)) = x$$

$$\Rightarrow g'(f(x)) = (f'(x)) \\ = f'(x) = 1$$

$$f'(x) = \frac{1}{g'(f(x))}$$

$$f'(2) = \frac{1}{g'(f(2))}$$

But $f(2)$ is the value of x for which $g(x) = 2$, as they are inverse of each other hence $f(2) = 0$

$$\Rightarrow f'(2) = \frac{1}{g'(0)} = \frac{1}{3}$$

142. Let $g(x) = x - [x] - 1$ and $f(x) = \begin{cases} -1, & x < 0 \\ 0, & x = 0 \\ 1, & x > 0 \end{cases}$, where $[\cdot]$ represents the greatest integer function, then for all x ,

$$f(g(x)) =$$

(1) 2 (2) 1 (3) 0 (4) -1

Sol. Answer (4)

$$g(x) = x - [x] - 1 = \{x\} - 1 < 0$$

$$f(g(x)) = -1$$

143. Let $f: R \rightarrow R$ be defined by $f(x) = x^2 - 3x + 4$ for all $x \in R$, then $f^{-1}(2)$ is

(1) 2 (2) 1 (3) Not defined (4) $\frac{1}{2}$

Sol. Answer (3)

Let $x \in R$ be such that

$$x \in f^{-1}(\{2\}) \Rightarrow f(x) \in \{2\} \Rightarrow f(x) = 2$$

$$\Rightarrow x^2 - 3x + 4 = 2 \Rightarrow x^2 - 3x + 2 = 0 \Rightarrow x = 1, 2$$

144. Let $f: R \rightarrow R$, $g: R \rightarrow R$ be two functions given by $f(x) = 2x - 3$, $g(x) = x^3 + 5$. Then $(fog)^{-1}(x)$ is equal to

(1) $\left(\frac{x+7}{2}\right)^{1/3}$ (2) $\left(x - \frac{7}{2}\right)^{1/3}$ (3) $\left(\frac{x-2}{7}\right)^{1/3}$ (4) $\left(\frac{x-7}{2}\right)^{1/3}$

Sol. Answer (4)

$$\text{As, } f(g(x)) = f(x^3 + 5) = 2(x^3 + 5) - 3 = 2x^3 + 7$$

$$\text{So, its inverse is given by } \left(\frac{x-7}{2}\right)^{1/3}.$$

145. If $f(x) = \frac{x}{x-1} = \frac{1}{y}$, then the value of $f(y)$ is

(1) x

(2) $x+1$

(3) $x-1$

(4) $1-x$

Sol. Answer (4)

We have,

$$f(y) = f\left(\frac{x-1}{x}\right) = \frac{\frac{x-1}{x}}{\frac{x-1}{x} - 1} = 1-x$$

146. If $f: R - \{-1\} \rightarrow R - \{1\}$, $f(x) = \frac{x-3}{x+1}$, then $f^{-1}(x)$ equals

(1) $\frac{x+1}{x+3}$

(2) $\frac{x-3}{x+1}$

(3) $\frac{3+x}{1-x}$

(4) $\frac{3+x}{1+x}$

Sol. Answer (3)

$$\text{Let } f^{-1}(x) = z \Rightarrow f(z) = x \Rightarrow \frac{z-3}{z+1} = x \Rightarrow z = \frac{x+3}{1-x}$$

147. If function $f: R^+ \rightarrow R^+$, $f(x) = 2^x$, then $f^{-1}(x)$ will be equal to

(1) $\log_x 2$

(2) $\log_2(1/x)$

(3) $\log_2 x$

(4) x

Sol. Answer (3)

$$\text{Let } f^{-1}(x) = z \Rightarrow f(z) = x \Rightarrow 2^z = x \Rightarrow z = \log_2 x$$

148. If the function $f: [0, \infty) \rightarrow [2, \infty)$ is defined by $f(x) = x^2 + 2$, then $f^{-1}(x)$ is

(1) $\frac{1}{x^2+2}$

(2) x^2-2

(3) $-\sqrt{x-2}$

(4) $\sqrt{x-2}$

Sol. Answer (4)

$$\text{Let } y = f(x) = x^2 + 2$$

$$\Rightarrow x = \pm\sqrt{y-2}$$

$$\text{Since } x \in [0, \infty), \text{ hence } x = \sqrt{y-2}$$

$$\Rightarrow f^{-1}(x) = \sqrt{x-2}$$

149. If $f: [1, \infty) \rightarrow [2, \infty)$ is given by $f(x) = x + \frac{1}{x}$, then $f^{-1}(x)$ equals:

(1) $\frac{x + \sqrt{x^2-4}}{2}$

(2) $\frac{x}{1+x^2}$

(3) $\frac{x - \sqrt{x^2-4}}{2}$

(4) $1 + \sqrt{x^2-4}$

Sol. Answer (1)

$$\text{Let } f^{-1}(x) = z \Rightarrow f(z) = x \Rightarrow z + \frac{1}{z} = x \Rightarrow z^2 - xz + 1 = 0$$

$$\text{So, } z = \frac{x + \sqrt{x^2-4}}{2}$$

150. Select the correct option

$$(1) f: [1, \infty) \rightarrow [1, \infty), f(x) = 2^{x(x-1)} \Rightarrow f^{-1}(x) = \frac{1}{2}(1 \pm \sqrt{1+4\log_2 x})$$

$$(2) f: [-1, \infty) \rightarrow [-1, \infty), f(x) = (x+1)^2 - 1 \text{ and } f(x) = f^{-1}(x) \text{ then } x = \{0, 1\} \text{ only}$$

$$(3) f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = 3x + 5 \text{ then } f^{-1}(x) = \frac{x-5}{3}$$

$$(4) f: [1, \infty) \rightarrow [2, \infty), f(x) = x + \frac{1}{x}, \text{ then } f^{-1}(x) = \frac{x - \sqrt{x^2 - 4}}{2}$$

Sol. Answer (3)

We have

$$y = f(x) = 2^{x(x-1)}$$

$$\Rightarrow x^2 - x + \log_2 y = 0$$

$$\Rightarrow x = \frac{1 \pm \sqrt{1 - 4\log_2 y}}{2}$$

Since the range of inverse function is $[1, \infty)$ hence neglecting negative sign, we get

$$f^{-1}(x) = \frac{1 + \sqrt{1 - 4\log_2 x}}{2}$$

Similarly (2), (4) are false but (3) is true.

151. The function $f(x) = |\sin x|$ has an inverse if its domain is

$$(1) [0, \pi] \quad (2) \left[0, \frac{\pi}{2}\right] \quad (3) \left[-\frac{\pi}{4}, \frac{\pi}{4}\right] \quad (4) [0, 2\pi]$$

Sol. Answer (2)

Range of $|\sin x|$ is $[0, 1]$

Hence it is invertible in $\left[0, \frac{\pi}{2}\right]$.

152. Let $f: [2, 4) \rightarrow [1, 3)$ be a function defined by $f(x) = x - \left[\frac{x}{2}\right]$ (where $[.]$ denotes the greatest integer function). Then f^{-1} equals

$$(1) x \quad (2) x + 1 \quad (3) x + \left[\frac{x}{2}\right] \quad (4) x + 2$$

Sol. Answer (2)

We have,

$$2 \leq x < 4$$

$$\Rightarrow 1 \leq \frac{x}{2} < 2 \Rightarrow \left[\frac{x}{2}\right] = 1 \text{ or } 2.$$

$$\Rightarrow f(x) = x - 1 \text{ or } f(x) = x - 2$$

Since $f: [2, 4) \rightarrow [1, 3)$

Hence $f(x) = x - 1$

$$\Rightarrow f^{-1}(x) = x + 1.$$

153. The graph of the function $y = \log_a (x + \sqrt{x^2 + 1})$ is not symmetric about the origin.

- (1) True (2) False (3) Can't say (4) Insufficient data

Sol. Answer (2)

Clearly the function $y = \log_a (x + \sqrt{x^2 + 1})$ is an ODD function, so it is symmetric about the origin.

154. Let $f(x) = \frac{x}{1+|x|}$, $x \in R$, then f is

- (1) One-one (2) Even (3) Decreasing (4) Many one

Sol. Answer (1)

Given $f(x) = \frac{x}{1+|x|}$, $x \in R$ is one-one.

155. Identify the correct option

- (1) If $f(x) = \frac{1}{1-x}$, $x > 0$, then the graph of $y = f(f(f(x)))$ is a parabola
 (2) If $g(f(x)) = |\sin x|$ and $f(g(x)) = (\sin \sqrt{x})^2$, then $f(x) = \sin^2 x$ and $g(x) = \sqrt{x} + 1$
 (3) Let $f(x) = (a - x^n)^{\frac{1}{n}}$, $x > 0, n \in N$ and $f \circ f(x) = f^2(x)$, then $f^{2006}(x) = x$
 (4) Even functions are one-one

Sol. Answer (3)

(1) $f(x) = \frac{1}{1-x}$

$y = f(f(f(x)))$

$$y = f\left\{f\left(\frac{1}{1-x}\right)\right\} = f\left\{\frac{1}{1-\frac{1}{1-x}}\right\} = f\left\{\frac{x-1}{x}\right\} = \frac{1}{1-\frac{x-1}{x}}$$

$y = x$, which represents a straight line.

(2) $g(f(x)) = |\sin x|$

$f(g(x)) = (\sin \sqrt{x})^2$

$\Rightarrow f(x) = \sin^2 x; g(x) = \sqrt{x}$

(3) $f(x) = (a - x^n)^{1/n}$

$\Rightarrow f(f(x)) = x$

$\Rightarrow f^2(x) = x$

Hence $f^{2006}(x) = x$

(4) Even function is symmetrical about y axis hence it is many one.

156. Which of the following function is an even function?

(1) $f(x) = \sqrt{1+x^3} - \sqrt{1-x^3}$

(2) $f(x) = x|x| + x^2 \sin x$

(3) $f(x) = \sqrt{1+x^3} + \sqrt{1-x^3}$

(4) $f(x) = \log\left(\frac{2-\sin x}{2+\sin x}\right)$

Sol. Answer (3)

For a function to be even,

$$f(x) = f(-x)$$

Only option (3) satisfies this condition.

157. Let $f(x+y) + f(x-y) = 2f(x)f(y)$ for all $x, y \in R$ and $f(0) \neq 0$. Then $f(x)$ must be

(1) One-one function

(2) Onto function

(3) Even function

(4) Odd function

Sol. Answer (3)

$$f(x+y) + f(x-y) = 2f(x)f(y)$$

$$x = 0, y = 0$$

$$f(0) + f(0) = 2f(0)^2$$

$$\Rightarrow f(0) = 1 \text{ as } f(0) \neq 0$$

Put $y = x$ and $x = 0$

$$f(x) + f(-x) = 2f(0)f(x)$$

$$\Rightarrow f(x) + f(-x) = 2f(x)$$

$$\Rightarrow f(x) = f(-x)$$

Hence the given function f is even.

158. Let $f: R \rightarrow R$ be a function defined as $f(x) = [(x+1)^2]^{1/3} + [(x-1)^2]^{1/3} + \frac{x}{2^x-1} + \frac{x}{2} + 1$

(1) Even

(2) Odd

(3) Neither even nor odd

(4) Even and odd both

Sol. Answer (1)

$$f(x) = [(x+1)^2]^{1/3} + [(x-1)^2]^{1/3} + \frac{x}{2^x-1} + \frac{x}{2} + 1$$

$$f(-x) = [(-x+1)^2]^{1/3} + [(-x-1)^2]^{1/3} + \frac{(-x)}{2^{-x}-1} + \frac{(-x)}{2} + 1$$

$$f(x) - f(-x) = [(x+1)^2]^{1/3} + [(x-1)^2]^{1/3} - [(1-x)^2]^{1/3} - [(x+1)^2]^{1/3} + \frac{x}{2^x-1} + \frac{x}{2} + 1 + \frac{x}{2^{-x}-1} + \frac{x}{2} - 1$$

$$= \frac{x}{2^x-1} + \frac{x \cdot 2^x}{1-2^x} + x$$

$$= x \left[\frac{1}{2^x-1} - \frac{2^x}{2^x-1} \right] + x$$

$$= x \left[\frac{1-2^x}{2^x-1} \right] + x$$

$$= -x \left[\frac{2^x-1}{2^x-1} \right] + x = 0$$

$\therefore f(x) - f(-x) = 0$. Hence function is even.

159. Let $f(x) = 3\sin^3 x + 4x - \sin |x| + \log(1 + |x|)$ be defined on the interval $[0, 1]$. The even extension of $f(x)$ to the interval $[-1, 0]$ is

- (1) $3\sin^3 x + 4x - \sin |x| + \log(1 + |x|)$ (2) $-3\sin^3 x - 4x + \sin |x| - \log(1 + |x|)$
 (3) $-3\sin^3 x - 4x - \sin |x| + \log(1 + |x|)$ (4) $3\sin^3 x + 4x + \sin |x| - \log(1 + |x|)$

Sol. Answer (3)

Even extension of $f(x)$ is obtained by changing sign of odd terms present in the function while keeping sign of even terms same.

Hence, required extension of $f(x) = 3\sin^3 x + 4x - \sin |x| + \log(1 + |x|)$ will be

$$= -3\sin^3 x - 4x - \sin |x| + \log(1 + |x|)$$

160. Select the correct option

- (1) $f(x) = e^{3-2x} \cdot \tan^2 x + |x| - \tan x$ is an even function
 (2) $f(x) = h(x) + h(-x) + x^2 \cot x$ is an even function
 (3) $f(x) = \begin{cases} x, & x \in \mathbb{Q} \\ -x, & x \notin \mathbb{Q} \end{cases}$ is an odd function
 (4) $f(x) = g(x) + g(-x) + |\operatorname{sgn}(x)|$ is neither even nor odd function

Sol. Answer (3)

Option (2) $f(x)$ is neither even nor odd

Option (3) $f(x)$ is odd

Option (4) $f(x)$ is even

161. If the function $f(x) = [4.8 + a \sin x]$ (where $[.]$ denotes the greatest integer function) is an even function then a belongs to

- (1) $(-0.8, 0.2)$ (2) $(-0.8, 0.8)$ (3) $(-0.2, 0.2)$ (4) $(0, 0.2)$

Sol. Answer (3)

According to question

Function is an even function if $a \in (-0.2, 0.2)$

162. If $f(x) = \begin{cases} x^2 \sin \frac{\pi x}{2}, & |x| < 1 \\ x|x|, & |x| \geq 1 \end{cases}$, then $f(x)$ is

- (1) Even function (2) Odd function (3) Periodic function (4) Neither even nor odd

Sol. Answer (2)

We have,

$$f(x) = \begin{cases} x^2 \sin \frac{\pi x}{2}, & -1 < x < 1 \\ x^2, & x \geq 1 \\ -x^2, & x \leq -1 \end{cases}$$

Clearly $f(x)$ is odd.

163. Let a real valued function f satisfy $f(x+y) = f(x)f(y) \forall x, y \in R$ and $f(0) \neq 0$ Then $g(x) = \frac{f(x)}{1+[f(x)]^2}$ is

- (1) An even function (2) An odd function
(3) Neither even nor odd function (4) Periodic function

Sol. Answer (1)

$$f(x+y) = f(x)f(y), \forall x, y \in R$$

$$\Rightarrow f(x) = a^{kx}, a \neq 0$$

$$g(x) = \frac{a^{kx}}{1+a^{2kx}}$$

$$g(-x) = \frac{a^{-kx}}{1+a^{-2kx}} = \frac{a^{kx}}{1+a^{2kx}}$$

Hence $g(x)$ is even function.

164. Let $f(x) = |\sin x| + |\cos x|$, $g(x) = \cos(\cos x) + \cos(\sin x)$

$h(x) = \left\{ -\frac{x}{2} \right\} + \sin \pi x$, where $\{ \}$ represents the fractional function, then the period of

- (1) $f(x) + g(x)$ is π (2) $f(x) - g(x)$ is π
(3) $f(x) + g(x) + h(x)$ is 2π (4) $f(x) + g(x) + h(x)$ is non-existent

Sol. Answer (4)

Period of $f(x)$, $g(x)$, $h(x)$ is $\frac{\pi}{2}, \frac{\pi}{2}, 2$ respectively.

165. Identify the correct statement

- (1) The period of $f(x) = \sin \cos \left(\frac{x}{2} \right) + \cos(\sin x)$ is 2π
(2) The period of $f(x) = \cos x \cos 2x \cos 3x$ is 2π
(3) Let $n \in Z$ and the period of $f(x) = \frac{\sin nx}{\sin \left(\frac{x}{n} \right)}$ is 4π then $n = 2$
(4) If the period of $f(x) = \cos \sqrt{(a)}x$ is π and $()$ denotes the least integer function then $a \in [2, 4)$

Sol. Answer (3)

$$(1) f(x) = \sin \left(\cos \frac{x}{2} \right) + \cos(\sin x)$$

Period of $\cos \frac{x}{2}$ is 4π hence period of $\sin \left(\cos \frac{x}{2} \right)$ is 4π period of $\sin x$ is 2π but $\cos x$ is even hence period of $\cos(\sin x)$ will be π .

Hence period of complete function will be LCM of $(4\pi, \pi) \Rightarrow 4\pi$.

$$(2) f(x) = \cos x \cos 2x \cos 3x$$

$$\text{We have } f(x + \pi) = \cos(\pi + x) \cos(2\pi + 2x) \cos(3\pi + 3x) = \cos x \cos 2x \cos 3x$$

$$\text{Hence period} = \pi$$

$$(3) \text{ The fundamental period of } \frac{\sin nx}{\sin\left(\frac{x}{n}\right)} = \text{L.C.M. of } \left\{ \frac{2\pi}{n}, 2\pi n \right\} = 2\pi n$$

$$\Rightarrow 2\pi n = 4\pi \Rightarrow n = 2$$

$$(4) f(x) = \cos \sqrt{a}x$$

$$\text{Period of } f(x) \text{ is } \frac{2\pi}{\sqrt{a}} = \pi$$

$$\therefore \sqrt{a} = 2 \Rightarrow a = 4$$

$$\Rightarrow a \in (3, 4]$$

166. Consider that the graph of $y = f(x)$ is symmetric about the lines $x = 2$ and $x = 4$ then the period of $f(x)$ is

$$(1) 1$$

$$(2) 2$$

$$(3) 3$$

$$(4) 4$$

Sol. Answer (4)

$$f(2 + x) = f(2 - x) \quad \dots (i)$$

$$f(4 + x) = f(4 - x) \quad \dots (ii)$$

$$\text{By (i), (ii) we get } f(x) = f(x + 4)$$

$$\text{Hence period} = 4$$

167. If $f(x)$ is a real valued function defined as $f(x) = \ln(1 - \sin x)$ then graph of $f(x)$ is

$$(1) \text{ Symmetric about line } x = \pi$$

$$(2) \text{ Symmetric about } y \text{ axis.}$$

$$(3) \text{ Symmetric about line } x = \frac{\pi}{2}$$

$$(4) \text{ Symmetric about origin}$$

Sol. Answer (3)

$$\therefore f(x) = \ln(1 - \sin x)$$

$$\therefore \sin\left(\frac{\pi}{2} + x\right) = \sin\left(\frac{\pi}{2} - x\right) \quad \forall x \in \mathbb{R}$$

$$\therefore \text{Hence graph will be symmetric about line } x = \frac{\pi}{2}$$

168. If $\tan ax + \cot ax$ and $|\tan x| + |\cot x|$ are periodic functions of the same fundamental period then a equals

$$(1) 4$$

$$(2) 2$$

$$(3) 1$$

$$(4) 3$$

Sol. Answer (2)

$$\text{Fundamental period of } |\tan x| + |\cot x| \text{ is } \frac{\pi}{2}$$

$$\text{Fundamental period of } \tan ax + \cot ax \text{ is } \frac{\pi}{a}$$

$$\therefore a = 2$$

169. If $f(x + 10) + f(x + 4) = 0$, there $f(x)$ is a periodic function with period

- (1) 2 (2) 4 (3) 6 (4) 12

Sol. Answer (4)

$$\text{Given } f(x + 10) + f(x + 4) = 0$$

Replacing x by $x + 2$, we get

$$f(x + 12) + f(x + 6) = 0 \quad \dots(i)$$

again, replacing x by $x - 4$, we get

$$f(x + 6) + f(x) = 0 \quad \dots(ii)$$

solving (i) and (ii), we get

$$f(x + 12) = f(x)$$

170. If f is a function such that $f(0) = 2$, $f(1) = 3$, $f(x + 2) = 2f(x) - f(x + 1)$, then $f(5)$ is

- (1) -3 (2) -5 (3) 7 (4) 13

Sol. Answer (4)

171. If $f(x) = \frac{x-1}{x+1}$, then $f(2x)$ is equal to

- (1) $\frac{f(x)+2}{f(x)+3}$ (2) $\frac{f(x)+3}{f(x)+2}$ (3) $\frac{3f(x)+1}{f(x)+3}$ (4) $\frac{f(x)+3}{3f(x)+1}$

Sol. Answer (3)

172. If $f(x + 3) = x^2 - 1$, then $f(x)$ is equal to

- (1) $x^3 - x$ (2) $(x - 3)^2 - 1$ (3) $(x - 1)^2 - 3$ (4) $(x + 1)^2 - 3$

Sol. Answer (2)

173. Let $f(x^3) = x^4 + x^5 + x + 1$, then the value of $f(8)$ is

- (1) 51 (2) 52 (3) 53 (4) 54

Sol. Answer (1)

$$f(x^3) = x^4 + x^5 + x + 1$$

$$f(8) = 16 + 32 + 2 + 1 = 51$$

174. If $f(x) = \frac{x-1}{x+1}$, then $f(ax)$ in term of $f(x)$ is given by

- (1) $\frac{f(x)+a}{1+af(x)}$ (2) $\frac{(a-1)f(x)+a+1}{(a+1)f(x)+a-1}$ (3) $\frac{(a+1)f(x)+a-1}{(a-1)f(x)+a+1}$ (4) $\frac{af(x-1)+1}{af(x-1)+a}$

Sol. Answer (3)

$$f(x) = \frac{x-1}{x+1} \quad \dots(i)$$

$$\Rightarrow \frac{f(x)+1}{f(x)-1} = \frac{x-1+x+1}{x-1-x-1}$$

$$\Rightarrow x = \frac{1+f(x)}{1-f(x)} \quad \dots(ii)$$

$$\text{Now from (i) } f(ax) = \frac{ax-1}{ax+1}$$

Putting the value of x from (ii) in (i), we get

$$f(ax) = \frac{a\left(\frac{1+f(x)}{1-f(x)}\right)-1}{a\left(\frac{1+f(x)}{1-f(x)}\right)+1} = \frac{a(1+f(x))-1-f(x)}{a(1+f(x))+1-f(x)} = \frac{(a-1)+(a+1)f(x)}{(a+1)+(a-1)f(x)} = \frac{(a+1)f(x)+(a-1)}{(a-1)f(x)+(a+1)}$$

175. If $f(x+y, x-y) = xy$ then the arithmetic mean of $f(x, y)$ and $f(y, x)$ is

- (1) x (2) y (3) 0 (4) $(x^2 - y^2)$

Sol. Answer (3)

$$f(x+y, x-y) = xy$$

$$\text{Let } x+y = u \text{ and } x-y = v$$

$$\Rightarrow x = \frac{1}{2}(u+v) \text{ and } y = \frac{1}{2}(u-v)$$

$$\text{So, } f(u, v) = \frac{1}{4}(u^2 - v^2)$$

$$\Rightarrow f(x, y) = \frac{1}{4}(x^2 - y^2)$$

$$\text{A.M. of } f(x, y) \text{ and } f(y, x) = \frac{f(x, y) + f(y, x)}{2} = 0$$

176. If $2f(x^2) + 3f\left(\frac{1}{x^2}\right) = x^2 - 1$, then $f(x^2)$ is

- (1) $\frac{1-x^4}{5x^2}$ (2) $\frac{1-x^2}{5x}$ (3) $\frac{5x^2}{1-x^4}$ (4) $\frac{3-x^2-2x^4}{5x^2}$

Sol. Answer (4)

$$2f(x^2) + 3f\left(\frac{1}{x^2}\right) = x^2 - 1 \quad \dots(i)$$

Replacing x by $\frac{1}{x}$, we get

$$3f(x^2) + 2f\left(\frac{1}{x^2}\right) = \frac{1}{x^2} - 1 \quad \dots(ii)$$

Multiplying equation (i) by 2 and equation (ii) by 3 and subtracting, we get

$$4f(x^2) - 9f(x^2) = 2x^2 - 2 - \frac{3}{x^2} + 3$$

$$-5f(x^2) = 2x^2 - \frac{3}{x^2} + 1$$

$$-5f(x^2) = \frac{2x^4 - 3 + x^2}{x^2}$$

$$f(x^2) = \frac{2x^4 + x^2 - 3}{-5x^2} = \frac{3 - x^2 - 2x^4}{5x^2}$$

177. Let $y = \operatorname{sgn}(x)$, then

$$(1) |x| = x \operatorname{sgn}(x)$$

$$(2) \operatorname{sgn}(\operatorname{sgn}(x)) = \operatorname{sgn}(x)$$

$$(3) x = |x| \operatorname{sgn}(x)$$

$$(4) \text{ All of these}$$

Sol. Answer (4)

$$y = \operatorname{sgn}(x)$$

$$|x| = \begin{cases} x & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -x & \text{if } x < 0 \end{cases}, x \operatorname{sgn}(x) = \begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x < 0 \\ 0 & \text{if } x = 0 \end{cases} \text{ and also } |x| = x \operatorname{sgn}(x)$$

(2) & (3) also true

178. Let $g(x)$ be a polynomial function satisfying $g(x) \cdot g(y) = g(x) + g(y) + g(xy) - 2$ for all $x, y \in \mathbb{R}$ and $g(1) \neq 1$. If $g(3) = 10$ then $g(5)$ equals

$$(1) -24$$

$$(2) 16$$

$$(3) 26$$

$$(4) 34$$

Sol. Answer (3)

We have,

$$g(x) \cdot g(y) = g(x) + g(y) + g(xy) - 2$$

Let us put $x = y = 1$

$$\Rightarrow g(1)^2 = 3g(1) - 2$$

$$\Rightarrow g(1)^2 - 3g(1) + 2 = 0$$

$$\Rightarrow g(1) = 2 \text{ as } g(1) \neq 1$$

$$\text{Put } y = \frac{1}{x}$$

$$g(x) \cdot g\left(\frac{1}{x}\right) = g(x) + g\left(\frac{1}{x}\right) + g(1) - 2$$

$$\Rightarrow g(x) \cdot g\left(\frac{1}{x}\right) = g(x) + g\left(\frac{1}{x}\right)$$

$$\Rightarrow g(x) = \pm x^n + 1$$

$$\Rightarrow g(3) = x^n + 1 = 3^n + 1 = 3^2 + 1$$

$$\therefore n = 2 \text{ and } g(x) = x^2 + 1$$

$$\Rightarrow g(5) = 5^2 + 1 = 25 + 1 = 26$$

179. A real valued function f satisfy $f(x+y) = f(x) + f(y)$, $\forall x, y \in R$ and $f(1) = 1$ then $f(1) + f(2) + f(3) + \dots + f(2009)$ equals

- (1) 2009×2010 (2) 2008×2009 (3) 2009×1005 (4) 2009×1004

Sol. Answer (3)

$$f(x+y) = f(x) + f(y) \quad \forall x, y \in R$$

$$f(x+y) = f(x) + f(y)$$

$$\Rightarrow f(x) = ax$$

$$\text{As } f(1) = 1 \Rightarrow a = 1$$

$$\text{Now, } f(1) + f(2) + \dots + f(2009)$$

$$= \frac{2009 \times 2010}{2} = 2009 \times 1005$$

180. If $2^{f(x)} = \frac{2+x}{2-x}$, $x \in (-2, 2)$ and

$$f(x) = \lambda f\left(\frac{8x}{4+x^2}\right) \text{ then value of '}\lambda\text{' will be}$$

- (1) 2 (2) $\frac{1}{2}$ (3) 1 (4) -1

Sol. Answer (2)

$$2^{f(x)} = \frac{2+x}{2-x}$$

$$f(x) = \log_2 \left(\frac{2+x}{2-x} \right)$$

$$\text{Now } f\left(\frac{8x}{4+x^2}\right) = \log_2 \left[\frac{2 + \frac{8x}{4+x^2}}{2 - \frac{8x}{4+x^2}} \right] = \log_2 \left[\frac{8+2x^2+8x}{8+2x^2-8x} \right] = \log_2 \left[\frac{4+x^2+4x}{4+x^2-4x} \right]$$

$$= \log_2 \left(\frac{2+x}{2-x} \right)^2 = 2 \cdot \log_2 \left(\frac{2+x}{2-x} \right) = 2f(x)$$

$$\therefore f\left(\frac{8x}{4+x^2}\right) = 2f(x)$$

$$\Rightarrow \lambda = \frac{1}{2}$$

181. If $\{x\}$ and $[x]$ represent fractional and integral part of x , then $[x] + \sum_{r=1}^{1090} \frac{\{x+r\}}{1090} =$

- (1) x (2) $1090x$ (3) $\frac{x}{1090}$ (4) 1090

Sol. Answer (1)

$\because \{x+r\} = \{x\}$, as $r \in \text{integer}$

$$[x] + \sum_{r=1}^{1090} \frac{\{x\}}{1090} = [x] + \frac{1090\{x\}}{1090} = x$$

182. If $y = 3[x] + 5$ and $y = 2[x - 1] + 9$ then value of $[x + y]$ is (where $[.]$ denotes greatest integer function)

- (1) 11 (2) 4 (3) 14 (4) 13

Sol. Answer (4)

$$3[x] + 5 = 2[x - 1] + 9$$

$$\Rightarrow 3[x] + 5 = 2[x] + 7$$

$$\Rightarrow [x] = 2 \Rightarrow 2 \leq x < 3 \text{ or } x = 2 + f, f \rightarrow \text{fraction}$$

$$y = 3[x] + 5 = 11$$

$$\text{Hence, } [x + y] = [2 + f + 11] = 13$$

183. $\left[\frac{4}{5}\right] + \left[\frac{4}{5} + \frac{1}{1000}\right] + \left[\frac{4}{5} + \frac{2}{1000}\right] + \dots + \left[\frac{4}{5} + \frac{999}{1000}\right] =$, where $[.]$ denotes greatest integer function

- (1) 998 (2) 980 (3) 800 (4) 801

Sol. Answer (3)

$$\underbrace{\left[\frac{4}{5}\right] + \left[\frac{4}{5} + \frac{1}{1000}\right] + \dots + \left[\frac{4}{5} + \frac{199}{1000}\right]}_{200 \text{ terms are zero}} + \underbrace{\left[\frac{4}{5} + \frac{200}{1000}\right] + \left[\frac{4}{5} + \frac{201}{1000}\right] + \dots + \left[\frac{4}{5} + \frac{999}{1000}\right]}_{800 \text{ terms are equal to 1}}$$

$$\Rightarrow 800$$

184. The graph of the function $y = f(x)$ is symmetrical about $x = 5$, then

- (1) $f(x+5) = f(x-5)$ (2) $f(5+x) = f(5-x)$ (3) $f(x) = f(-x)$ (4) $f(x) = -f(-x)$

Sol. Answer (2)

If the graph is symmetrical about $x = 5$ then $f(5-t) = f(5+t)$

$$\therefore f(5+x) = f(5-x)$$

185. If $f: R \rightarrow R$ satisfies, $f(x+y) = f(x) + f(y)$, $\forall x, y \in R$ and $f(1) = 4$, then $\sum_{r=1}^n f(r)$ is

- (1) $\frac{n(n+1)}{2}$ (2) $n(n+1)$ (3) $2n(n+1)$ (4) $4n(n+1)$

Sol. Answer (3)

$$f(x + y) = f(x) + f(y)$$

$$\text{Putting } x = y = 1, f(2) = 2f(1)$$

$$\text{Putting } x = 2, y = 1, f(3) = f(2) + f(1) = 3f(1)$$

$$\text{In general, } f(n) = nf(1)$$

$$\therefore \sum_{r=1}^n f(r) = (1+2+3+\dots+n)f(1) = \frac{n(n+1)}{2}f(1) = \frac{n(n+1)}{2} \cdot 4 = 2n(n+1)$$

186. If $f(x) = a(x^n + 3)$; $f(1) = 12$, $f(3) = 36$; then $f(2)$ is equal to

(1) 21

(2) 18

(3) 24

(4) 27

Sol. Answer (1)

$$f(1) = 12 \Rightarrow 12 = a(1 + 3) \Rightarrow a = 3$$

$$f(3) = 36 \Rightarrow 36 = 3(3^n + 3) \Rightarrow n = 2$$

$$\therefore f(x) = 3(x^2 + 3)$$

$$\text{and } f(2) = 3(4 + 3) = 21$$

187. If $f\left(x + \frac{1}{y}\right) + f\left(x - \frac{1}{y}\right) = 2f(x)f\left(\frac{1}{y}\right)$ for all $x, y \in \mathbb{R} - \{0\}$ and $f(0) = \frac{1}{2}$, then $f(4)$ is

(1) 0

(2) 4

(3) -4

(4) 2

Sol. Answer (1)At $x = 0$,

$$f\left(\frac{1}{y}\right) + f\left(\frac{-1}{y}\right) - 2f(0)f\left(\frac{1}{y}\right)$$

$$\Rightarrow f\left(\frac{-1}{y}\right) = 0 \text{ since } f(0) = \frac{1}{2}$$

$$\text{at } y = \frac{-1}{4}, f(4) = 0$$

188. Let $f(x) = ||x - 1| + a| - 4$, if $f(x) = 0$ has three real solution, then the values of a lies in

(1) $a \in \{-4\}$

(2) $a \in (-\infty, -4)$

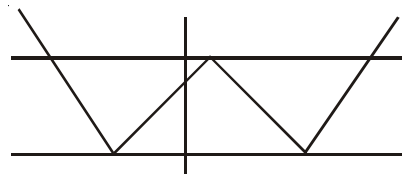
(3) $a \in [4, \infty)$

(4) $a \in [4, 10]$

Sol. Answer (1)

$$\text{Given } f(x) = 0$$

$$\Rightarrow ||x - 1| + a| = 4$$



Clearly, it has 3 solutions, from the graph

$$\therefore a = -4$$

189. Let '*' be the binary operation defined on the set Z of all integers as $a * b = a + b + 1$ for all $a, b \in Z$. The identity element w.r.t this operation is

- (1) -1 (2) -2 (3) 1 (4) 0

Sol. Answer (1)

If e is the identity element under '*' then

$$a * e = e * a = a \text{ for all } a \in Z$$

$$\Rightarrow a + e + 1 = e + a + 1 = a$$

$$\Rightarrow e + 1 = 0$$

$$\Rightarrow e = -1.$$

190. The binary operation '*' defined on the set of integers as $a * b = |a - b| - 1$ is

- (1) Commutative (2) Associative (3) Non-associative (4) Non-commutative

Sol. Answer (1)

$$\text{Here, } b * a = |b - a| - 1 = |(-1)(a - b)| - 1 = |(-1)||a - b| - 1 = 1|a - b| - 1$$

$$= |a - b| - 1 = a * b$$

$\therefore$ '*' is commutative.

191. If $A = \{a, b\}$, then the number of binary operations that can be defined on A is

- (1) 4 (2) 2 (3) 16 (4) 1

Sol. Answer (3)

Here, $O(A) = 2$. Hence, $O(A \times A) = 2 \times 2 = 4$. Number of binary operations on A is same as the number of relations that can be defined from $A \times A$ to A .

Hence, number of binary operations on $A = 2^4 = 16$.

($\because$ Any ordered pair of $A \times A$ can be associated to any one of the two elements of A)

192. Let A be the set of all real numbers except -1 and an operation 'o' be defined on A by $aob = a + b + ab$ for all $a, b \in A$, then identity element w.r.t 'o' is

- (1) a (2) b (3) 1 (4) 0

Sol. Answer (4)

If e is the identity element w.r.t. 'o', then $aoe = eoa = a$ for all $a \neq -1$.

$$\Rightarrow a + e + ae = a$$

$$\Rightarrow e(1 + a) = 0$$

$$\Rightarrow e = 0$$

193. A real valued function $f(x)$ satisfies the functional equation $f(x - y) = f(x)f(y) - f(a - x)f(a + y)$, where a is a given constant and $f(0) = 1, f(2a - x) =$

- (1) $-f(x)$ (2) $f(x)$ (3) $f(a - x)$ (4) $f(-x)$

Sol. Answer (1)Putting $x = y = 0$, then $x = a$, $y = x - a$

$$\text{Now, } f(x - y) = f(x)f(y) - f(a - x)f(a + y) \quad \dots(i)$$

$$f(0) = [f(0)]^2 - [f(a)]^2 \Rightarrow f(a) = 0, \therefore f(0) = 1$$

$$\text{Again, } f(2a - x) = f[a - (x - a)]$$

$$= f(a)f(x - a) - f(a - a)f(a + x - a)$$

$$= 0 - f(x) \Rightarrow f(2a - x) = -f(x)$$

194. Let W denotes the set of words in the English dictionary. Define the relation R by $R = \{(x, y) \in W \times W\}$, the words x and y have at least one letter in common, then R is

- (1) Reflexive, not symmetric and transitive
- (2) Not reflexive, symmetric and transitive
- (3) Reflexive, symmetric and not transitive
- (4) Reflexive, symmetric and transitive

Sol. Answer (3) R is reflexive as x and x have at least one letter common R is symmetric as if x and y have at least one letter common then y and x have at least one letter commonBut R is not transitive

Let $x = ABC$

$y = BCD$

$z = DEF$

Here $x R y$, $y R z$, but $x R z$ not possible.195. Let (x_0, y_0) be the solution of the following equations

$$(2x)^{\ln 2} = (3y)^{\ln 3}$$

$$3^{\ln x} = 2^{\ln y}$$

Then x_0 is

[IIT-JEE 2011]

(1) $\frac{1}{6}$

(2) $\frac{1}{3}$

(3) $\frac{1}{2}$

(4) 6

Sol. Answer (3)

We have

$$(2x)^{\ln 2} = (3y)^{\ln 3}$$

$$\Rightarrow (\ln 2)\log_3 2x = (\ln 3)\log_3 3y$$

$$\Rightarrow \log_3 2x = \log_3 3$$

$$\Rightarrow (2x) = 3^a, (3y) = 2^a, \text{ say}$$

Also $3^{\ln x} = 2^{\ln y}$

$$\Rightarrow \ln x = \ln y (\log_3 2)$$

$$\begin{aligned} \Rightarrow \log_y x &= \log_3 2 \\ \Rightarrow x &= 2^k, y = 3^k \\ \Rightarrow 3^a &= 2^{k+1} \\ \Rightarrow a &= 0 \text{ and } k + 1 = 0 \\ \Rightarrow 2x &= 3^a \\ \Rightarrow x &= \frac{1}{2} \end{aligned}$$

196. Let $f(x) = x^2$ and $g(x) = \sin x$ for all $x \in \mathbb{R}$. Then the set of all x satisfying $(f \circ g \circ g \circ f)(x) = (g \circ g \circ f)(x)$, where $(f \circ g)(x) = f(g(x))$, is [IIT-JEE 2011]

$$(1) \pm\sqrt{n\pi}, n \in \{0, 1, 2, \dots\}$$

$$(2) \pm\sqrt{n\pi}, n \in \{1, 2, \dots\}$$

$$(3) \frac{\pi}{2} + 2n\pi, n \in \{\dots, -2, -1, 0, 1, 2, \dots\}$$

$$(4) 2n\pi, n \in \{\dots, -2, -1, 0, 1, 2, \dots\}$$

Sol. Answer (1)

We have,

$$\begin{aligned} f(x) &= x^2 \text{ and } g(x) = \sin x, \forall x \\ \Rightarrow f(g(g(f(x)))) &= g(g(f(x))) \\ \Rightarrow g(f(x)) &= g(x^2) = \sin x^2 \\ \Rightarrow g(g(f(x))) &= g(\sin x^2) = \sin(\sin x^2) \\ \Rightarrow f(g(g(f(x)))) &= (\sin(\sin x^2))^2 \\ \Rightarrow (\sin \sin x^2)^2 &= \sin(\sin x^2) \\ \Rightarrow \sin(\sin x^2) &= 0 \text{ or } \sin(\sin x^2) = 1 \end{aligned}$$

But $\sin(\sin x^2) = 1$ is not possible hence $\sin x^2 = 0$

$$\Rightarrow x^2 = n\pi$$

$$\Rightarrow x = \pm\sqrt{n\pi}, n \in \{0, 1, 2, 3, \dots\}$$

197. The function $f: [0, 3] \rightarrow [1, 29]$, defined by $f(x) = 2x^3 - 15x^2 + 36x + 1$, is [IIT-JEE 2012]

(1) One-one and onto

(2) Onto but not one-one

(3) One-one but not onto

(4) Neither one-one nor onto

Sol. Answer (2)

$$f(x) = 2x^3 - 15x^2 + 36x + 1$$

$$f'(x) = 6x^2 - 30x + 36$$

$$= 6(x^2 - 5x + 6)$$

$$= 6(x - 2)(x - 3)$$

Clearly the derivative changes sign in $[0, 3]$ so, f is NOT one-one.

Now, the function is increasing in $[0, 2]$ and decreasing in $[2, 3]$

$$\text{Also, } f(0) = 1$$

$$f(2) = 29$$

$$f(3) = 8$$

Hence the range is $[1, 29]$ and so, the function is onto.

SECTION - B

Objective Type Questions (More than one options are correct)

1. Let $f : N \rightarrow R$ be a function such that $f(1) + 2f(2) + 3f(3) + \dots + nf(n) = n(n+1)f(n)$, for $n \geq 2$ and $f(1) = 1$ then

(1) $f(n) = \frac{1}{n}$

(2) $f(5) = \frac{1}{10}$

(3) $f(5) = \frac{1}{5}$

(4) $f(n) = \frac{1}{2n}$

Sol. Answer (2, 4)

$$f : N \rightarrow R$$

$$f(1) + 2f(2) + 3f(3) + \dots + nf(n) = n(n+1)f(n), \text{ for } n \geq 2, f(1) = 1$$

$$n = 2$$

$$f(1) + 2f(2) = 6f(2)$$

$$\Rightarrow 4f(2) = 1 \Rightarrow f(2) = \frac{1}{4}$$

$$n = 3$$

$$f(1) + 2f(2) + 3f(3) + 4f(4) = 3 \times 4 \times f(3)$$

$$1 + \frac{2}{4} + 3f(3) = 12f(3)$$

$$\Rightarrow 9f(3) = \frac{6}{4} = \frac{3}{2} \Rightarrow f(3) = \frac{1}{6}$$

$$n = 4$$

$$f(1) + 2f(2) + 3f(3) = 4 \times 5 \times f(4)$$

$$\Rightarrow f(4) = \frac{1}{8}$$

.....

$$f(n) = \frac{1}{2n}$$

$$\text{So } f(5) = \frac{1}{10},$$

$$f(n) = \frac{1}{2n}$$

2. Which of the following pairs are equal functions?

(1) $\ln x^2, 2\ln x; x > 0$

(2) $\frac{x^2}{x}, x$

(3) $|x|^2, x^2$

(4) $|x|^3, x^3$

Sol. Answer (1, 3)

$$\text{Clearly, } \ln x^2 = 2\ln x \text{ for } x > 0$$

$$\text{And } |x|^2 = x^2, \forall x \in R$$

3. Let $\{x\}$ and $[x]$ denote the fractional and integral part of a real number x respectively. The value(s) of x satisfying $4\{x\} = x + [x]$ is/are

(1) 0

(2) $\frac{5}{3}$

(3) $\frac{1}{3}$

(4) 1

Sol. Answer (1, 2)

We have

$$4\{x\} = x + [x]$$

$$3\{x\} = x - \{x\} + [x]$$

$$3\{x\} = 2[x]$$

$$\Rightarrow 0 \leq 2[x] < 3$$

$$\Rightarrow 0 \leq [x] < \frac{3}{2}$$

$$\Rightarrow [x] = 0, 1$$

$$\text{Also } 3\{x\} = 2[x] \Rightarrow 3(\{x\} + [x]) = 5[x]$$

$$\Rightarrow 3x = 5[x] \Rightarrow x = \frac{5}{3}[x] \Rightarrow x = 0, \frac{5}{3} \quad [\text{As } [x] = 0 \text{ and } 1 \text{ only}]$$

4. Let n_K be the number of real solutions of the equation $|x + 1| + |x - 3| = K$, then

$$(1) \quad n_K = 0, \text{ if } K < 4$$

$$(2) \quad n_K = 2 \text{ if } K > 4$$

$$(3) \quad n_K \text{ is infinitely many if } K = 4$$

$$(4) \quad \text{Minimum value of } f(x) = |x + 1| + |x - 3| \text{ is } 2$$

Sol. Answer (1, 2, 3)

If $K < 4$ obviously no solution

As $n_K = 2$ if $K > 4$

and infinitely many if $K = 4$

Minimum value of $|x + 1| + |x - 3|$ is 4

5. Let $f(x)$ be a function such that $f(x) \cdot f(y) = f(x + y)$, $f(0) = 1$, $f(1) = 4$. If $2g(x) = f(x)(1 - g(x))$ then

$$(1) \quad g(x) + g(1 - x) = 0 \quad (2) \quad g(x) = 1 - g(1 - x) \quad (3) \quad \sum_{K=1}^9 g\left(\frac{K}{10}\right) = \frac{9}{2} \quad (4) \quad \sum_{K=1}^{18} g\left(\frac{K}{19}\right) = 9$$

Sol. Answer (2, 3, 4)

$$2g(x) = f(x)(1 - g(x))$$

$$2g(1 - x) = f(1 - x)(1 - g(1 - x))$$

$$\Rightarrow g(x) + g(1 - x) = 1$$

$$\sum_{k=1}^9 g\left(\frac{k}{10}\right) = g\left(\frac{1}{10}\right) + g\left(\frac{2}{10}\right) + \dots + g\left(\frac{8}{10}\right) + g\left(\frac{9}{10}\right)$$

$$= \left(g\left(\frac{1}{10}\right) + g\left(\frac{9}{10}\right)\right) + \left(g\left(\frac{2}{10}\right) + g\left(\frac{8}{10}\right)\right) + \left(g\left(\frac{3}{10}\right) + g\left(\frac{7}{10}\right)\right) + \left(g\left(\frac{4}{10}\right) + g\left(\frac{6}{10}\right)\right) + g\left(\frac{5}{10}\right)$$

$$= 4 + g\left(\frac{1}{2}\right) = 4 + \frac{1}{2} = \frac{9}{2}$$

6. Which of the following is correct? ($[.]$ represents greatest integer function, $\{.\}$ represents fractional function)

(1) $\{[x]\} + \{[x]\} + \{[x^2 + x + 2]\} + \{[x^2 + x + 2]\} = 0$

(2) $\left(\left[\frac{200+1}{2}\right] + \left[\frac{200+2}{2^2}\right] + \dots + \infty\right) + \left([10] + \left[10 + \frac{1}{10}\right] + \left[10 + \frac{2}{10}\right] + \dots + \left[10 + \frac{9}{10}\right]\right) = 300$

(3) $[[[x]]] = [x]$

(4) $[|x|] = |[x]|$

Sol. Answer (1, 2, 3)

(i) Fractional part of any integer is zero and integral part of any fraction is zero

(ii) We have $\left[\frac{n+1}{2}\right] + \left[\frac{n+2}{2}\right] + \left[\frac{n+4}{2}\right] + \dots + \infty = n; n \in N$

And $[x] + \left[x + \frac{1}{n}\right] + \left[x + \frac{2}{n}\right] + \dots + \left[x + \frac{n-1}{n}\right] = [nx]; x \in N$

Hence $= 200 + 100 = 300$

7. Let f be a function from a set X to X , such that $(f(f(x))) = x$, for all $x \in X$, then

- (1) f is one-one (2) f is onto (3) f is many one (4) f is into

Sol. Answer (1, 2)

$f[f(x)] = x \Rightarrow f(x) = f^{-1}(x)$

Hence, function is bijective.

8. Which of the following statements is/are true?

(1) $f: R \rightarrow R, f(x) = \log(x + \sqrt{1+x^2})$ is an odd function

(2) $f: R \rightarrow R, f(0) = 3$, then $f(x)$ must not be an odd function

(3) $f: R \rightarrow R, f(x) = x \left(\frac{e^x - e^{-x}}{e^x + e^{-x}} \right)$ is an onto function

(4) Graph of $f(x) = x \sin x$ is bounded between lines $y = x, y = -x$

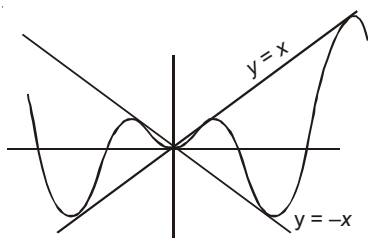
Sol. Answer (1, 2, 4)

(1) $f(-x) = -f(x)$

(2) A function can be odd only if its value at $x = 0$ is zero or undefined.

(3) Given function is even, continuous hence range cannot be ' R ' hence into in $R \rightarrow R$.

(4) True, see graph



9. Let $f(x) = \sin ax + \cos bx$ be a periodic function, then

$$(1) a = \frac{3\pi}{2}, b = \pi \quad (2) a = \sqrt{3}, b = 5\sqrt{3} \quad (3) a = 3\sqrt{2}, b = 2\sqrt{3} \quad (4) a, b \in R$$

Sol. Answer (1, 2)

$$f(x) = \sin ax + \cos bx \text{ is periodic with fundamental period} = \text{LCM} \left(\frac{2\pi}{|a|}, \frac{2\pi}{|b|} \right)$$

$$\text{From (1), if } a = \frac{3\pi}{2}, b = \pi$$

$$\text{Period} = \text{LCM} \left(\frac{4}{3}, 2 \right) = 4$$

$$\text{From (2), if } a = \sqrt{3}, b = 5\sqrt{3}$$

$$\text{Period} = \text{LCM} \left(\frac{2\pi}{\sqrt{3}}, \frac{2\pi}{5\sqrt{3}} \right) = \frac{2\pi}{\sqrt{3}}$$

$$\text{From (3), if } a = 3\sqrt{2}, b = 2\sqrt{3}$$

$$\text{Period} = \text{LCM} \left(\frac{2\pi}{3\sqrt{2}}, \frac{2\pi}{2\sqrt{3}} \right) = \frac{\pi\sqrt{6}}{3} \text{ but } f\left(x + \frac{\pi\sqrt{6}}{3}\right) \neq f(x)$$

From (4) if $a, b \in R$ such period of one of $\sin ax$ and $\cos bx$ is rational and other is irrational then LCM is not possible. Hence options (1) and (2) are correct.

10. Which of the following function is periodic?

$$(1) \operatorname{sgn}(e^{-x}) x > 0$$

$$(2) |\sin x| + \sin x$$

$$(3) \min(4\cos x, |x|)$$

$$(4) \left[x + \frac{1}{2} \right] + \left[x - \frac{1}{2} \right] + 2[-x], [\cdot] \text{ represents greatest integral function}$$

Sol. Answer (1, 2, 4)

$$(1) e^{-x} > 0$$

$$\Rightarrow \operatorname{Sgn}(e^{-x}) = 1 \text{ for all } x \in R$$

Hence, $\operatorname{Sgn}(e^{-x})$ is periodic.

$$(2) f(x) = |\sin x| + \sin x$$

$f(x)$ is periodic with fundamental period 2π .

11. If $f(x) = \sin^2 x$ and $g(x) = \{x\}$ are two real valued function then

$$(1) \text{ Period of } f[g(x)] \text{ will be '1'}$$

$$(2) \text{ Period of } g[f(x)] \text{ will be } \pi$$

$$(3) \text{ Period of } f[g(x)] + g[f(x)] \text{ will be } \pi$$

$$(4) \text{ Period of } f(g(x)) + g(f(x)) \text{ will be } 1$$

Sol. Answer (1, 2)

Period of $f(x)$ is π , period of $g(x)$ is 1

Hence period of $f[g(x)] = \sin^2\{x\}$ will be 1.

Period of $g[f(x)] = \{\sin^2 x\}$ will be π

$$\therefore \{\sin^2 x\} = \sin^2 x, \forall x \in \mathbb{R} - \left\{(2n+1)\frac{\pi}{2}\right\}$$

But period of $f(g(x)) + g(f(x))$ will not be defined since LCM of rational and irrational is not defined.

12. Let us consider a function $f(x) = \sin[x]$, where $[x]$ denotes the greatest integer function. Then

- (1) $f(x)$ is non-periodic
- (2) There does not exist x such that $\sin[x] = \cos[x]$
- (3) There exist infinitely many x for which $\sin[x] \neq \cos[x]$
- (4) There exist infinitely many x for which $\sin[x] = \tan[x]$

Sol. Answer (1, 2, 3, 4)

$f(x) = \sin[x]$ is non-periodic

For $\sin(x) = [0] [x]$

$$[x] = 2n\pi + \frac{\pi}{4} \text{ which is not possible.}$$

Also, for $0 \leq x < 1$

$$\sin[x] = \tan[x] = 0$$

13. Let $f(x) = \left[x^2 \left[\frac{1}{x^2} \right] \right]$, $x \in \mathbb{R} - \{0\}$. $[.]$ represents greatest integral function. Then

- (1) $f(x)$ is an even function
- (2) $f(1) = 0$
- (3) $f(x) = 0 \quad \forall x \in (1, \infty) \cup (-\infty, -1)$
- (4) $f(1) = 1$

Sol. Answer (1, 3, 4)

Clearly $f(x)$ is even function.

For $x \in (-\infty, -1) \cup (1, \infty)$

$$0 < \frac{1}{x^2} < 1$$

Hence $f(x) = 0$

Also $f(1) = 1$

14. Which of the following functions are bounded in the interval as indicated?

- (1) $f(x) = \sin x, x \in \mathbb{R}$
- (2) $g(x) = x \cos \frac{1}{x}$ on $(-\infty, \infty)$
- (3) $h(x) = xe^{-x}$ on $(0, \infty)$
- (4) $l(x) = \arctan 2^x$ on $(-\infty, \infty)$

Sol. Answer (1, 3, 4)

A function is called bounded if $|f(x)| \leq M$ (M is finite number)

(1) $f(x) = \sin x$, $|\sin x| \leq 1$, hence bounded.

(2) At $x = 0$ $x \cos \frac{1}{x}$ is undefined also as $x \rightarrow \infty$ $x \cos x \rightarrow \infty$ hence unbounded.

(3) If $x \in (0, \infty)$ $xe^{-x} \in (0, e^{-1})$ hence bound

(4) $x \in (-\infty, \infty)$ $\arctan 2^x \in \left(0, \frac{\pi}{2}\right)$

15. Let $f(x) = \frac{1}{x}$ and $g(x) = \frac{1}{\sqrt{x}}$, then

(1) $f[g(x)]$ and $g[f(x)]$ have different domain

(2) $f[g(x)]$ and $g[f(x)]$ have the same range

(3) $f[g(x)]$ is a one-one

(4) $g[f(x)]$ is neither odd nor even

Sol. Answer (2, 3, 4)

$$f(x) = \frac{1}{x} \quad \text{Dom}(f(x)) = R_0$$

$$g(x) = \frac{1}{\sqrt{x}} \quad \text{Dom}(g(x)) = R^+$$

$$\left. \begin{aligned} f(g(x)) &= \frac{1}{\frac{1}{\sqrt{x}}} = \sqrt{x}, \quad x \geq 0 \\ g(f(x)) &= \sqrt{x} \quad x \geq 0 \end{aligned} \right\} \text{domain is same.}$$

$\therefore f(g(x)) = g(f(x))$ hence range is $[0, \infty)$ for both.

16. Which of the following functions are periodic?

$$(1) f(x) = \begin{cases} |\operatorname{sgn}(x)|, & x \neq 0 \\ 1, & x = 0 \end{cases}$$

$$(2) g(x) = \sin^{-1}(\sin x)$$

$$(3) h(x) = \begin{cases} \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

$$(4) w(x) = \frac{e^x + e^{-x}}{2}, \quad \forall x \in R$$

Sol. Answer (1, 2)

(1) $f(x) = 1, \forall x \in R$ hence constant hence periodic.

(2) Periodic with period 2π .

(3) Graph of function oscillates but is not periodic.

(4) $f(x + 2\pi i) = f(x)$ function is not periodic since time period is imaginary.

17. If $f(x)$ is a polynomial function satisfying the condition $f(x) \cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$ and $f(2) = 9$ then

$$(1) 2f(4) = 3f(6)$$

$$(2) 14f(1) = f(3)$$

$$(3) 9f(3) = 2f(5)$$

$$(4) f(10) = f(11)$$

Sol. Answer (2, 3)

$$f(x) - f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$$

$$\Rightarrow f(x) = 1 \pm x^n$$

$$\therefore f(2) = 1 + 2^n = 9 \Rightarrow n = 3$$

Hence $f(x) = 1 + x^3$

18. $f : [0, \infty) \rightarrow [0, \infty)$, $f(x) = \frac{5x}{5+x}$ is

- (1) One-one (2) Many one (3) Onto (4) Into

Sol. Answer (1, 4)

$$y = \frac{5x}{5+x} = \frac{5}{\frac{5}{x}+1} \text{ for each value of } x \text{ there is a unique value of } y \text{ and hence the mapping is one-one from given}$$

$$x = \frac{5y}{5-y}, \text{ equation for } y = 5 \text{ there is no } x \text{ and hence it is not onto.}$$

19. Which of the following is/are periodic?

- (1) $f(x) = x - [x]$, where $[x]$ denotes the greatest integral function
 (2) $f(x) = c$ (where c is a constant)
 (3) $f(x) = \tan x^2$
 (4) $f(x) = 2 \tan \frac{1}{2}(x - \pi)$

Sol. Answer (1, 2, 4)

$$f(x) = x - [x], f(x) = c \text{ and } f(x) = 2 \tan \frac{1}{2}(x - \pi) \text{ are periodic functions.}$$

20. If $f(x) = \sin \theta \cdot x + a$ and the equation $f(x) = f^{-1}(x)$ is satisfied by every real value of x , then

- (1) $\theta = \frac{\pi}{2}$ (2) $\theta = \frac{3\pi}{2}$ (3) $a \in R$ (4) $a = 1, \theta = \frac{\pi}{2}$

Sol. Answer (2, 3)

$$f(x) = \sin \theta \cdot x + a \Rightarrow f^{-1}(x) = \frac{x}{\sin \theta} = \frac{a}{\sin \theta}$$

$$\text{Since, } f(x) = f^{-1}(x), \forall x \in R$$

$$\Rightarrow \frac{1}{\sin \theta} = \sin \theta \text{ and } a = \frac{-a}{\sin \theta}$$

$$\Rightarrow \theta = \frac{3\pi}{2} \text{ and } a \in R$$

21. Which pair of functions is/are not identical?

(1) $\log_e x^3, 3\log_e x$

(2) $\log_e e^x, e^{\log_e x}$

(3) $\sin(\sin^{-1}x), \sin^{-1}(\sin x)$

(4) $\sin^2 x + \cos^2 x, \sec^2 x - \tan^2 x$

Sol. Answer (2, 3, 4)

$\log_e x^3$ is defined for all $x \in \mathbb{R} - \{0\}$, while $3\log_e x$ is defined for $x > 0$

$\log_e e^x$ is defined for all x , while $e^{\log_e x}$ is defined for $x > 0$. $\sin(\sin^{-1}x)$ is defined for $x \in [-1, 1]$ and $\sin^{-1}(\sin x)$ is defined for $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. Domain of $\sin^2 x + \cos^2 x$ and $\sec^2 x - \tan^2 x$ are not equal.

22. If $f: \{x: -1 \leq x \leq 1\} \rightarrow \{x: -1 \leq x \leq 1\}$, then which is/are bijective?

(1) $f(x) = [x]$

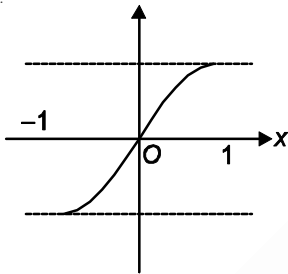
(2) $f(x) = \sin \frac{\pi x}{2}$

(3) $f(x) = |x|$

(4) $f(x) = x|x|$

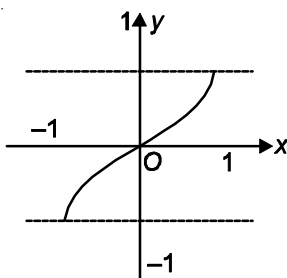
Sol. Answer (2, 4)

$$f(x) = \sin \frac{\pi x}{2}$$



Clearly, it is one-one and onto i.e., bijective.

$$f(x) = x|x|$$



Clearly, it is bijective

23. Let $f(x) = \begin{cases} 2 & , x \in \mathbb{Q} \\ -2 & , x \notin \mathbb{Q} \end{cases}$, then

(1) $f(f(\sqrt{2})) = 2$

(2) $f(f(\pi)) = 2$

(3) $f(x)$ is non-periodic

(4) $f(x)$ is periodic but fundamental period does not exist

Sol. Answer (1, 2, 4)

$$f(x) = \begin{cases} 2 & , x \in \mathbb{Q} \\ -2 & , x \notin \mathbb{Q} \end{cases}$$

Between two rational numbers at least one irrational number exists and between two irrational numbers at least one rational number exists. So, function is periodic but fundamental period does not exist.

24. Let $f : (0, 1) \rightarrow \mathbb{R}$ be defined by

$$f(x) = \frac{b-x}{1-bx}$$

where b is a constant such that $0 < b < 1$. Then

[IIT-JEE 2011]

- (1) f is not invertible on $(0, 1)$ (2) $f \neq f^{-1}$ on $(0, 1)$ and $f'(b) = \frac{1}{f'(0)}$
- (3) $f = f^{-1}$ on $(0, 1)$ and $f'(b) = \frac{1}{f'(0)}$ (4) f^{-1} is differentiable on $(0, 1)$

Sol. Answer (1)

Let $f : (0, 1) \rightarrow \mathbb{R}$ defined by

$$f(x) = \frac{b-x}{1-bx}, \text{ where } 0 < b < 1$$

We observe that

$$f'(x) = \frac{1+b^2}{(1-bx)^2} > 0$$

$\Rightarrow f(x)$ is strictly increasing $\forall x \in (0, 1)$

It is obvious that $f(x)$ does not take all real values for $0 < b < 1$

$\Rightarrow f : (0, 1) \rightarrow \mathbb{R}$ is into function, and hence its increase does not exist.

25. If $3^x = 4^{x-1}$, then $x =$

[JEE (Advanced) 2013]

- (1) $\frac{2\log_3 2}{2\log_3 2 - 1}$ (2) $\frac{2}{2 - \log_2 3}$ (3) $\frac{1}{1 - \log_4 3}$ (4) $\frac{2\log_2 3}{2\log_2 3 - 1}$

Sol. Answer (1, 2, 3)

$$3^x = 4^{x-1}$$

$$x \log_2 3 = (x-1) \log_2 4$$

$$x \log_2 3 = (x-1) \cdot 2$$

$$2x - x \log_2 3 = 2$$

$$x[2 - \log_2 3] = 2$$

$$x = \frac{2}{2 - \log_2 3} \Rightarrow (2)$$

$$\Rightarrow \frac{2}{2 - \frac{1}{\log_3 2}} = \frac{2 \log_3 2}{2 \log_3 2 - 1} \Rightarrow (1)$$

$$\Rightarrow \frac{1}{1 - \frac{1}{2} \log_2 3} = \frac{1}{1 - \log_4 3} \Rightarrow (3)$$

So, (1), (2), (3)

26. Let $f : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \rightarrow \mathbb{R}$ be given by $f(x) = (\log(\sec x + \tan x))^3$. Then

[JEE(Advanced)-2014]

- (1) $f(x)$ is an odd function (2) $f(x)$ is a one-one function
(3) $f(x)$ is an onto function (4) $f(x)$ is an even function

Sol. Answer (1, 2, 3)

$$f(x) = (\log(\sec x + \tan x))^3$$

$$f(-x) = (\log(\sec x - \tan x))^3 = \log \left(\frac{1}{\sec x + \tan x} \right)^3 = -f(x)$$

$\therefore f$ is odd.

$$\text{Also } f'(x) = 3(\log(\sec x + \tan x))^2 \cdot \frac{(\sec x \tan x + \sec^2 x)}{\sec x + \tan x} = 3 \sec x \cdot (\log(\sec x + \tan x))^2 > 0 \text{ in } \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$\therefore f$ is increasing on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$\therefore f$ is one-one

$$\lim_{x \rightarrow \frac{\pi}{2}^-} (\log(\sec x + \tan x))^3 \rightarrow \infty \text{ and } \lim_{x \rightarrow \frac{\pi}{2}^+} (\log(\sec x + \tan x))^3 \rightarrow -\infty$$

$\therefore$ Range is $\mathbb{R}$.

27. Let $f(x) = \sin\left(\frac{\pi}{6} \sin\left(\frac{\pi}{2} \sin x\right)\right)$ for all $x \in \mathbb{R}$ and $g(x) = \frac{\pi}{2} \sin x$ for all $x \in \mathbb{R}$. Let $(fog)(x)$ denote $f(g(x))$ and $(gof)(x)$ denote $g(f(x))$. Then which of the following is (are) true? [JEE(Advanced)-2015]

(1) Range of f is $\left[-\frac{1}{2}, \frac{1}{2}\right]$

(2) Range of fog is $\left[-\frac{1}{2}, \frac{1}{2}\right]$

(3) $\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = \frac{\pi}{6}$

(4) There is an $x \in \mathbb{R}$ such that $(gof)(x) = 1$

Sol. Answer (1, 2, 3)

$$f(x) = \sin\left(\frac{\pi}{6} \sin\left(\frac{\pi}{2} \sin x\right)\right)$$

$$-\frac{\pi}{2} \leq \frac{\pi}{2} \sin x \leq \frac{\pi}{2}$$

$$-1 \leq \sin\left(\frac{\pi}{2} \sin x\right) \leq 1$$

$$\sin\left(\frac{-\pi}{6}\right) \leq \sin\left(\frac{\pi}{6} \left(\sin\left(\frac{\pi}{2} \sin x\right)\right)\right) \leq \sin\frac{\pi}{6}$$

$$\frac{-1}{2} \leq \sin\left(\frac{\pi}{6} \left(\sin\left(\frac{\lambda}{2} \sin x\right)\right)\right) \leq \frac{1}{2}$$

$$f(g(x)) = \sin\left(\frac{\pi}{6} \left(\sin\left(\frac{\pi}{2} \sin\left(\frac{\pi}{2} \sin x\right)\right)\right)\right)$$

$$-1 \leq \sin\left(\frac{\pi}{2} \sin x\right) \leq 1$$

$$\frac{-\pi}{2} \leq \frac{\pi}{2} \sin\left(\frac{\pi}{2} \sin x\right) \leq \frac{\pi}{2}$$

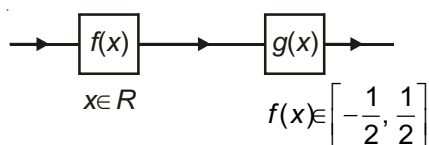
$$-1 \leq \sin\left(\frac{\pi}{2} \left(\sin\left(\frac{\pi}{2} \sin x\right)\right)\right) \leq 1$$

$$-\frac{\pi}{6} \leq \frac{\pi}{6} \sin\left(\frac{\pi}{2} \left(\sin\left(\frac{\pi}{2} \sin x\right)\right)\right) \leq \frac{\pi}{6}$$

$$\frac{-1}{2} \leq \sin\left(\frac{\pi}{6} \sin\left(\frac{\pi}{2} \sin\left(\frac{\pi}{2} \sin x\right)\right)\right) \leq \frac{1}{2}$$

$$\lim_{x \rightarrow 0} \frac{\sin\left(\frac{\pi}{6} \sin\left(\frac{\pi}{2} \sin x\right)\right)}{\frac{\pi}{2} \sin x} \times \frac{\frac{\pi}{6} \sin\left(\frac{\pi}{2} \sin x\right)}{\frac{\pi}{6} \sin\left(\frac{\pi}{2} \sin x\right)}$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{\pi}{6} \frac{\sin\left(\frac{\pi}{2} \sin x\right)}{\frac{\pi}{2} \sin x} = \frac{\pi}{6}$$



$$\text{Range of } g(f(x)) \text{ is } \left[\frac{\pi}{2} \sin\left(\frac{-1}{2}\right), \frac{\pi}{2} \sin\frac{1}{2}\right]$$

$$\left[-\frac{\pi}{2} \sin\frac{1}{2}, \frac{\pi}{2} \sin\frac{1}{2}\right]$$

Hence, 1 does not belong to this range.

SECTION - C

Linked Comprehension Type Questions

Comprehension-I

Let $f(x)$ be a real valued function such that the area of an equilateral triangle with two of its vertices at $(0, 0)$ and

$(x, f(x))$ is $\frac{\sqrt{3}}{4}$ square units. Then

1. Perimeter of the equilateral triangle is

- (1) 1 (2) 3 (3) 6 (4) $3\sqrt{3}$

Sol. Answer (2)

$\sqrt{x^2 + (f(x))^2}$ be the length of the side of the given equilateral triangle

Now area of equilateral triangle = $\frac{\sqrt{3}}{4} \times (\text{side})^2$

$$\frac{\sqrt{3}}{4} = \frac{\sqrt{3}}{4} [x^2 + (f(x))^2]$$

$$\Rightarrow (f(x))^2 + x^2 = 1 \Rightarrow \{f(x)\}^2 = 1 - x^2$$

$$\Rightarrow f(x) = \pm \sqrt{1 - x^2}$$

And since area = $\frac{\sqrt{3}}{4}$ (given)

$$\frac{\sqrt{3}}{4} \times a^2 = \frac{\sqrt{3}}{4} \Rightarrow a = 1$$

Perimeter = 3

2. $f(x)$ is given by

- (1) $\sqrt{2 - x^2}$ (2) $\sqrt{1 + x^2}$ (3) $\sqrt{1 - x^2}$ (4) $\pm \sqrt{1 - x^2}$

Sol. Answer (3)

$\sqrt{x^2 + (f(x))^2}$ be the length of the side of the given equilateral triangle

Now area of equilateral triangle = $\frac{\sqrt{3}}{4} \times (\text{side})^2$

$$\frac{\sqrt{3}}{4} = \frac{\sqrt{3}}{4} [x^2 + (f(x))^2]$$

$$\Rightarrow (f(x))^2 + x^2 = 1 \Rightarrow \{f(x)\}^2 = 1 - x^2$$

$$\Rightarrow f(x) = \pm \sqrt{1 - x^2}$$

3. The domain of the function is

(1) $[1, \infty)$

(2) $(-\infty, 1)$

(3) $(-1, 1)$

(4) $[-1, 1]$

Sol. Answer (4)

$$\text{Domain of } f(x) : 1 - x^2 \geq 0$$

$$\Rightarrow x^2 - 1 \leq 0 \quad \Rightarrow x \in [-1, 1]$$

Comprehension-II

Let $f(x)$ and $g(x)$ be two real valued functions then $|f(x) - g(x)| \leq |f(x)| + |g(x)|$

Let $f(x) = x - 3$ and $g(x) = 4 - x$, then

1. The number of solution(s) of the above inequality when $x \geq 4$ is

(1) 0

(2) 1

(3) 2

(4) Infinitely many

Sol. Answer (4)

$$\text{If } x \geq 4$$

$$|f(x)| = |x - 3| = x - 3$$

$$|g(x)| = |4 - x| = x - 4$$

$$\text{Now } |f(x) - g(x)| \leq |f(x)| + |g(x)|$$

$$\Rightarrow |(x - 3) - (4 - x)| \leq |x - 3| + |4 - x| \quad \Rightarrow |2x - 7| \leq |x - 3| + |x - 4|$$

$$\Rightarrow 2x - 7 \leq x - 3 + x - 4 \quad \Rightarrow 2x - 7 \leq 2x - 7, \text{ equality holds}$$

This inequality has infinitely many solutions for $x \geq 4$.

2. The number of solution(s) of the above inequality for $3 < x < 4$ is

(1) 0

(2) 1

(3) 2

(4) Infinitely many

Sol. Answer (4)

$$\text{In the region } 3 < x < 4 \text{ we get that } |x - 3| = x - 3, |x - 4| = 4 - x$$

$$|f(x) - g(x)| = 2x - 7, |f(x)| + |g(x)| = 1$$

$$\Rightarrow |2x - 7| \leq 1$$

$$\Rightarrow 3 \leq x \leq 4$$

Clearly the given inequality has infinitely many solutions.

Comprehension-III

The absolute valued function f is defined as $f(x) = \begin{cases} x; & x \geq 0 \\ -x; & x < 0 \end{cases}$ and fractional part function $g(x)$ as $g(x) = x - [x]$,

graphically, the number of real solution(s) of the equation $f(x) = g(x)$ is obtained by finding the point(s) of intersection of the graph of $y = f(x)$ and $y = g(x)$.

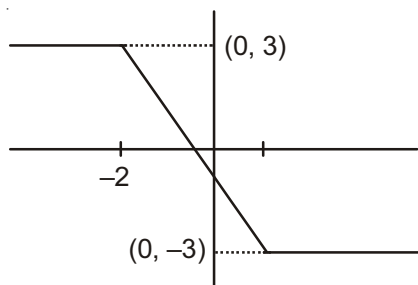
1. The number of solution(s) of $|x - 1| - |x + 2| = k$, when $-3 < k < 3$, is

- (1) 0 (2) 1 (3) 2 (4) 3

Sol. Answer (2)

Consider $f(x) = |x - 1| - |x + 2|$

$$f(x) = \begin{cases} -(x-1) + x + 2 = 3 & ; x \leq -2 \\ -(x-1) - (x+2) = -2x - 1 & ; -2 < x < 1 \\ (x-1) - (x+2) = -3 & ; x \geq 1 \end{cases}$$

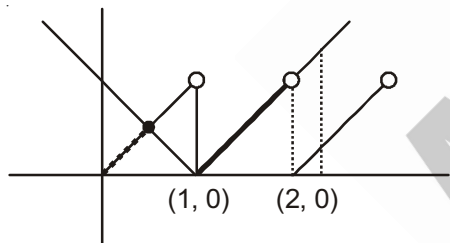


Clearly when $-3 < k < 3$, then there is only one solution

2. The number of solution(s) of $|x - 1| = \{x\}$, $x \in [-1, 2]$ is

- (1) 1 (2) 2 (3) 3 (4) Infinitely many

Sol. Answer (4)



Comprehension-IV

Let $f(x)$ and $g(x)$ be a function defined on $[-2, 2]$ such that $f(x) = -1$, $-2 \leq x \leq 0$; $f(x) = x - 1$, $0 < x \leq 2$ and $g(x) = |x|$. Let $h(x)$ be a function defined as $h(x) = fog(x) + gof(x)$

1. The range of $h(x)$ is

- (1) $[0, 1]$ (2) $[-2, 2]$ (3) $[0, 2]$ (4) $[1, 2]$

2. The function $h(x)$ is

- (1) One-one (2) One-one on $[-1, 1]$
(3) A linear function on $[-2, 1]$ (4) A linear function on $[1, 2]$

3. The function $h(x)$

- (1) Decrease in $[-2, 2]$ (2) Decreases strictly in $[-2, 1]$
(3) Increases in $[-2, 2]$ (4) Increases in $[1, 2]$

Solution of Comprehension-IV

$$f(x) = \begin{cases} -1, & -2 \leq x \leq 0 \\ x-1, & 0 \leq x \leq 2 \end{cases}$$

$$g(x) = |x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

$$(fog)(x) = \begin{cases} |x|-1 & 0 \leq |x| \leq 2 \end{cases}$$

$$\Rightarrow (fog)(x) = \begin{cases} x-1, & 0 \leq x \leq 2 \\ -x-1, & -2 \leq x \leq 0 \end{cases}$$

$$\text{Also } (gof)(x) = \begin{cases} |-1|, & -2 \leq x \leq 0 \\ |x-1|, & 0 \leq x \leq 2 \end{cases}$$

$$\Rightarrow (gof)(x) = \begin{cases} 1, & -2 \leq x \leq 0 \\ -(x-1), & 0 \leq x \leq 1 \\ x-1, & 1 \leq x \leq 2 \end{cases}$$

$$h(x) = fog + gof = \begin{cases} -x, & -2 \leq x \leq 0 \\ 0, & 0 \leq x \leq 1 \\ 2(x-1), & 1 \leq x \leq 2 \end{cases}$$

Graph of $h(x)$

1. Answer (3)

From graph range $[0, 2]$

2. Answer (4)

Linear in $[1, 2]$

3. Answer (4)

From graph $f(x)$ increases in $[1, 2]$

Comprehension-V

Consider that $f: A \rightarrow B$

(i) If $f(x)$ is one-one $f(x_1) = f(x_2) \Leftrightarrow x_1 = x_2$ or $f'(x) \geq 0$ or $f'(x) \leq 0$.

(ii) If $f(x)$ is onto the range of $f(x) = B$.

(iii) If $f(x)$ and $g(x)$ are inverse of each other then $f(g(x)) = g(f(x)) = x$.

Now, consider the answer of the following questions.

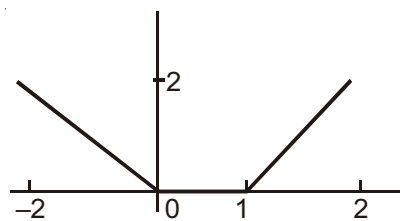
1. If $f(x)$ and $g(x)$ are mirror image of each other through $y = x$ and such that $f(x) = e^x + x$ then the value of $g'(1)$ is

(1) $\frac{1}{2}$

(2) 2

(3) 1

(4) 3



Sol. Answer (1)

$$fg(x) = gf(x) = x \Rightarrow f'(g(x)) \cdot g'(x) = g'f(x) \cdot f'(x) = 1$$

$$\Rightarrow g'(x) = \frac{1}{f'(g(x))} \Rightarrow g'(1) = \frac{1}{f'(g(1))} = \frac{1}{f'(0)} = \frac{1}{2}$$

2. Let f be one-one function with domain $\{x, y, z\}$ and range $\{1, 2, 3\}$. It is given that exactly one of the following statement is true and the remaining two are false. $f(x) = 1, f(y) \neq 1, f(z) \neq 2$, then the value of $f^{-1}(1)$ is

- (1) x (2) z (3) y (4) Does not exist

Sol. Answer (3)

$$f^{-1}(1) = y$$

3. Let $f(x) = \frac{kx}{x+1}$ then the value of k such that $f(x)$ is inverse of itself is

- (1) 0 (2) -1 (3) 1 (4) 2

Sol. Answer (2)

$$f(f(x)) = x$$

$$\Rightarrow \frac{\alpha^2 x}{\alpha x + x + 1} = x$$

$$\Rightarrow (\alpha^2 - 1)x = x^2(\alpha + 1) \Rightarrow \alpha = -1$$

Comprehension-VI

Let f be a real valued function with domain R satisfying

$$f(x+3) = 1 + \left[1 - 3f(x) + 3(f(x))^2 - (f(x))^3\right]^{1/3}, \forall x \in R. \text{ Then}$$

1. The period of $f(x)$ is

- (1) 2 (2) 3 (3) 6 (4) 9

Sol. Answer (3)

2. The value of $f(3) - f(9) + f(15) - f(21) + f(27) - f(33) + f(39) - f(45) + f(51) - f(57)$

- (1) 0 (2) $f(3)$ (3) $5 + (3)$ (4) $-3f(3)$

Sol. Answer (1)

3. The value of $f(x+3) + f(x)$ is

- (1) 2 (2) 0 (3) 3 (4) 4

Sol. Answer (1)

$$f(x+3) = 1 + [1 - 3f(x) + 3f(x)^2 - (f(x))^3]^{1/3}, \forall x \in R$$

$$= 1 + (1 - f(x))^{3 \times 1/3}$$

$$= 2 - f(x)$$

$$\Rightarrow f(x+3) + f(x) = 2 \quad \dots (1)$$

Replacing x by $(x-3)$

$$\Rightarrow f(x) + f(x-3) = 2 \quad \dots (2)$$

$$(1) - (2)$$

$$f(x+3) = f(x-3)$$

$\Rightarrow$ Period is '6'.

$$\begin{aligned} \text{Now } f(3) - f(9) + f(15) - f(21) + f(27) - f(33) + f(39) - f(45) + f(51) - f(57) \\ = 0 \quad [\text{as } f(3) = f(9), f(15) = f(21), \dots] \end{aligned}$$

$$\text{Also } f(x+3) + f(x) = 2$$

Comprehension-VII

Let $f: [-3, 3] \rightarrow R$ defined by $f(x) = \left[\frac{x^2}{a} \right] \tan ax + \sec ax$. then

1. If $f(x)$ is an even function, then

$$(1) \ a > 3$$

$$(2) \ a < 3$$

$$(3) \ a > 9$$

$$(4) \ a < 9$$

Sol. Answer (3)

$$f(x) = \left[\frac{x^2}{a} \right] \tan ax + \sec ax$$

$$f(-x) = - \left[\frac{x^2}{a} \right] \tan ax + \sec ax$$

For even function $f(x) = f(-x)$

$$\Rightarrow 2 \left[\frac{x^2}{a} \right] \tan ax = 0$$

$$\Rightarrow \left[\frac{x^2}{a} \right] = 0$$

$$\Rightarrow 0 \leq \frac{x^2}{a} < 1$$

$$\Rightarrow a > 9$$

2. If $f(x)$ is an odd function, then

$$(1) \ a > 3$$

$$(2) \ a < 3$$

$$(3) \ a > 9$$

$$(4) \ f(x) \text{ can't be an odd function for any real value of } a$$

Sol. Answer (4)

For odd function $f(x) = -f(-x)$

$$\left[\frac{x^2}{a} \right] \tan ax + \sec ax = - \left(- \left[\frac{x^2}{a} \right] \tan ax + \sec ax \right)$$

$$\Rightarrow 2 \sec ax = 0$$

$$\Rightarrow \text{No real value of } a \text{ exist}$$

3. The fundamental period of $f(x)$ when $a = 10$

(1) π

(2) 2π

(3) $\frac{\pi}{10}$

(4) $\frac{\pi}{5}$

Sol. Answer (4)

For $a = 10$

$$F(x) = \sec 10x$$

$$\text{Fundamental period of } f(x) = \frac{2\pi}{10} = \frac{\pi}{5}$$

Comprehension-VIII

A function $y = f(x)$ is said to be an explicit function of x if the dependent variable y can be expressed in terms of independent variable x and it is said to be an implicit function of x if y cannot be expressed in terms of x only

For example $y = x^2$ and $xy = \tan(x + y)$ are explicit and implicit functions respectively. Then express the implicit function into explicit functions.

1. If $2008^x + 2008^y = 2008$, then explicit function is

(1) $y = \log_{2008}(2008 - 2008^x)$

(2) $y = \log(2008 - 2008^x)$

(3) $y = \log_{2008}(2008^x - 2008)$

(4) $y = \log_{2008}(2008 + 2008^x)$

Sol. Answer (1)

$$2008^x + 2008^y = 2008$$

$$2008^y = 2008 - 2008^x$$

$$y = \log_{2008}(2008 - 2008^x)$$

2. If $x^2 - \sin^{-1}y = \frac{\pi}{2}$, then explicit function is

(1) $\cos x^2$

(2) $\sin x^2$

(3) $-\cos x^2$

(4) $-\sin x^2$

Sol. Answer (3)

$$\sin^{-1}y = -\frac{\pi}{2} + x^2$$

$$y = -\cos x^2$$

3. If $e^y - e^{-y} = 2x$, then

(1) $y = \log_e\left(x + \sqrt{x^2 + 1}\right), x \geq 0$

(2) $\log_e\left(x - \sqrt{x^2 + 1}\right), x \geq 0$

(3) $-\log_e\left(x + \sqrt{x^2 + 1}\right)$

(4) $-\log_e\left(x - \sqrt{x^2 + 1}\right)$

Sol. Answer (1)

$$e^y - e^{-y} = 2x$$

$$e^{2y} - 2e^y x - 1 = 0$$

$$e^y = \frac{2x \pm \sqrt{4x^2 + 4}}{2}$$

$$e^y = x \pm \sqrt{x^2 + 1} \quad [\text{as } e^y > 0 \quad \therefore e^y = x + \sqrt{x^2 + 1}]$$

$$y = \log(x + \sqrt{x^2 + 1})$$

Comprehension-IX

$$\text{Let } f(x) = f_1(x) - 2f_2(x)$$

$$\text{where } f_1(x) = \min(x^2, |x|) \text{ for } -1 \leq x \leq 1$$

$$= \max(x^2, |x|) \text{ for } |x| > 1$$

$$f_2(x) = \max(x^2, |x|) \text{ for } -1 \leq x \leq +1$$

$$= \min(x^2, |x|) \text{ for } |x| > 1$$

$$g(x) = \min(f(t) : -3 \leq t \leq x, -3 \leq x \leq 0)$$

$$= \max(f(t) : 0 \leq t \leq x, 0 \leq x \leq 3)$$

1. For $-3 \leq x \leq -1$, range of $g(x)$ is

(1) $[-1, 3]$

(2) $[-1, -15]$

(3) $[-1, 9]$

(4) $\{-1\}$

2. Number of critical points of $f(x)$ is

(1) 1

(2) 2

(3) 3

(4) 4

3. For $x \in (-1, 0)$; $f(x) - g(x)$ is

(1) $x^2 - 2x + 1$

(2) $x^2 + 2x - 1$

(3) $x^2 + 2x + 1$

(4) $x^2 - 2x - 1$

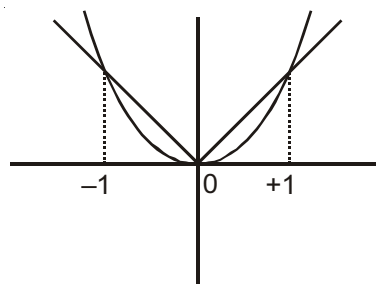
Solution of Comprehension-IX

$$f_1(x) = \begin{cases} \min(x^2, |x|) & \text{for } -1 \leq x \leq 1 \\ \max(x^2, |x|) & \text{for } |x| > 1 \end{cases}$$

$$\text{Hence } f_1(x) = x^2$$

$$f_2(x) = \begin{cases} \max(x^2, |x|), & -1 \leq x \leq +1 \\ \min(x^2, |x|) & \text{if } |x| > 1 \end{cases}$$

$$f_2(x) = |x|$$



Hence $f(x) = f_1(x) - 2f_2(x)$

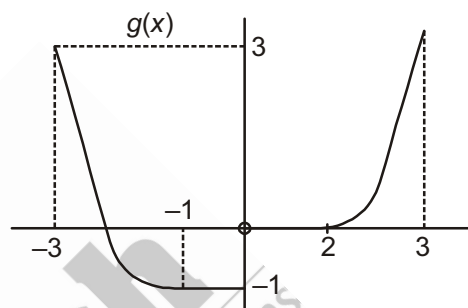
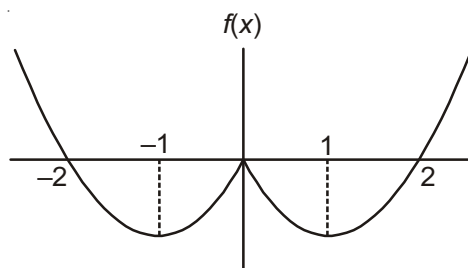
$$= x^2 - 2|x|$$

$$= \begin{cases} x^2 + 2x, & x \leq 0 \\ x^2 - 2x, & x \geq 0 \end{cases}$$

$$g(x) = \begin{cases} \min. f(x); & -3 \leq t \leq x, -3 \leq x \leq 0 \\ \max. f(x); & 0 \leq t \leq x, 0 \leq x \leq 3 \end{cases}$$

$$= \begin{cases} f(x) & ; -3 \leq x \leq -1 \\ f(-1) & ; -1 \leq x \leq 0 \\ f(0) & ; 0 < x \leq 2 \\ f(x) & ; 2 < x \leq 3 \end{cases}$$

$$= \begin{cases} x^2 + 2x & ; -3 \leq x \leq -1 \\ -1 & ; -1 < x \leq 0 \\ 0 & ; 0 < x \leq 2 \\ x^2 - 2x & ; 2 < x \leq 3 \end{cases}$$



1. Answer (1)

If $-3 \leq x \leq -1$, then

$$\text{Range} = [-1, 3]$$

2. Answer (3)

$$\text{Critical point} = 3$$

3. Answer (3)

$$x \in (-1, 0), f(x) = x^2 - 2|x| = x^2 + 2x$$

$$x \in (-1, 0), g(x) = -1$$

$$f(x) - g(x) = x^2 + 2x + 1$$

Comprehension-X

Let $f(x) = x^2 - 3x + 2$ be a function defined from $R \rightarrow R$

and $|x| = \begin{cases} x, & x \geq 0 \\ -x, & x \leq 0 \end{cases}$ then answer the following

1. No. of solutions of the equation $|f(x)| = 2$

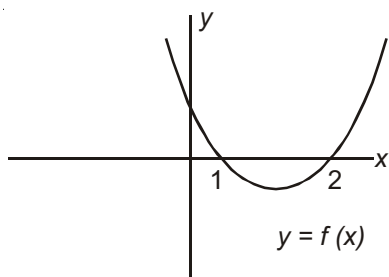
(1) 2

(2) 3

(3) 4

(4) 8

Sol. Answer (2)



Line $y = 2$ cuts the graph at exactly '3' points.

2. Value of k for which equation $|f(x)| = k$ has six solutions

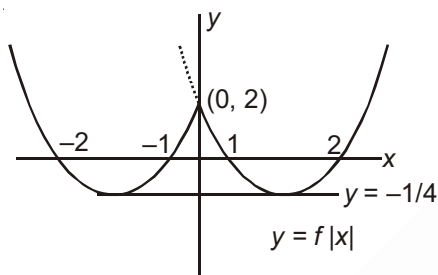
(1) $\frac{1}{2}$

(2) $\frac{1}{4}$

(3) Zero

(4) $\frac{1}{8}$

Sol. Answer (2)



$y = \frac{1}{4}$ cuts the graph at six points.

3. Range of value of x for which $|f(x)| - f(x) = 0$ has no solution

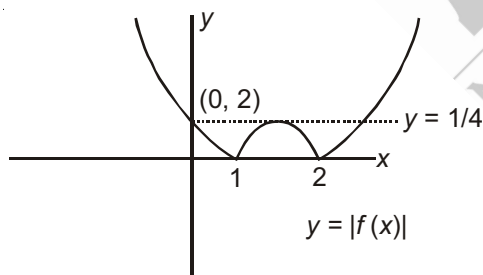
(1) $[1, 2]$

(2) $[-2, -1] \cup [1, 2]$

(3) $(-2, -1) \cup (1, 2)$

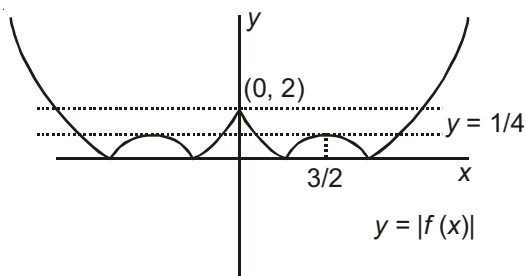
(4) $(-\infty, 0) \cup (1, 2)$

Sol. Answer (4)



$|f(x)| = f(x)$

Graph coincides for all value of x . Exclusion $(-\infty, 0) \cup (1, 2)$, hence equation will have no solution in this zone.



Comprehension-XI

Three students A, B, C applied for admission in three universities P, Q, R where eligibility criteria is min 60%. Form processing software of university P, Q and R use three functions $[x]$, (x) and $\{x\}$ respectively for conversion of percentage to nearest integer. Percentage marks of A, B and C are respectively 59.4, 59.5, 60.1. Hence due to rounding of all the three qualified for university P. Only 'C' qualified for university 'Q' but B and C both qualified for university 'R' since software 'R' rounds off as per normal calculator.

1. Domain of the function $f(x) = \sqrt{2\{\sin x\} - 1}$, if $n \in I$

(1) $\left[2n\pi + \frac{\pi}{6}, 2n\pi + \frac{5\pi}{6}\right]$

(2) $\left(2n\pi + \frac{\pi}{6}, 2n\pi + \frac{5\pi}{6}\right)$

(3) $R - \left[2n\pi + \frac{\pi}{6}, 2n\pi + \frac{5\pi}{6}\right]$

(4) $R - \left(2n\pi + \frac{\pi}{6}, 2n\pi + \frac{5\pi}{6}\right)$

2. Solution of equation $\{x\} = [x]$ will lie in the interval. (I represent set of integer)

(1) $\left[I, I + \frac{1}{2}\right]$

(2) $\left(I, I + \frac{1}{2}\right)$

(3) $\left[I - \frac{1}{2}, I\right]$

(4) R

3. Which of the following statement will be true for all real value of x ?

(1) Solution of equation $\{x\} = x$ will be all integers

(2) $\{x\} \leq x, \forall x \in R$

(3) Both (1) & (2)

(4) $[x] < (x), \forall x \in R - I$

Solution of Comprehension-XI

Symbols $[x]$, (x) , $\{x\}$ do not have their usual meaning and have been redefined as follow.

$[x] \rightarrow$ represent least integer function.

$(x) \rightarrow$ represent greatest integer function

$\{x\} \rightarrow$ represent round off function which will convert any number to next integer if decimal part of it is 0.5 or more otherwise to previous integer.

1. Answer (1)

$$f(x) = \sqrt{\{ \sin x \} - 1}$$

$$\Rightarrow \{ \sin x \} - 1 \geq 0 \Rightarrow \{ \sin x \} \geq 1$$

$$\Rightarrow \sin x \geq \frac{1}{2}$$

$$\Rightarrow x \in \left[2n\pi + \frac{\pi}{6}, 2n\pi + \frac{5\pi}{6}\right]$$

2. Answer (3)

$$\text{If } x \in \left[I - \frac{1}{2}, I + \frac{1}{2}\right) \rightarrow \{x\} = I$$

$$x \in (I - 1, I] \rightarrow [x] = I$$

Taking intersection of intervals

$$x \in \left[I - \frac{1}{2}, I\right]$$

3. Answer (1)

$\{x\} \in I$, hence $\{x\} = x$ only for $x \in I$.

Comprehension-XII

A five degree expression $f(x)$ satisfies the condition

$$f(x) + f\left(\frac{1}{x}\right) = f(x) \cdot f\left(\frac{1}{x}\right)$$

1. $f(x) =$

(1) x^5

(2) x^{-5}

(3) $x^5 \pm 1$

(4) $1 \pm x^5$

Sol. Answer (4)

$$f(x) \cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$$

$$\therefore f(x) \left(f\left(\frac{1}{x}\right) - 1 \right) = f\left(\frac{1}{x}\right) - 1 + 1$$

$$\text{or } \left(f\left(\frac{1}{x}\right) - 1 \right) (f(x) - 1) = 1$$

$\therefore f(x) - 1$ and $f\left(\frac{1}{x}\right) - 1$ are reciprocal of each other. It is possible when $f(x) - 1 = x^n$ or $-x^n$

$$\text{So, that } f\left(\frac{1}{x}\right) - 1 = \frac{1}{x^n} \text{ or } \frac{1}{x^n}$$

Hence, $f(x) = 1 + x^n$ or $1 - x^n$, i.e., $1 \pm x^n$.

Here $f(x)$ is a five degree expression

$$\therefore f(x) = 1 \pm x^5$$

2. If $f(1) = 0$, then $f(2) =$

(1) 30

(2) -31

(3) 0

(4) 2

Sol. Answer (2)

$$\text{Given that } f(1) = 0 = 1 - 1^5 = 1 - x^5$$

$$\therefore f(2) = 1 - 2^5 = -31$$

3. $f(x) + f(-x) =$

(1) A constant function

(2) An identity function

(3) An even function

(4) An odd function

Sol. Answer (1)

$$f(x) + f(-x) \Rightarrow 2 = \text{It is constant function.}$$

SECTION - D

Assertion-Reason Type Questions

1. STATEMENT-1 : The number of functions from $A = \{1, 2, 3\}$ to $B = \{2008, 2009\}$ is 8.

and

STATEMENT-2 : The number of all possible functions from $A = \{1, 2, 3\}$ to $B = \{2008, 2009\}$ is 9.

Sol. Answer (3)

The number of functions from $A = \{1, 2, 3\}$ to $B = \{2008, 2009\}$ is $|B|^{|A|} = 2^3 = 8$

$\Rightarrow$ Statement-1 is true

Statement-2 is not true

2. STATEMENT-1 : The function f defined as $f(x) = a^x$ satisfies the inequality $f(x_1) < f(x_2)$ for $x_1 > x_2$ when $0 < a < 1$.

and

STATEMENT-2 : The function f defined as $f(x) = a^x$ satisfies the inequality $f(x_1) < f(x_2)$ for $x_1 < x_2$ when $a > 1$.

Sol. Answer (2)

From the graph of the two functions it is clear that both statements are true and statement-2 is not the correct explanation of statement-1

3. STATEMENT-1 : If function $f : [-3, 3] \rightarrow \mathbb{R}$ be defined as $f(x) = \left[\frac{x^2}{a} \right]$, then $f(x) = 0, \forall x \in D_f$, iff $a \in (9, \infty)$. ($[x]$ denotes the greatest integer function)

and

STATEMENT-2 : $[x] = 0, \forall 0 \leq x < 1$.

Sol. Answer (1)

Clearly if $a \in (9, \infty)$

$$0 \leq \frac{x^2}{a} < 1$$

$$\text{Hence } \left[\frac{x^2}{a} \right] = 0$$

4. If $f(x) = \sqrt{x+2\sqrt{x-1}} + \sqrt{x-2\sqrt{x-1}}; x \geq 1$.

STATEMENT-1 : Then, $f(x) = 2$, when $x \in [1, 2]$

and

$$\text{STATEMENT-2 : } \sqrt{(f(x))^2} = \begin{cases} f(x), & f(x) \geq 0 \\ -f(x), & f(x) < 0 \end{cases}$$

Sol. Answer (1)

$$x - 1 \geq 0 \Rightarrow x \geq 1$$

$$\text{And if } x \geq 1 \quad x + 2\sqrt{x-1} \geq 1$$

$$x - 2\sqrt{x-1} \geq 0$$

$$x^2 \geq 4(x-1)$$

$$(x^2 - 4x + 4) \geq 0$$

$$(x - 2)^2 \geq 0 \quad \forall x \in \mathbb{R}$$

Put $\sqrt{x-1} = t$; where $t \geq 0$

$$f(t) = \sqrt{t^2 + 2t + 1} + \sqrt{t^2 - 2t + 1} = \sqrt{(t+1)^2} + \sqrt{(t-1)^2}$$

$$= |t+1| + |t-1| \quad \text{Now as } t \geq 0$$

$$= t+1 + 1-t \quad \Rightarrow |1+t| = 1+t$$

$$= 2 \text{ As } 1 \leq x \leq 2$$

$$t \leq 1 \text{ Hence } |t-1| = 1-t$$

5. STATEMENT-1 : Domain of $f(x) = \frac{1}{\sqrt{[x]-x}}$ is ϕ .

and

STATEMENT-2 : $[x] \leq x \quad \forall x \in \mathbb{R}$.

Sol. Answer (1)

$$[x] - x > 0$$

$$\Rightarrow x \in \phi \text{ as } [x] \leq x \quad \forall x \in \mathbb{R}$$

6. STATEMENT-1 : If two sets A and B are having 99 elements in common, then the number of elements common to each of the sets $(A \times B)$ and $(B \times A)$ are $99 + 99 = 198$.

and

STATEMENT-2 : If number of common elements in set A and B are a , the number of element common in $(A \times B)$ and $(B \times A)$ are a^2 .

Sol. Answer (4)

If A and B has a elements in common. The number of common elements in $(A \times B)$ and $(B \times A)$ are a^2 so answer is $(99)^2$ so statement 1 is false.

7. STATEMENT-1 : $f: [-1, 1] \rightarrow \left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$ $f(x) = \sin^{-1}x$ then $\sin^{-1}x_1 > \sin^{-1}x_2 \quad \forall x_1 > x_2$.

and

STATEMENT-2 : For principle values $\sin^{-1}x$ is an increasing function.

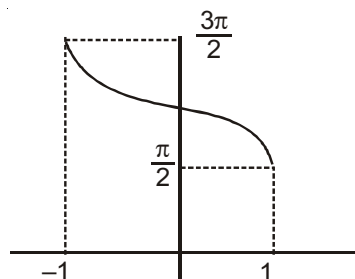
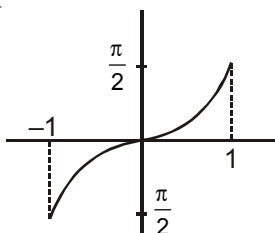
Sol. Answer (4)

Statement 1:

From graph clearly if $x_1 > x_2 \Rightarrow \sin^{-1}x_1 < \sin^{-1}x_2$

Statement 2: For principle value

Clearly $\sin^{-1}x$ is increasing



Statement-1 is false, statement-2 is true.

8. STATEMENT-1 : $f(x) = x^7 + x^6 - x^5 + 3$ is an onto function.

and

STATEMENT-2 : $f(x)$ is a continuous function.

Sol. Answer (2)

Statement 1 : Range of $f(x)$ is R .

Statement 2 : $f(x)$ is polynomial function of degree 7 hence it is continuous, but it is not necessary that continuous function is onto hence

Statement 1 is true statement 2 is true but statement 2 is not correct explanation.

9. STATEMENT-1 : If $f(x)$ and $g(x)$ are one-one functions then $f(g(x))$ and $g(f(x))$ is also a one-one function.

and

STATEMENT-2 : The composite function of two one-one function may or may not be one-one.

Sol. Answer (3)

$f(x)$ and $g(x)$ are one-one functions

Thus $f[g(x_1)] = f[g(x_2)]$

$\Rightarrow g(x_1) = g(x_2)$ as f is one-one

$\Rightarrow x_1 = x_2$ i.e, g is one-one

$\Rightarrow f(g(x))$ is also one-one

Now $g[f(x_1)] = g[f(x_2)]$

$\Rightarrow f(x_1) = f(x_2)$

$\Rightarrow x_1 = x_2, \forall x_1, x_2,$

Hence $f \circ g$ and $g \circ f$ are one-one functions

$\Rightarrow$ Statement 1 is true.

Statement 2 is false as composite function of two one-one function is one-one.

10. Assuming $f(x)$ is an invertible function.

STATEMENT-1 : $f(x)$ is a one-one function $\Leftrightarrow f^{-1}(x)$ is a one-one function.

and

STATEMENT-2 : $f^{-1}(x)$ is the reflection of the function $f(x)$ with respect to $y = x$.

Sol. Answer (1)

$f(x)$ is one-one then $f^{-1}(x)$ is also one-one.

Statement 1 is true.

$f^{-1}(x)$ is the reflection of the function $f(x)$ with respect to $y = x$.

Hence Statement 1 is true statement 2 is true and statement 2 is a correct explanation of statement 1.

11. STATEMENT-1 : Let $f : [1, \infty) \rightarrow [1, \infty)$ be a function such that $f(x) = x^x$ then the function is an invertible function.

and

STATEMENT-2 : The bijective functions are always invertible.

Sol. Answer (1)

Statement 1 : $f(x) = x^x$

$$f'(x) = x^x [1 + \ln x]$$

$f'(x) > 0$ hence $f(x)$ is one-one.

Range of $f(x) = [1, \infty)$

Hence $f(x)$ is onto.

i.e., $f(x)$ is invertible.

Statement 2 is true and statement 2 is a correct explanation of statement 1.

12. STATEMENT-1 : $f \circ g = g \circ f \Rightarrow (f \circ g)(x) = (g \circ f)(x) = x$.

and

STATEMENT-2 : $f \circ g = g \circ f \Rightarrow$ either $f^{-1} = g$ or $g^{-1} = f$.

Sol. Answer (1)

Statement 1 : Let $f : A \rightarrow B$ and $g : B \rightarrow A$

$$g \circ f = I_A \text{ and } f \circ g = I_B$$

$$\text{If } g \circ f = f \circ g$$

$$\Rightarrow A = B$$

$$\text{Hence } g \circ f(x) = f \circ g(x) = x$$

Statement 1 is true.

Statement 2 :

$$\text{Let } f : A \rightarrow B \text{ and } g : B \rightarrow A$$

$$g \circ f = I_A \text{ and } f \circ g = I_B$$

Then f and g are bijection $g = f^{-1}$ or $f = g^{-1}$

Statement 2 is true and is a correct explanation of statement 1.

13. STATEMENT-1 : $f(x) = [\{x\}]$ is a periodic function with no fundamental period.

and

STATEMENT-2 : $f(g(x))$ is periodic if $g(x)$ is periodic.

Sol. Answer (2)

If $g(x)$ is periodic then $g(x + T) = g(x)$

Also $f[g(x + T)] = f(g(x))$ hence $f(g(x))$ is periodic.

$f(x) = [\{x\}] = 0$ is a constant function, hence periodic but its fundamental period cannot be defined.

14. STATEMENT-1 : $f(x) = \log_{10}(\log_{1/x} x)$ will not be defined for any value of x .

and

STATEMENT-2 : $\log_{1/x} x = -1, \forall x > 0, x \neq 1$

Sol. Answer (1)

Statement-2 is true since $\log_{1/x} x = -1$ for $x > 0, x \neq 1$ hence $\log_{10}(\log_{1/x} x)$ will not be defined for any value of x .

15. STATEMENT-1 : $y = \log_e[x^{1001} - x^{101} + x - 1]$ is an onto function in $R \rightarrow R$.

and

STATEMENT-2 : If $f : R \rightarrow R^+$ is an onto function then $y = \log(f(x))$ will also be an onto function in $R \rightarrow R$.

Sol. Answer (4)

Since $f(x)$ is an onto function in $R \rightarrow R^+$, hence range of $f(x)$ is R^+ whereas $\log(f(x))$ need only $f(x) > 0$ to show its complete range ' R ' hence range of $\log(f(x))$ will be ' R ' hence onto.

Statement-1 is false since domain of given function is not R .

16. STATEMENT-1 : $f : R \rightarrow R, f(x) = x^2 \log(|x| + 1)$ is an into function.

and

STATEMENT-2 : $f(x) = x^2 \log(|x| + 1)$ is a continuous even function.

Sol. Answer (1)

$$f(x) = x^2 \log(|x| + 1)$$

$$f(-x) = (-x)^2 \log(|-x| + 1) = f(x) \text{ hence even domain of } f(x) \text{ is 'R' and it is continuous function.}$$

Hence statement-2 is true.

$\therefore$ Range of any even continuous function cannot be ' R ' hence any even continuous function cannot be onto in co-domain R .

17. STATEMENT-1 : $f(x) = x^4 - 3x^2 + 4x - 1$ is many one into in $R \rightarrow R$.

and

STATEMENT-2 : If $f : R \rightarrow R$ is a polynomial of even degree it will neither be injective nor surjective.

Sol. Answer (1)

Statement-1: $f(x) = 4x^3 - 6x + 1$, may be positive as well as negative, hence $f(x)$ is many one

Statement-2 is clearly true

18. STATEMENT-1 : $f(x) = \tan 3x + \{2x\}$, where $\{x\}$ is fractional part of x , $f(x)$ is a periodic function.

and

STATEMENT-2 : LCM of a rational and irrational number is not possible.

Sol. Answer (4)

Statement-1 is false but Statement-2 is correct.

19. STATEMENT-1 : $fog(x) = gof(x) = x \Rightarrow f^{-1} = g$ or $g^{-1} = f$.

and

STATEMENT-2 : $fog \neq gof$.

Sol. Answer (2)

Statement-1 and Statement-2 both correct but Statement-2 is not a correct explanation of Statement-1.

SECTION - E

Matrix-Match Type Questions

1. Match the following functions satisfying particular functional relationship.

Column-I

(A) $f(x + y) = f(x) + f(y)$

(B) $f(xy) = f(x) + f(y)$

(C) $f(x + y) = f(x) \cdot f(y)$

(D) $f(x - y) = \frac{f(x)}{f(y)}$

Column-II

(p) $(2011)^x$

(q) $2011x$

(r) $\log_{2011} x$

(s) $\frac{x}{2011}$

Sol. Answer A(q, s), B(r), C(p), D(p)

(A) Clearly $f(x)$ is a linear function in x

(B) $f(x)$ is a logarithmic function

(C) $f(x)$ is an exponential function

(D) $f(x)$ is an exponential function

2. Let $f(x)$, $g(x)$ and $h(x)$ be three real valued invertible functions, $\forall x \in R$, then match the following

Column-I

(A) $f(g(x))$,

(B) $f(g(h(x)))$

(C) $f(x) + g(x) + h(x)$

(D) $\frac{f(x) \cdot g(x)}{h(x)}$, $(h(x) \neq 0)$

Column-II

(p) Always one-one

(q) Always onto

(r) May not be one-one

(s) May not be onto

Sol. Answer A(p, q), B(p, q), C(r, s), D(r, s)

(A) As $f(x)$ and $g(x)$ are one-one onto.

$\therefore f(g(x))$ is one-one onto

(B) $fg(h(x))$ is also one-one onto.

(C) $f(x) + g(x) + h(x)$ may not be one-one

(D) $\frac{f(x) \cdot g(x)}{h(x)}$ may not be one-one onto.

3. Let $A = \{1, 2, 3, 4\}$ and $f: A \rightarrow A$, then match the following

Column-I**Column-II**

- | | |
|--|---------|
| (A) The number of functions from A to A which are one-one | (p) 3 |
| (B) The number of possible functions from A to A | (q) 9 |
| (C) The number of onto functions from A to A | (r) 256 |
| (D) The number of all one-one onto functions such that $f(1) = 1, f(2) \neq 2, f(4) \neq 4$ | (s) 24 |
| (E) The number of all one-one functions such that $f(1) \neq 1, f(2) \neq 2, f(3) \neq 3, f(4) \neq 4$ | |

Sol. Answer A(s), B(r), C(s), D(p), E(q)

- (A) No. of one-one functions = $4! = 24$
 (B) The number of possible functions from A to A
 $= 4 \times 4 \times 4 \times 4 = 256$
 (C) The number of onto functions = $4! = 24$
 (D) **Case (i),** $f(2) \neq 2, f(4) \neq 4, f(n) \neq n$

Number of such functions

$$= 3! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} \right)$$

$$= 3 - 1 = 2$$

Case (ii), $f(3) = 3, f(2) \neq 2, f(4) \neq 4$

No. of such functions = 1

Total no. of such functions = 3.

- (E) No. of such one-one functions

$$= 4! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} \right)$$

$$= 12 - 4 + 1 = 9$$

4. Let $f(x) = -1 + |x - 1|$, $-1 \leq x \leq 3$ and $g(x) = 2 - |x + 1|$; $-2 \leq x \leq 2$

The composite functions gof and fog are defined as $(gof)(x) = g(f(x))$ and $(fog)(x) = f(g(x))$ respectively, then match the following.

Column-I**Column-II**

- | | |
|------------------------|--------|
| (A) $(gof)(0)$ equals | (p) 1 |
| (B) $(fog)(0)$ equals | (q) -1 |
| (C) $(gof)(3)$ equals | (r) 0 |
| (D) $(fog)(-1)$ equals | (s) 3 |

Sol. Answer A(p), B(q), C(r), D(r)

$$f(x) = -1 + |x - 1|, -1 \leq x \leq 3$$

$$= \begin{cases} -x & , -1 \leq x < 1 \\ x - 2 & , 1 \leq x \leq 3 \end{cases}$$

$$g(x) = 2 - |x + 1|; -2 \leq x \leq 2$$

$$= \begin{cases} 3 + x & , -2 \leq x < -1 \\ 1 - x & , -1 \leq x \leq 2 \end{cases}$$

$$\text{Now } f(0) = 0, g(0) = 1, f(3) = 1, g(-1) = 2$$

$$gof(0) = g(0) = 1$$

$$fog(0) = f(1) = 1 - 2 = -1$$

$$gof(3) = g(1) = 1 - 1 = 0$$

$$fog(-1) = f(2) = 0$$

5. Let us consider a real valued function f defined as $f(x) = \frac{1-x}{1+x}, x \neq -1$. Then match the following.

Column-I

Column-II

(A) $f(x) + f\left(\frac{1}{x}\right); x \neq 0$ equals

(p) 1

(B) $\frac{1}{x} f(f(x)); x \neq 0$ equals

(q) 0

(C) $f(f(x)) + f\left(f\left(\frac{1}{x}\right)\right)$ when $x > 0$ or $x < 0$ may be

(r) 2

(D) $\frac{1}{x} f(f(f(\dots f(x)))) \dots; x \neq 0$ 2008 times equals

(s) -2

Sol. Answer A(q), B(p), C(r), D(p)

$$f(x) = \frac{1-x}{1+x}, x \neq -1$$

(A) $f(x) + f\left(\frac{1}{x}\right) = \frac{1-x}{1+x} + \frac{x-1}{x+1} = 0$

(B) $\frac{1}{x} f(f(x)) = \frac{1}{x} \times \frac{1 - \frac{1-x}{1+x}}{1 + \frac{1-x}{1+x}} = \frac{1}{x} \frac{1+x-1+x}{1+x+1-x} = \frac{2x}{2x} = 1$

$$(C) f(f(x)) + f\left(f\left(\frac{1}{x}\right)\right) = x + f\left(\frac{x-1}{x+1}\right) = x + \frac{1 - \frac{x-1}{x+1}}{1 + \frac{x-1}{x+1}} = x + \frac{1}{x}$$

$$x + \frac{1}{x} \geq 2 \text{ for } x > 0 \text{ or } x + \frac{1}{x} \leq -2 \text{ for } x < 0$$

$$(D) \frac{1}{x} f(f(f(f \dots f(x)))) \dots = 1$$

$$\text{As } \frac{1}{x} f(f(x)) = 1 \quad f$$

6. Match the following

Column-I

$$(A) y = x - \frac{x^3}{6} + \frac{x^5}{120}$$

$$(B) y = \frac{x}{a^x - 1}$$

$$(C) y = x \frac{a^x - 1}{a^x + 1}$$

$$(D) y = \frac{a^x + 1}{a^x - 1}$$

Column-II

(p) Even function

(q) Odd function

$$(r) \lim_{x \rightarrow 0} f(x) = \frac{1}{\ln a}$$

(s) Neither even nor odd

Sol. Answer A(q), B(r, s), C(p), D(q)

$$(A) y = x - \frac{x^3}{6} + \frac{x^5}{120}$$

$$f(x) = x - \frac{x^3}{6} + \frac{x^5}{120}$$

$$f(-x) = -x + \frac{x^3}{6} - \frac{x^5}{120} = -\left(x - \frac{x^3}{6} + \frac{x^5}{120}\right) = -f(x)$$

which is an odd function

$$(B) y = \frac{x}{a^x - 1}$$

$$f(x) = \frac{x}{a^x - 1}$$

$$f(-x) = \frac{-x}{a^{-x} - 1} = \frac{-x}{\frac{1}{a^x} - 1} \Rightarrow \frac{-x}{1 - a^x} (a^x)$$

which is neither even nor odd

$$f(0) = \lim_{x \rightarrow 0} \frac{x}{a^x - 1} = \frac{1}{\ln a}$$

$$(C) f(x) = x \frac{a^x - 1}{a^x + 1}$$

$$f(-x) = (-x) \frac{\frac{1}{a^x} - 1}{\frac{1}{a^x} + 1} = (-x) \frac{1 - a^x}{1 + a^x}$$

$$f(-x) = x \frac{(a^x - 1)}{a^x + 1} = f(x)$$

which is an even function.

$$(D) f(x) = \frac{a^x + 1}{a^x - 1}$$

$$f(-x) = \frac{a^{-x} + 1}{a^{-x} - 1} = \frac{1 + a^x}{1 - a^x} = -\left(\frac{a^x + 1}{a^x - 1}\right)$$

$f(-x) = -f(x) \Rightarrow f(x)$ is an odd function

7. The period of the function

Column-I

$$(A) \sin 2\theta - \sqrt{3} \cos 2\theta$$

$$(B) \operatorname{cosec}^2 \frac{3}{4}x + \cot x$$

$$(C) 2\sin 3x + 3\sin 2x$$

$$(D) \sin \frac{\pi t}{3} + \sin \frac{\pi t}{4}$$

Column-II

$$(p) 4\pi$$

$$(q) \pi$$

$$(r) 24$$

$$(s) 2\pi$$

Sol. Answer A(q), B(p), C(s), D(r)

$$(A) f(\theta) = \sin 2\theta - \sqrt{3} \cos 2\theta$$

Period of $\sin 2\theta = \pi$

Period of $\cos 2\theta = \pi$

Hence required period LCM. $(\pi, \pi) = \pi$

$$(B) f(x) = \operatorname{cosec}^2 \frac{3}{4}x + \cot x$$

$$\text{Period of } \operatorname{cosec}^2 \frac{3}{4}x = \frac{4\pi}{3}$$

And period of $\cot x = \pi$

$$\text{Required period LCM} \left(\frac{4\pi}{3}, \pi \right) = 4\pi$$

(C) $f(x) = 2 \sin 3x + 3 \sin 2x$

Period of $\sin 3x = \frac{2\pi}{3}$

Period of $\sin 2x = \frac{2\pi}{2}$

Required period LCM $\left(\frac{2\pi}{3}, \pi\right) = 2\pi$

(D) $\sin \frac{\pi t}{3} + \sin \frac{\pi t}{4}$

Period of $\left(\sin \frac{\pi t}{3}\right) = \frac{2\pi}{\pi} \times 3 = 6$

Period of $\left(\sin \frac{\pi t}{4}\right) = \frac{2\pi}{\pi} \times 4 = 8$

Required period LCM = (6, 8) = 24

8. Match the following with their fundamental periods (where $[.]$ denotes greatest integer function and $\{.\}$ denotes fractional part function)

Column-I

(A) $f(x) = e^{\cos\{x\}} + \sin\pi[x]$

(B) $f(x) = \sin^4\pi x + \cos^3\pi x + \tan^2\pi x + \sin^2\pi x$

(C) $f(x) = e^{\sin\{x\}} + \sin\left(\frac{\pi}{2}[x]\right)$

(D) $f(x) = e^{x-[x]} + \cos 2\pi x$

Column-II

(p) 2

(q) 4

(r) 24

(s) 1

Sol. Answer A(s), B(p), C(q), D(s)

(A) $f(x) = e^{\cos\{x\}} + 0 \quad (\sin\pi[x] = 0)$

Fundamental period is 1

(B) $f(x) = \sin^4\pi x + \cos^3\pi x + \tan^2\pi x + \sin^2\pi x$

Fundamental period of $\sin^4\pi x = 1$

Fundamental period of $\cos^3\pi x = 2$

Fundamental period of $\tan^2\pi x = 1$

Fundamental period of $\sin^2\pi x = 1$

$\therefore$ L.C.M. of 1, 2, 1, 1 is 2.

$$(C) f(x) = e^{\sin\{x\}} + \sin\left(\frac{\pi}{2}[x]\right)$$

$$f(x) = e^{\sin\{x\}} \text{ if } (x) \text{ is even.}$$

$$f(x) = e^{\sin\{x\}} \pm 1 \text{ if } (x) \text{ is odd}$$

$\therefore$ Fundamental period is 1.

$$(D) f(x) = e^{x-[x]} + \cos 2\pi x$$

$$= e^{\{x\}} + \cos 2\pi x$$

Fundamental period of $e^{\{x\}}$ is 1

Fundamental period of $\cos 2\pi x$ is 1

$\therefore$ Fundamental period of $f(x)$ is 1

9. Match the following

Column-I

Column-II

(A) If function $f(x)$ is defined in $[-3, 3]$, then domain of $f([x+1])$ is where $[.]$ represents greatest integer function

(p) $\left[\frac{1}{4}, \frac{3}{4}\right]$

(B) Range of function $f(x) = \frac{(\sin^{-1} x + \cos^{-1} x + \tan^{-1} x)}{\pi}$

(q) $[-4, 3]$

(C) Range of function $3|\sin x| - 4|\cos x|$ is

(r) $[-4, 3]$

(D) Range of $f(x) = \sin^{-1} x \cdot \sin x$

(s) $\left[0, \frac{\pi}{2} \sin 1\right]$

Sol. Answer A(r), B(p), C(q), D(s)

(A) Domain of $f(x)$ is $[-3, 3]$

For $f([x] + 1)$ to be defined $-3 \leq [x+1] \leq 3$

$$\Rightarrow -4 \leq [x] \leq 2$$

$$\Rightarrow x \in [-4, 3)$$

(B) $f(x) = \frac{\sin^{-1} x + \cos^{-1} x + \tan^{-1} x}{\pi}$

Domain of $f(x)$ is $[-1, +1]$

$$f(x) = \frac{\frac{\pi}{2} + \tan^{-1} x}{\pi} = \frac{1}{2} + \frac{1}{\pi} \tan^{-1} x$$

$$\therefore -1 \leq x \leq 1$$

$$\therefore -\frac{\pi}{4} \leq \tan^{-1} x \leq \frac{\pi}{4}$$

$$\Rightarrow -\frac{1}{4} \leq \frac{\tan^{-1} x}{\pi} \leq \frac{1}{4}$$

$$\frac{1}{4} \leq \frac{1}{2} + \frac{\tan^{-1} x}{\pi} \leq \frac{3}{4} \text{ hence range of } f(x) \text{ is } \left[\frac{1}{4}, \frac{3}{4} \right]$$

$$(C) \because 0 \leq |\sin x| \leq 1, 0 \leq |\cos x| \leq 1$$

$$\therefore 0 \leq 3|\sin x| \leq 3, 0 \leq 4|\cos x| \leq 4$$

$$\Rightarrow 0 \geq -4|\cos x| \geq -4$$

$$\therefore 0 + (-4) \leq 3(\sin x) + (-4|\cos x|) \leq 3 + 0$$

$$-4 \leq 3|\sin x| - 4|\cos x| \leq 3$$

Hence range of function is $[-4, 3]$

$$(D) \because f(x) = \sin^{-1} x \cdot \sin x$$

$$\because f(-x) = f(x) \text{ hence function is even.}$$

$$\because x \in [0, 1]$$

$$0 \leq \sin^{-1} x \leq \frac{\pi}{2} \text{ and } 0 \leq \sin x \leq \sin 1$$

$\therefore$ Product of two increasing function will be increasing if both are increasing and positive.

Hence $f(x)$ will be increasing in $[0, 1)$. Hence range $\left[0, \frac{\pi}{2} \sin 1 \right]$

10. Match the following

Column-I

(A) Number of solution of $2[x] = x + 2\{x\}$

(B) Number of solution of $\{x\} = e^{x^2}$

(C) Number of solution of $\sin^{-1} x = \text{Sgn}(x)$

(D) Fundamental period of function $f(x) = \left\{ \frac{2}{3}x \right\} + \sin 6\pi x$

where $\{.\}$ represents fractional part function

Column-II

(p) 0

(q) 3

(r) 1

(s) Not defined

Sol. Answer A(q), B(p), C(p), D(q)

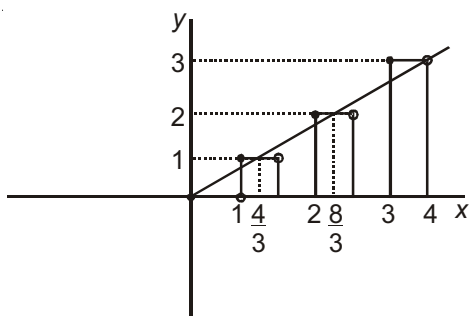
(A) $2[x] = x + 2\{x\}$

$$2[x] = x + 2[x - [x]]$$

$$4[x] = 3x$$

$$[x] = \frac{3}{4}x$$

Hence three solution $x = 0, \frac{4}{3}, \frac{8}{3}$



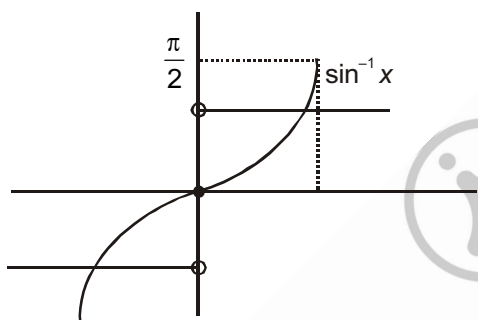
(B) $\because 0 \leq x^2$ and $0 < \{x\} < 1$

$$\therefore e^0 \leq e^{x^2}$$

$$1 \leq e^{x^2}$$

Hence equation $\{x\} = e^{x^2}$ will have zero solution.

(C) See graph



No. of solution are '3'

(D) $f(x) = \left\{ \frac{2}{3}x \right\} + \sin 6\pi x$

Fundamental period of $\left\{ \frac{2}{3}x \right\} \rightarrow \frac{1}{\frac{2}{3}} \Rightarrow \frac{3}{2}$

Fundamental period of $\sin 6\pi x \rightarrow \frac{2\pi}{6\pi} \Rightarrow \frac{1}{3}$

Fundamental period of $f(x) = \text{LCM of } \left(\frac{3}{2}, \frac{1}{3} \right) = 3$

11. Match the following

Column-I

(A) Continuous domain of $f(x) = \sqrt{x^x + \sqrt{x^x + \sqrt{x^x}}}$

(B) Range of $f(x) = \sqrt{2 - [\sin x] - [\sin x]^2}$

(C) Solution of the equation $1 + \sin \frac{\pi}{\sqrt{2}} x = x^2 - \sqrt{2}x + 1$

(D) Domain of $f(x) = \sqrt{\log_{\{x\}}[x]}$

Column-II

(p) R^+

(q) $\{0, \sqrt{2}\}$

(r) $(1, 2)$

(s) R

Sol. Answer A(p), B(q), C(q), D(r)

$$(A) f(x) = \sqrt{x^x + \sqrt{x^x} + \sqrt{x^x}}$$

x^x is defined for $x > 0$ and I^- and some other negative value of x . But continuous domain will be only R^+

$$(B) f(x) = \sqrt{2 - [\sin x] - [\sin x]^2}$$

$$\therefore -1 \leq \sin x \leq +1$$

$$[\sin x] = \{-1, 0, +1\}$$

$$\text{Hence range} = \{\sqrt{2}, 0\}$$

$$(C) 1 + \sin \frac{\pi}{\sqrt{2}} x = x^2 - \sqrt{2}x + 1$$

$$\sin \frac{\pi}{\sqrt{2}} x = x(x - \sqrt{2})$$

By observation at $x = 0$, $x = \sqrt{2}$

$$(D) f(x) = \sqrt{\log_{\{x\}}[x]}$$

For $\log_{\{x\}}[x]$ to be defined $[x] > 0$ $0 < \{x\} < 1$ $\{x\} \neq 1$

$$\Rightarrow [x] \geq 1 \quad x \notin I \quad x \in R$$

$$\Rightarrow x \in [1, \infty) \quad \dots(i)$$

$$\text{Also } \log_{\{x\}}[x] \geq 0 \Rightarrow [x] \leq 1 \quad \dots(ii)$$

From (i) and (ii)

$$[x] = 1 \Rightarrow 1 \leq x < 2 \quad x \notin I$$

Hence $x \in (1, 2)$

12. Let us consider two functions $f(x) = \ln(2x - x^2) + \sin \frac{\pi x}{2}$ and $g(x) = \log_{|x-1|} \frac{|x|}{x}$, where $[.]$ denotes G.I.F.

Match the items of Column I with those of Column II

Column I

- (A) Graph of f is symmetrical about the line
 (B) Maximum value of f occurs at
 (C) Domain of g is not equal to
 (D) Range of g is not equal to

Column II

- (p) $x = 1$
 (q) $x = 2$
 (r) $[3, \infty)$
 (s) $\{0\}$
 (t) $\{0, 1\}$

Sol. Answer A(p), B(p), C(p, q, s, t), D(p, q, r, t)

$$f(1 + \alpha) = f(1 - \alpha), \forall \alpha \in (0, 1)$$

Domain of f is $(0, 2)$

Maxima of $\ln(2x - x^2)$ as well as that of $\sin\left(\frac{\pi x}{2}\right)$ occurs at $x = 1$

$$g(x) = \log_{|x-1|} \text{sgn}(x)$$

$$\text{Domain } (g) = [3, \infty)$$

$$\text{Range } (g) = \{0\}$$

13. Match the following

Column I

- (A) $\sin(\pi[x])$, where $[.]$ is G.I.F.
 (B) $\sin\{\pi(x - [x])\}$, where $[.]$ is G.I.F.
 (C) $\tan x, x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
 (D) $1 + x^{2009}$

Column II

- (p) Differentiable everywhere
 (q) Nowhere differentiable
 (r) Not differentiable at 1 and -1
 (s) One-one
 (t) Both one-one and onto

Sol. Answer A(p), B(r), C(p, s, t), D(p, s, t)

(A) $\because [x] \in I \forall x \in R$, therefore $\sin(\pi[x]) = 0 \forall x \in R$

(B) Again $f(x) = \sin\{\pi(x - [x])\} = \sin(\pi\{x\})$

Which is not differentiable at $x \in I$

(C) $y = \tan x, x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ is differentiable everywhere in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ and from the graph of $y = \tan x$, it is clear that $\tan x$ is one-one & onto

(D) $y = 1 + x^{2009}$ is a polynomial, so differentiable everywhere as well as one-one & onto.

14. Match the following

Column-I

- (A) Let A and B be two finite sets having 8 and 2 elements respectively. Then total number of mappings from A to B is
 (B) The total number of injective mappings from a set with 2 elements to a set with 8 elements, is not equal to
 (C) Let A be a set containing 4 distinct elements then the total number of distinct functions from A to A is not equal to
 (D) Let A and B be two sets containing 8 and 2 elements respectively then the total number of surjective mapping from A to B is

Column-II

- (p) 56
 (q) 254
 (r) 2^8
 (s) 8^2
 (t) 512

Sol. Answer A(r), B(q, r, s, t), C(p, q, s, t), D(q)

(A) The number of mapping is $2 \times 2 \times 2 \times \dots \times 2$ 8 times $= 2^8$

(B) The required number of injective mappings $= \frac{8!}{(8-2)!} = 56$

(C) The total number of distinct functions from A to A is $4^4 = 256$

(D) The number of surjective mappings are $2^8 - 2 = 254$

15. Let $f_1 : \mathbb{R} \rightarrow \mathbb{R}$, $f_2 : [0, \infty) \rightarrow \mathbb{R}$, $f_3 : \mathbb{R} \rightarrow \mathbb{R}$ and $f_4 : \mathbb{R} \rightarrow [0, \infty)$ be defined by [JEE(Advanced)-2014]

$$f_1(x) = \begin{cases} |x| & \text{if } x < 0, \\ e^x & \text{if } x \geq 0; \end{cases}$$

$$f_2(x) = x^2;$$

$$f_3(x) = \begin{cases} \sin x & \text{if } x < 0, \\ x & \text{if } x \geq 0 \end{cases}$$

$$\text{and } f_4(x) = \begin{cases} f_2(f_1(x)) & \text{if } x < 0, \\ f_2(f_1(x)) - 1 & \text{if } x \geq 0. \end{cases}$$

Column-I

- (A) f_4 is
(B) f_3 is
(C) $f_2 \circ f_1$ is
(D) f_2 is

A B C D

(1) r p s q

(2) p r s q

(3) r p q s

(4) p r q s

Sol. Answer A(p), B(r), C(q), D(s)

$$(A) f_4(x) = \begin{cases} f_2[f_1(x)], & x < 0 \\ f_2[f_1(x)] - 1, & x \geq 0 \end{cases}$$

$$\text{Onto but not one-one} = \begin{cases} x^2, & x < 0 \\ e^{2x} - 1, & x \geq 0 \end{cases}$$

Now $f(x)$ is not differentiable at $x = 0$, not one-one but continuous.

(A) $\rightarrow$ p

$$(B) \text{ Now, } f_3 = \begin{cases} \sin x, & x < 0 \\ x, & x \geq 0 \end{cases}$$

Differentiable but not one-one.

(B) $\rightarrow$ r

$$(C) f_2 \circ f_1 = \begin{cases} x^2, & x < 0 \\ e^{2x}, & x \geq 0 \end{cases}$$

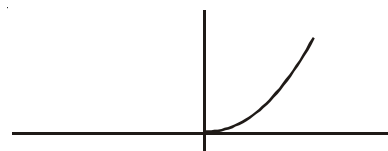
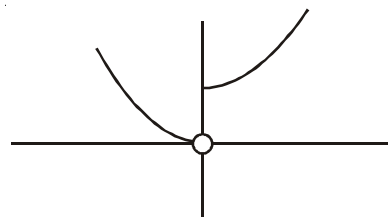
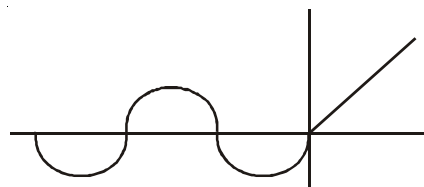
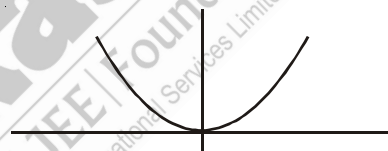
(C) $\rightarrow$ q

$$(D) f_2 : [0, \infty) \rightarrow \mathbb{R}, f_2(x) = x^2$$

(D) $\rightarrow$ s

Column-II

- (p) Onto but not one-one
(q) Neither continuous nor one-one
(r) Differentiable but not one-one
(s) Continuous and one-one



SECTION - F

Integer Answer Type Questions

1. The length of the interval in which the function f defined as $f(x) = \log_2 \left\{ -\log_{1/2} \left(1 + \frac{1}{\sqrt[6]{x}} \right) - 1 \right\}$ is $(0, k)$, then the value of k _____.

Sol. Answer (1)

In order that the given function is well defined

$$-\log_{1/2} \left(1 + \frac{1}{\sqrt[6]{x}} \right) - 1 > 0 \text{ and } x > 0$$

$$\Rightarrow \log_{1/2} \left(1 + \frac{1}{\sqrt[6]{x}} \right) < -1$$

$$\Rightarrow 1 + \frac{1}{\sqrt[6]{x}} > \left(\frac{1}{2} \right)^{-1}$$

$$\Rightarrow \frac{1}{\sqrt[6]{x}} > 1$$

$$\Rightarrow \frac{1}{x} > 1 \Rightarrow x < 1$$

Thus required interval = $(0, 1)$ whose length = 1

2. Value of $-\frac{4}{5}f(2,3)$ if $f(x+y, x-y) = xy$, is _____.

Sol. Answer (1)

Put $x+y=2$, $x-y=3$

$$\Rightarrow x = \frac{5}{2}, y = \frac{-1}{2}$$

$$\therefore -\frac{4}{5}f(2,3) = -\frac{4}{5} \times \frac{5}{2} \times \frac{-1}{2} = 1$$

3. Number of integral values of x in the domain of $f(x) = \sqrt{-[x]^2 + 3[x] - 2}$ is _____.

Sol. Answer (2)

$$-[x]^2 + 3[x] - 2 \geq 0$$

$$[x]^2 - 3[x] + 2 \leq 0$$

$$\Rightarrow 1 \leq [x] \leq 2$$

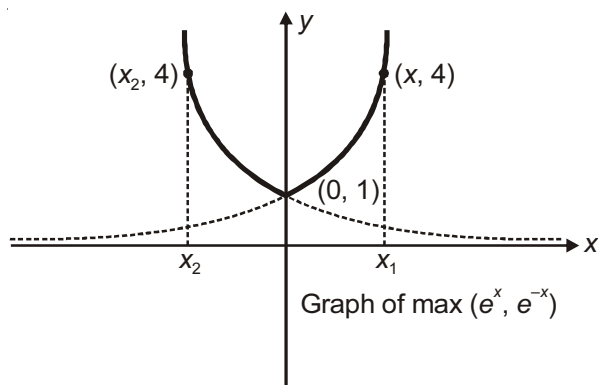
$$1 \leq x \leq 3$$

Integral $x = 1, 2$

Number of integral values of x is 2

4. Sum of all the values of x satisfying maximum $\{e^x, e^{-x}\} = 4$ is _____.

Sol. Answer (0)



Clearly, $x_1 + x_2 = 0$ (from symmetry)

5. $f(x) = |x - 1| + |x - 2|$; $g(x) = x + \frac{1}{x}, x > 0$ if $m_1 = \min(f(x))$ and $m_2 = \min(g(x))$, then $\frac{m_1 + m_2}{m_2 - m_1}$ is equal to _____.

Sol. Answer (3)

$$f(x) = |x - 1| + |x - 2|$$

$$\text{Min } f(x) = 1 \Rightarrow m_1 = 1$$

$$g(x) = x + \frac{1}{x}; x > 0$$

$$\text{Min } g(x) = 2 \Rightarrow m_2 = 2$$

$$\Rightarrow \frac{m_1 + m_2}{m_2 - m_1} = 3$$

6. Let $Y = \{1, 2, 3, 4, 5\}$, $A = \{1, 2\}$, $B = \{3, 4, 5\}$. If $(A \times B)$ denotes Cartesian product of the set A and B , then number of elements in $(Y \times A) \cap (Y \times B)$ is _____.

Sol. Answer (0)

$$A = \{1, 2\}, B = \{3, 4, 5\}, Y = \{1, 2, 3, 4, 5\}$$

$$(Y \times A) = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 1), (3, 2), (4, 1), (4, 2), (5, 1), (5, 2)\}$$

$$(Y \times B) = \{(1, 3), (1, 4), (1, 5), (2, 3), (2, 4), (2, 5), \dots, (5, 4), (5, 5)\}$$

$$(Y \times A) \cap (Y \times B) = \phi \text{ so no element into this set.}$$

7. The number of possible values of k if fundamental period of $\sin^{-1}(\sin kx)$ is $\frac{\pi}{2}$, is

Sol. Answer (2)

$$\text{Fundamental period of } \sin^{-1}(\sin kx) \text{ is } \frac{2\pi}{|k|} = \frac{\pi}{2}$$

$$\text{i.e. } |k| = 4 \quad \text{i.e. } k = \pm 4$$

8. Number of elements in the domain of $f(x) = \tan^{-1}x + \sin^{-1}x + \sec^{-1}x$ is,

Sol. Answer (2)

Domain of $\tan^{-1}x$ is R

Domain of $\sin^{-1}x$ is $[-1, 1]$

And domain of $\sec^{-1}x$ is $(-\infty, -1] \cup [1, \infty)$

$\therefore$ domain of $f(x)$ is $\{-1, 1\}$

9. Period of the function $f(x) = \sin\left(\frac{\pi x}{2}\right)\cos\left(\frac{\pi x}{2}\right)$ is

Sol. Answer (2)

$$f(x) = \frac{2}{2}\sin\left(\frac{\pi x}{2}\right)\cos\left(\frac{\pi x}{2}\right) = \frac{1}{2}\sin\pi x$$

Period of $f(x)$ is 2.

10. If the range of $f(x) = \cos^{-1}[5x]$ is $\{a, b, c\}$ and $a + b + c = \frac{\lambda\pi}{2}$, then λ is equal to ([.] denotes G.I.F.)

Sol. Answer (3)

$$f(x) = \cos^{-1}[5x]$$

$[5x]$ can take the values $-1, 0, 1$

$$\therefore \text{range} = \left\{\pi, \frac{\pi}{2}, 0\right\}$$

$$\therefore a + b + c = \pi + \frac{\pi}{2} + 0 = \frac{3\pi}{2}$$

11. If $f(x) = {}^{8-x}P_{x-5}$, then number of integral values in the domain of $f(x)$ is _____.

Sol. Answer (2)

$$f(x) = {}^{8-x}P_{x-5} \Rightarrow x \leq 8, x \geq 5 \text{ and } 8 - x \geq x - 5$$

$\therefore$ Domain consists of integers $\{5, 6\}$

$\therefore$ Number of integers is 2.

12. If $f: R \rightarrow R$ is defined by $f(x) = x^3 + 1$, then $f^{-1}(28) =$ _____.

Sol. Answer (3)

$$f(x) = x^3 + 1 \Rightarrow f^{-1}(x) = (x - 1)^{1/3} \Rightarrow f^{-1}(28) = (28 - 1)^{1/3} = 3$$

13. Period of the function $f(x) = \cos(\cos\pi x) + e^{\{4x\}}$, where $\{.\}$ denotes the fractional part of x , is _____.

Sol. Answer (1)

$$\text{Period of } \cos(\cos\pi x) \text{ is } \left|\frac{\pi}{\pi}\right| = 1 \text{ and period of } e^{\{4x\}} \text{ is } \frac{1}{4}$$

$$\therefore \text{Period of } f(x) = \text{LCM of } \left\{1, \frac{1}{4}\right\} = 1$$

SECTION - G

Multiple True-False Type Questions

1. STATEMENT-1 : If $n(A) = 3$, $n(B) = 6$, then minimum number of elements in $A \cup B$ is 6.

STATEMENT-2 : $A = \{x \mid x \in R, |x| < 2\}$. and $B = \{x \mid x \in R, |x| \geq 2\}$. If $A \cup B = C - R$, then $C = \{x \mid x \in R, 2 \leq x < 3\}$.

STATEMENT-3 : The range of $f(x) = 2^{(x^2-2)^3+8}$ is $[1, \infty)$

(1) T F T

(2) T T F

(3) T T T

(4) T F F

Sol. Answer (1)

2. STATEMENT-1 : If $f(x) = \frac{1}{2}(3^x + 3^{-x})$, then $f(x) \geq 1 \quad \forall x \in R$.

STATEMENT-2 : The domain of $f(x) = \log_3 |x| + \sqrt{x^2 - 1} + \frac{1}{|x|}$ is $R - [-1, 1]$.

STATEMENT-3 : If $f(x^2 - 2x + 3) = x - 1$, then $f(2) + f(0)$ is equal to -2 .

(1) T T F

(2) T T T

(3) T F F

(4) F F F

Sol. Answer (3)

(i) Use A.M $\geq$ G.M

(ii) $x \neq 0, \quad |x - 1| \geq 1$
 $\Rightarrow x \neq 1$

(iii) $f(x - 1)^2 + 2 = (x - 1)$

$f(x) = -\sqrt{x - 2}$

$f(2) = 0; f(6) = -2$

3. STATEMENT-1 : A, B, C, D are four sets such that $n(A \cap C) = 3$ and $n(B \cap D) = 2$, then $n(P(A \cap C)) \neq n(P(B \cap D))$.

STATEMENT-2 : If $A \subseteq B$ and $C \subseteq D$, then $A \times C \subseteq B \times D$.

STATEMENT-3 : If $A \cup B = A \cap B$, then it is NOT necessary that $A = B$.

(1) T T T

(2) T T F

(3) T F T

(4) F T T

Sol. Answer (2)

Clearly $n(A \cap C) = 3$

$n(B \cap D) = 2$

$n(P(A \cap C)) = 2^3; n(P(B \cap D)) = 2^2$

4. STATEMENT-1 : If $f(x)$ is a constant function, then $f^{-1}(x)$ is also a constant function.

STATEMENT-2 : If graphs of $f(x)$ and $f^{-1}(x)$ are intersecting then they always intersect on the line $y = x$.

STATEMENT-3 : The inverse of $f(x) = \frac{x}{1 + |x|}$ is $\frac{x}{1 - |x|}$

(1) F T T

(2) T F F

(3) F F T

(4) T F T

Sol. Answer (3)

Statement-1 : Constant function is not invertible as it is many one.

Statement-2 : $f(x)$ and $f^{-1}(x)$ intersect on $y = x$ or $y = x + c$

$$\text{Statement-3 : } f(x) = \begin{cases} \frac{x}{1+x}; & x \geq 0 \\ \frac{x}{1-x}; & x < 0 \end{cases}; f^{-1}(x) = \begin{cases} \frac{x}{1-x}; & x \geq 0 \\ \frac{x}{1+x}; & x < 0 \end{cases}$$

5. STATEMENT-1 : For real values of x and y the relation $y^2 = 2x - x^2 - 1$ represents y as a function of x .STATEMENT-2 : If $f(x) = \log(x-2)(x-3)$ and $g(x) = \log(x-2) + \log(x-3)$ then $f = g$ STATEMENT-3 : If $f(x+2) = 2x-5$ then $f(x) = 2x-9$.

(1) F T F

(2) F T T

(3) T F T

(4) T T F

Sol. Answer (3)Statement-1 : $y^2 = -(x-1)^2$, y is a function of x with domain $\{1\}$ and range $\{0\}$ Statement-2 : $f \neq g \because D(f) \in (-\infty, 2) \cup (3, \infty)$ and $D(g) \in (3, \infty)$ Statement-3 : Replace x by $x-2$, $f(x) = 2(x-2) - 5 = 2x-9$.**SECTION - H****Aakash Challengers Questions**

1. Let $f(x) = \frac{2008^x + 2008^{-x}}{2}$, $g(x) = \frac{2008^x - 2008^{-x}}{2}$ then prove that $f(x+y) = f(x)f(y) + g(x)g(y)$.

Sol. Let $f(x) = \frac{2008^x + 2008^{-x}}{2}$,

$$g(x) = \frac{2008^x - 2008^{-x}}{2}$$

The given function satisfies $f(x+y) = f(x)f(y) + g(x)g(y)$ 2. Let $g(x)$ be a function satisfying $g(0) = 2$, $g(1) = 3$, $g(x+2) = 2g(x) - g(x+1)$, then find $g(5)$.**Sol.** $g(0) = 2$

$$g(1) = 3$$

$$g(x+2) = 2g(x) - g(x+1)$$

$$g(2) = 2g(0) - g(1)$$

$$= 4 - 3$$

$$= 1$$

$$g(2) = 1$$

$$g(3) = 2g(1) - g(2)$$

$$= 6 - 1$$

$$= 5$$

$$g(4) = 2g(2) - g(3)$$

$$= 2 - 5$$

$$= -3$$

$$g(5) = 2g(3) - g(4)$$

$$= 10 - (-3)$$

$$= 13$$

3. Let $f : W \rightarrow W$ be a given function satisfying $f(x) = f(x-1) + f(x-2)$ for $x \geq 2$. If $f(0) = 0$ and $f(1) = 1$, then find the value of $f(2) + f(3) + f(4) + f(5) + f(6)$.

Sol. $f : W \rightarrow W$ be a given function satisfying $f(x) = f(x-1) + f(x-2)$ for $x \geq 2$

If $f(0) = 0$ and $f(1) = 1$

$$f(2) = f(1) + f(0) = 1 + 0 = 1$$

$$f(3) = f(2) + f(1) = 1 + 1 = 2$$

$$f(4) = f(3) + f(2) = 2 + 1 = 3$$

$$f(5) = f(4) + f(3) = 3 + 2 = 5$$

$$f(6) = f(5) + f(4) = 5 + 3 = 8$$

$$\text{Now } f(2) + f(3) + f(4) + f(5) + f(6) = 1 + 2 + 3 + 5 + 8 = 19$$

4. Let $f(x) = \ln\left(\frac{1-x}{1+x}\right)$. Find x, y for which $f(x) + f(y) = f\left(\frac{x+y}{1+xy}\right)$.

Sol. Let $f(x) = \ln\left(\frac{1-x}{1+x}\right)$

$$\text{Now } f(x) + f(y) = f\left(\frac{x+y}{1+xy}\right)$$

$$\Rightarrow \ln\left(\frac{1-x}{1+x}\right) + \ln\left(\frac{1-y}{1+y}\right) = \ln\left(\frac{1-\frac{x+y}{1+xy}}{1+\frac{x+y}{1+xy}}\right)$$

$$\Rightarrow \frac{(1-x)(1-y)}{(1+x)(1+y)} = \frac{1+xy-x-y}{1+xy+x+y}$$

$\Rightarrow$ Which is an identity

Clearly $f(x)$ is well-defined for $-1 < x < 1$, and in this case $-1 < y < 1$

Hence $x, y \in (-1, 1)$

5. If $f(x+y) = f(x)f(y) \quad \forall x, y \in R$, and $f(0) \neq 0$. Let $\phi(x) = \frac{f(x)}{1+(f(x))^2}$. Prove that $\phi(x) - \phi(-x) = 0 \quad \forall x \in R$.

Sol. $f(0) = 1$

$$f(x)f(-x) = f(0) = 1$$

Hence,

$$\phi(x) = \frac{f(x)}{1+(f(x))^2}$$

$$\phi(-x) = \frac{f(-x)}{1+(f(-x))^2}$$

$$= \frac{\frac{1}{f(x)}}{\frac{1+1}{(f(x))^2}}$$

$$= \frac{f(x)}{1+(f(x))^2}$$

$$\boxed{\phi(x) = \phi(-x)}$$

6. The function $f(x)$ satisfies the equation $f(x+1) + f(x-1) = \sqrt{3} f(x) \quad \forall x \in R$, then the period of $f(x)$ is.....

Sol. We have

$$f(x+1) + f(x-1) = \sqrt{3} f(x), \quad \forall x \in R.$$

$$f(x+1) = \sqrt{3} f(x) - f(x-1)$$

Let $x+1 = r$

$$f(r) = \sqrt{3} f(r-1) - f(r-2)$$

$$\Rightarrow f(r) = \sqrt{3} [\sqrt{3} f(r-2) - f(r-3)] - f(r-2) \Rightarrow f(r) = 3f(r-2) - \sqrt{3} f(r-3) - f(r-2)$$

$$\Rightarrow f(r) = 2f(r-2) - \sqrt{3} f(r-3)$$

$$\Rightarrow f(r) = 2[\sqrt{3} f(r-3) - f(r-4)] - \sqrt{3} f(r-3)$$

$$\Rightarrow f(r) = \sqrt{3} f(r-3) - 2f(r-4)$$

$$\Rightarrow f(r) = \sqrt{3} [\sqrt{3} f(r-4) - f(r-5)] - 2f(r-4)$$

$$\Rightarrow f(r) = f(r-4) - \sqrt{3} f(r-5)$$

$$\Rightarrow f(r) = \sqrt{3} [\sqrt{3} f(r-5) - f(r-6)] - \sqrt{3} f(r-5)$$

$$\Rightarrow f(r) + f(r-6) = 0$$

$$\Rightarrow f(r) = -f(r+6) = f(r+12)$$

Period of the given function is = 12

7. Let $g : R \rightarrow R$ be given by $g(x) = 3 + 4x$. If $g^n(x) = g \circ g \circ \dots \circ g(x)$ n times, then $g^4(1)$ equals.....

Sol. $g : R \rightarrow R$ $g(x) = 3 + 4x$.

$$g(x) = 3 + 4x$$

$$g^2(x) = g(g(x)) = 3 + 4g(x) = 3 + 4(3 + 4x) = 4^2x + 4.3 + 3$$

$$g^4(x) = g[g\{g\{g(x)\}\}] = 255 + 256x$$

$$g^4(1) = 511.$$

8. Let $f(x)$ satisfies the relation $f(x + y) = f(x) + f(y) \forall x, y \in R$ and $f(1) = 2$ then the value of $\sum_{r=1}^{50} f(r)$ is.....

Sol. $f(x + y) = f(x) + f(y)$

$$\Rightarrow f(1) = 2, f(2) = 4, f(3) = 6 \dots, f(50) = 100$$

$$\sum_{r=1}^{50} f(r) = f(1) + f(2) + f(3) + \dots + f(50)$$

$$= 2 + 4 + 6 + 8 + \dots + 100$$

$$= 25[4 + 98] = 25 \times 102 = 2550.$$

9. Let $f(x) = \frac{e^x - e^{-x}}{2}$ and if $g(f(x)) = x$, then $g\left(\frac{e^{1002} - 1}{2e^{501}}\right)$ equals.....

Sol. $f(x) = \frac{e^x - e^{-x}}{2} = \frac{e^x - \frac{1}{e^x}}{2} = \frac{e^{2x} - 1}{2e^x}$

$$g\{f(x)\} = x$$

$$\therefore g\left(\frac{e^{2x} - 1}{2e^x}\right) = x \Rightarrow g\left(\frac{e^{1002} - 1}{2e^{501}}\right) = 501$$

10. Let $f(x)$ be a function such that $f(x + y) = f(x) + f(y) \forall x, y \in N$ and $f(1) = 4$. If $\sum_{K=1}^n f(a + K) = 2n(33 + n)$, then 'a' equals.....

Sol. $f(x + y) = f(x) + f(y)$

$$f(1) = 4$$

$$\Rightarrow f(2) = f(1) + f(1) = 4 + 4 = 8$$

$$\Rightarrow f(3) = 12$$

$$\dots$$

$$\dots$$

$$\Rightarrow f(a) = 4a$$

$$\text{Now } \sum_{k=1}^n f(a + k) = 2n(33 + n) \Rightarrow \sum_{k=1}^n [f(a) + f(k)] = 2n(33 + n)$$

$$\Rightarrow 4an + [f(1) + f(2) + \dots + f(n)] = 2n(33 + n) \Rightarrow 4an + 2n(n + 1) = 2n(33 + n)$$

$$\Rightarrow 2a + n + 1 = 33 + n \Rightarrow a = 16.$$

11. Let $g(x) = f^{-1}(x)$, where $f(x) = \begin{cases} x, & x < 1 \\ x^2, & 1 \leq x \leq 4 \\ 8\sqrt{x}, & 4 < x \end{cases}$ then $g(256)$ equals

Sol. Clearly $g(x) = f^{-1}(x) = \frac{x^2}{64}$ for $x > 4$

$$g(256) = \frac{(256)^2}{64} = 1024.$$

12. Let $f(x) = \frac{1}{2} + \sqrt{x - \frac{3}{4}}$ and $\phi(x) = f^{-1}(x)$ then $\phi(30)$ equals.....

Sol. $f(x) = \frac{1}{2} + \sqrt{x - \frac{3}{4}}$

$$f(x) = \frac{1}{2} + \sqrt{x - \frac{3}{4}} = y \Rightarrow \sqrt{x - \frac{3}{4}} = y - \frac{1}{2} \Rightarrow x - \frac{3}{4} = \left(y - \frac{1}{2}\right)^2$$

$$\Rightarrow x = y^2 + \frac{1}{4} - y + \frac{3}{4} \Rightarrow 4x = 4y^2 + 1 - 4y + 3$$

$$\Rightarrow 4x = 4y^2 - 4y + 4 \Rightarrow x = y^2 - y + 1$$

$$\Rightarrow f^{-1}(x) = x^2 - x + 1 \Rightarrow f^{-1}(30) = (30)^2 - 30 + 1 \Rightarrow 871$$

$$\therefore f^{-1}(30) = 871. \Rightarrow \phi(30) = 871$$

13. Let $f: R \rightarrow R$ and $g: R \rightarrow R$ be two given functions defined as $f(x) = 3x^2 + 2$ and $g(x) = 3x - 1$, $\forall x \in R$. Then find $[(g \circ f)(x)][(f \circ g)(x)]$ at $x = 1$.

Sol. $f(x) = 3x^2 + 2$

$$g(x) = 3x - 1 \quad \forall x \in R$$

$$f \circ g(x) = 3(3x - 1)^2 + 2$$

$$g \circ f(x) = 3(3x^2 + 2) - 1$$

$$f \circ g(x) \cdot g \circ f(x) = (3(3x - 1)^2 + 2)(9x^2 + 5) = 196 \text{ and } x = 1$$

14. If the function $y = (\cot \alpha)x^2 + 2(\sqrt{\sin \alpha})x + \frac{1}{2} \tan \alpha$, $\alpha \in [0, 2\pi]$ can take negative value at all $x \in R$, then the value of ' α ' must lie in the interval _____.

Sol. $f(x) = (\cot \alpha)x^2 + 2(\sqrt{\sin \alpha})x + \frac{1}{2} \tan \alpha$

$f(x)$ can take negative value for all ' x ' means $f(x) < 0$, $\forall x \in R$

$$\therefore a < 0, b < 0$$

$$\cot \alpha < 0, \quad 4 \sin \alpha - 4^2 \times \frac{1}{2} \times \tan \alpha \cdot \cot \alpha < 0$$

$$2 \sin \alpha - 1 < 0$$

$$\sin \alpha < \frac{1}{2}$$

$$\therefore \sin \alpha \geq 0 \text{ for } \sqrt{\sin \alpha} \text{ to be define therefore } x \in \left(\frac{5\pi}{6}, \pi\right)$$

15. The range of value of a such that $f(x) = \frac{ax^2 + 2(a+1)x + 9a + 4}{x^2 - 8x + 32}$ is always negative is _____.

Sol. $f(x) = \frac{ax^2 + 2(a+1)x + 9a + 4}{x^2 - 8x + 32} < 0$

$$\therefore x^2 - 8x + 32 > 0 \text{ as } D < 0$$

$$\therefore ax^2 + 2(a+1)x + 9a + 4 < 0, \forall x \in \mathbb{R}$$

$$\therefore a < 0 \text{ and } (4(a+1)^2 - 4a(9a+4)) < 0$$

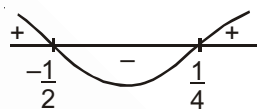
$$a^2 + 2a + 1 - 9a^2 - 4a < 0$$

$$8a^2 + 2a - 1 > 0$$

$$a^2 + \frac{a}{4} - \frac{1}{8} > 0$$

$$\left(a + \frac{1}{2}\right)\left(a - \frac{1}{4}\right) > 0$$

$$\therefore a \in \left(-\infty, -\frac{1}{2}\right) \cup \left(\frac{1}{4}, \infty\right) \text{ and } a < 0$$



16. Domain of the function $f(x) = \frac{1}{\sqrt{\sin(\cos x)}} + \sin^{-1}\left(\frac{2x}{\pi}\right) + \frac{1}{\{-x\}} + \frac{1}{\ln\left(1 - \left[\tan \frac{x}{2}\right] - \left[-\tan \frac{x}{2}\right]\right)}$ is _____.
- (where $[.]$ represents greatest integer function.)

Sol. $f(x) = \frac{1}{\sqrt{\sin(\cos x)}} + \sin^{-1}\left(\frac{2x}{\pi}\right) + \frac{1}{\{-x\}} + \frac{1}{\ln\left(1 - \left[\tan \frac{x}{2}\right] - \left[-\tan \frac{x}{2}\right]\right)}$

$$\begin{array}{l|l|l|l} \sin(\cos x) > 0 & -1 \leq \frac{2x}{\pi} \leq 1 & \{-x\} \neq 0 & 1 - \left[\tan \frac{x}{2}\right] - \left[-\tan \frac{x}{2}\right] > 0 \\ \hline \therefore 0 < \cos x < \pi & -\frac{\pi}{2} \leq x \leq \frac{\pi}{2} & n \notin \mathbb{Z} & \therefore [-x] = -1 - [x], x \notin \mathbb{Z} \\ \hline \Rightarrow x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) & & & \end{array}$$

Case-I

If $x = 0$

$$\left[\tan \frac{x}{2}\right] = 0$$

$$\Rightarrow \ln\left(1 - \left[\tan \frac{x}{2}\right] - \left[-\tan \frac{x}{2}\right]\right) = \ln 1 = 0$$

Case-II

$$x \in \left(-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right) \ln\left(1 - \left[\tan \frac{x}{2}\right] - \left[-\tan \frac{x}{2}\right]\right) = \ln 2$$

$$\text{Hence domain } \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) - \{0, -1, 1\}$$

17. For what integral values of n the number 3π is a period of the function $f(x) = \cos(nx) \cdot \sin\left(\frac{5}{n}\right)x$?

Sol. $n = \pm 1, \pm 3, \pm 5, \pm 15$

$$f(x) = \sin\left(\frac{5}{n}\right)x \cdot \cos(nx)$$

$$f(x) = \frac{1}{2} \left[2 \sin\left(\frac{5}{n}\right)x \cdot \cos(nx) \right] = \frac{1}{2} \left[\sin\left(\frac{5}{n} + n\right)x + \sin\left(\frac{5}{n} - n\right)x \right]$$

$$\text{Period of } \sin\left(\frac{5}{n} + n\right)x \text{ is } \frac{2\pi}{\frac{5}{n} + n} = \frac{2n\pi}{5 + n^2}$$

$$\text{Period of } \sin\left(\frac{5}{n} - n\right)x \text{ is } \frac{2\pi}{\frac{5}{n} - n} = \frac{2n\pi}{5 - n^2}$$

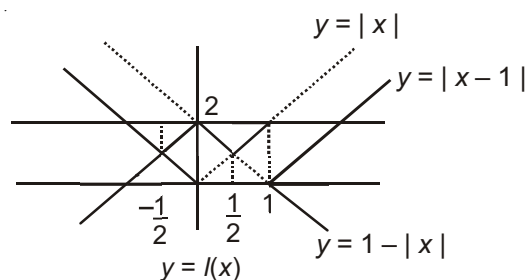
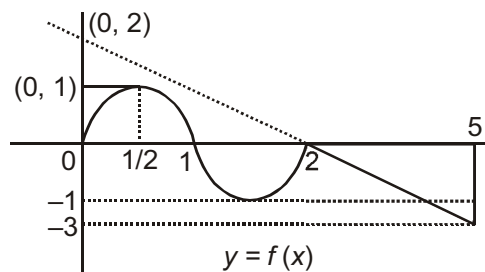
$$\text{LCM of } \left(\frac{2n\pi}{5 + n^2}, \frac{2n\pi}{5 - n^2} \right) \text{ period will be } 3\pi \text{ if } n = \pm 1, \pm 3, \pm 5, \pm 15$$

18. Let $f(x) = \begin{cases} \sin \pi x & 0 \leq x \leq 2 \\ 2 - x & 2 < x \leq 5 \end{cases}$, $g(x) = \begin{cases} \max(|x|, 1 - |x|, |x - 1|) & \text{if } x \in (0, 1) \\ \min(|x|, 1 - |x|, |x - 1|) & \text{otherwise} \end{cases}$. Then find $f(g(x))$.

Sol. $f(x) = \begin{cases} \sin \pi x & , 0 \leq x \leq 2 \\ 2 - x & , 2 < x \leq 5 \end{cases}$

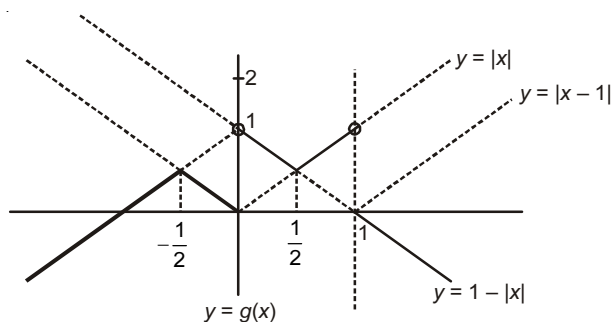
$$g(x) = \begin{cases} \max(|x|, 1 - |x|, |x - 1|) & \text{if } x \in (0, 1) \\ \min(|x|, 1 - |x|, |x - 1|) & , \text{otherwise} \end{cases}$$

$$g(x) = \begin{cases} 1 + x & , x \leq -\frac{1}{2} \\ -x & , -\frac{1}{2} \leq x < 0 \\ 1 - x & , 0 \leq x < \frac{1}{2} \\ x & , \frac{1}{2} \leq x < 1 \\ x - 1 & , x \geq 1 \end{cases}$$



$$f(g(x)) = \begin{cases} \sin \pi(g(x)) & 0 \leq g(x) \leq 2 \\ 2 - g(x) & 2 < g(x) \leq 5 \end{cases}$$

$$f(g(x)) = \begin{cases} 2 - (-x) & x \leq -2 \\ \sin \pi(-x) & -2 \leq x \leq -\frac{1}{2} \\ \sin \pi(1+x) & -\frac{1}{2} \leq x < 0 \\ \sin \pi(1-x) & 0 \leq x \leq \frac{1}{2} \\ \sin \pi(x) & \frac{1}{2} \leq x \leq 1 \\ \sin \pi(x-1) & 1 \leq x \leq 2 \\ 2 - (x-1) & 2 \leq x \end{cases}$$



$$f(g(x)) = \begin{cases} \sin \pi(1+x) & ; x \leq -\frac{1}{2} \\ \sin \pi(-x) & ; -\frac{1}{2} < x \leq 0 \\ \sin \pi(1-x) & ; 0 < x \leq \frac{1}{2} \\ \sin \pi(x) & ; \frac{1}{2} < x < 1 \end{cases}$$

19. Find the interval of c for which $f(x) = \frac{x^2 + 2x + c}{x^2 + 4x + 3c}$ attains any real value.

Sol. $f(x) = \frac{x^2 + 2x + c}{x^2 + 4x + 3c}$

$$y = \frac{x^2 + 2x + c}{x^2 + 4x + 3c}$$

$$x^2(y-1) + 2x(2y-1) + c(3y-1) = 0$$

$$\therefore x \in \mathbb{R}$$

$$D \geq 0$$

$$4(2y-1)^2 - 4(y-1)(3y-1)c \geq 0$$

$$4y^2 - 4y + 1 - 3cy^2 + 4cy - c \geq 0$$

$$y^2[4-3c] + 4y[c-1] + 1-c \geq 0$$

We need range of function 'R' hence

$$D = 0$$

$$16(c-1)^2 - 4(4-3c)(1-c) = 0$$

$$4c^2 - 8c + 4 - 4 + 7c - 3c^2 = 0$$

$$c^2 - c = 0$$

$$c = 0, 1$$

20. Let $f(x) = x + 5$ and $g(x) = x - 5$, $x \in R$. Find $(fog)(5)$.

Sol. $(fog)(x) = f(g(x)) = f(x - 5) = x - 5 + 5 = x \Rightarrow (fog)(5) = 5$

21. Let $f: R \rightarrow R: f(x) = \frac{2x-3}{4}$ be an invertible function. Find f^{-1} .

Sol. $y = \frac{2x-3}{4} \Rightarrow x = \frac{4y+3}{2} \Rightarrow f^{-1}(y) = \frac{4y+3}{2}$

$$\therefore f^{-1}(x) = \frac{4x+3}{2}$$

22. Show that the relation R defined on the set $A = \{1, 2, 3, 4, 5\}$, given by $R = \{(a, b) : |a - b| \text{ is even}\}$ is an equivalence relation.

Sol. Let $a \in A$. Then, $|a - a| = 0$, which is even $\Rightarrow aRa \forall a \in A$. It is reflexive $(a, b) \in R \Rightarrow |a - b| \text{ is even} \Rightarrow |-(a - b)| \text{ is even} \Rightarrow |b - a| \text{ is even}$.

$\Rightarrow (b, a) \in R$. It is symmetric.

$(a, b) \in R$ and $(b, c) \in R$.

$\Rightarrow (a - b) = \pm 2\alpha_1$ and $(b - c) = \pm 2\alpha_2$ for some $\alpha_1, \alpha_2 \in N$

$\Rightarrow [(a - b) + (b - c)] = \pm 2(\alpha_1 \pm \alpha_2)$

$\Rightarrow (a - c) = \pm 2\alpha$ for some $\alpha \in N$

$\Rightarrow (a, c) \in R$, it is transitive

$\therefore$ It is an equivalence relation

Set of elements related to 1

$$= \{b \in A : |1 - b| \text{ is even}\}$$

$$= \{1, 3, 5, 7, 9, 11\}$$

23. Let $f: A \rightarrow B$ and $g: B \rightarrow A$ such that $(gof) = I_A$. Show that f is one-one and g is onto.

Sol. We have, $f(x_1) = f(x_2) \Rightarrow g(f(x_1)) = g(f(x_2))$

$$\Rightarrow (gof)(x_1) = (gof)(x_2)$$

$$\Rightarrow I_A(x_1) = I_A(x_2)$$

$$\Rightarrow x_1 = x_2$$

$\therefore f$ is one-one

Now, show that g is onto, let $a \in A$ and let $f(a) = b \in B$. Then $g(b) = g(f(a)) = (gof)(a) = I_A(a) \because gof = I_A$

Thus, for each $a \in A$, there exists $b \in B$ such that $g(b) = a$

$\therefore g$ is onto.

24. Find the domain and range of the real function $f(x) = \sqrt{16 - x^2}$.

Sol. $f(x) = \sqrt{16 - x^2}, 16 - x^2 \geq 0 \Rightarrow x^2 \leq 16 \Rightarrow -4 \leq x \leq 4$

$$\therefore \text{dom}(f) = \{x \in R : -4 \leq x \leq 4\}$$

$$\text{Also, } y = \sqrt{16 - x^2} \Rightarrow y^2 = 16 - x^2 \Rightarrow x = \sqrt{16 - y^2}$$

$$16 - y^2 \geq 0 \Rightarrow y^2 \leq 16 \Rightarrow -4 \leq y \leq 4$$

$$\therefore \text{Range}(f) = \{y \in R : 0 \leq y \leq 4\}$$

25. Let $f: R \rightarrow R$ be a function is defined by $f(x) = x^2 - \frac{x^2}{1+x^2}$, then

- (1) f is one-one but not onto (2) f is onto but not one-one
(3) f is both one-one and onto (4) f is neither one-one nor onto

Sol. Answer (4)

$$\text{Given } f(x) = x^2 - \frac{x^2}{1+x^2} = x^2 \left(1 - \frac{1}{1+x^2}\right) = x^2 \left(\frac{1+x^2-1}{1+x^2}\right) = \frac{x^4}{1+x^2}$$

Let $1, -1 \in R$

$$\therefore f(1) = \frac{1}{2}, f(-1) = \frac{1}{2}$$

$\Rightarrow f(x)$ is not one-one

$$\therefore R_f = [0, \infty)$$

$\Rightarrow f(x)$ is not onto.

Thus, $f(x)$ is neither one-one nor onto function.

26. Let $S = \{1, 2, 3, 4\}$. The number of functions $f: S \rightarrow S$. Such that $f(i) \leq 2i$ for all $i \in S$ is

- (1) 32 (2) 64 (3) 128 (4) 256

Sol. Answer (3)

Given $S = \{1, 2, 3, 4\}$

$$\therefore f(i) \leq 2i, \forall i \in S$$

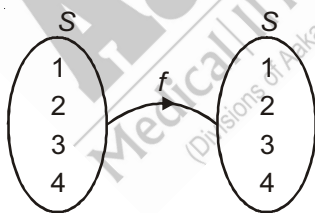
For $i = 1$, $f(1)$ can be 1 or 2

$i = 2$, $f(2)$ can be 1, 2, 3 or 4.

$i = 3$, $f(3)$ can be 1, 2, 3 or 4

$i = 4$, $f(4)$ can be 1, 2, 3 or 4

$$\therefore \text{Total number of such functions} = 2 \times 4 \times 4 \times 4 = 128$$



27. Let $\phi: [0, 1] \rightarrow [0, 1]$ be a continuous and one-one function. Let $\phi(0) = 0$, $\phi(1) = 1$, $\phi\left(\frac{1}{2}\right) = p$, and $\phi\left(\frac{1}{4}\right) = q$, then

- (1) $p > q$ (2) $p < q$
(3) $p = q$ (4) No comparison between p & q

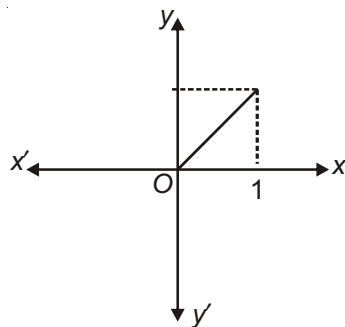
Sol. Answer (1)

Given $\phi : [0, 1] \rightarrow [0, 1]$

$$\therefore \phi(0) = 0, \phi(1) = 1$$

$$\phi\left(\frac{1}{2}\right) = p, \phi\left(\frac{1}{4}\right) = q$$

Here, $\phi(x) = x$



Now, $\phi(0) = 0, \phi(1) = 1$

$$\Rightarrow \phi\left(\frac{1}{2}\right) = \frac{1}{2}$$

$$\Rightarrow p = \frac{1}{2}$$

$$\text{Also, } \phi\left(\frac{1}{4}\right) = \frac{1}{4}$$

$$\Rightarrow q = \frac{1}{4}$$

From (i) and (ii), we get

$$p > q$$

28. Determine all functions $f : \mathbb{R} \setminus \{0, 1\} \rightarrow \mathbb{R}$, which satisfy the equation $f(x) + f\left(\frac{1}{1-x}\right) = \frac{2(1-2x)}{x(1-x)}$ valid for all $x \neq 0$ and $x \neq 1$.

$$\text{Sol. } \therefore f(x) + f\left(\frac{1}{1-x}\right) = \frac{2}{x} - \frac{2}{1-x} \quad \dots(i)$$

Replace x by $\frac{1}{1-x}$

$$\therefore f\left(\frac{1}{1-x}\right) + f\left(\frac{x-1}{x}\right) = 2(1-x) + 2\left(\frac{1-x}{x}\right) \quad \dots(ii)$$

Replace x by $\frac{x-1}{x}$ (ii), we get

$$f\left(\frac{x-1}{x}\right) + f(x) = \frac{2x}{x-1} - 2x \quad \dots(iii)$$

Equation (iii) to Equation (ii), we get

$$f(x) - f\left(\frac{1}{1-x}\right) = \frac{2x}{x-1} - \frac{2}{x} \quad \dots(\text{iv})$$

Adding (i) and (iv), we get,

$$f(x) = \frac{x+1}{x-1}$$

29. Let $A = \{1, 2, 3, 4\}$. The number of functions $f: A \rightarrow A$ satisfying $f(f(i)) = 1$ for all $1 \leq i \leq 4$ is

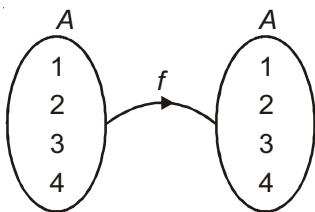
(1) 1

(2) 6

(3) 9

(4) 10

Sol. Answer (1)



$$\therefore f(f(i)) = 1$$

$$\Rightarrow f(i) = f^{-1}(1)$$

Clearly, number of function = 10

30. Let $f: [-2, 2] \rightarrow B$ defined as $f(x) = x^6 - 3x^2 - 10$. Set B for which f is onto is

(1) $[-12, -10]$

(2) $[-10, 42]$

(3) $[-12, 42]$

(4) $[-10, 12]$

Sol. Answer (3)

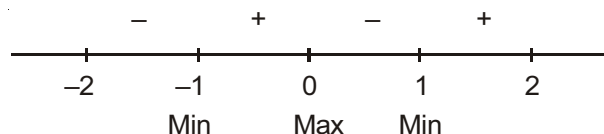
$$f(x) = x^6 - 3x^2 - 10$$

$$\Rightarrow f'(x) = 6x^5 - 6x$$

$$= 6x(x^4 - 1)$$

$$= 6x(x^2 - 1)(x^2 + 1)$$

$$= 6x(x+1)(x-1)(x^2+1)$$



$$\therefore f(-1) = 1 - 3 - 10 = -12$$

$$f(1) = 1 - 3 - 10 = -12$$

$$f(0) = 0 - 0 - 10 = -10$$

$$f(2) = 64 - 12 - 10 = 42$$

$$f(-2) = 64 - 12 - 10 = 42$$

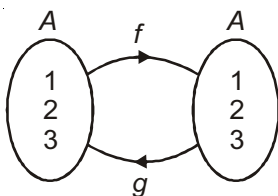
$$\therefore R_f = [-12, 42]$$

$$\Rightarrow B = [-12, 42]$$

31. Let $f: \{1, 2, 3\} \rightarrow \{1, 2, 3\}$ be a function. Then the number of functions $g: \{1, 2, 3\} \rightarrow \{1, 2, 3\}$. Such that $f(x) = g(x)$ for at least one $x \in \{1, 2, 3\}$ is

(1) 11 (2) 19 (3) 23 (4) 27

Sol. Answer (2)



$$\text{Number of function} = 3^3 - 2^3 = 27 - 8 = 19$$

32. Let $f: (-2, 2) \rightarrow (-2, 2)$ be a continuous function such that $f(x) = f(x^2) \forall x \in D_f$ and $f(0) = \frac{1}{2}$, then the value of $4f\left(\frac{1}{4}\right)$ is equal to

(1) 4 (2) 2 (3) -2 (4) 1

Sol. Answer (2)

$$\text{Given } f: (-2, 2) \rightarrow (-2, 2)$$

$$\therefore f(x) = f(x^2) \forall x \in D_f \text{ and } f(0) = \frac{1}{2}$$

$$\Rightarrow f(0) \text{ is a rational number}$$

$$\Rightarrow f(x) \text{ is a constant function}$$

$$\therefore 4f\left(\frac{1}{4}\right) = 4 \times \frac{1}{2} = 2$$

33. Let $f: R \rightarrow B$ is given by $f(x) = \frac{2x^8 + 6x^4 + 4x^2 + 3}{x^8 + 3x^4 + 2x^2 + 1}$. Interval of B for which f is onto is

(1) $[1, \infty)$ (2) $[0, \infty)$ (3) $[-\infty, \infty)$ (4) $(2, 3]$

Sol. Answer (4)

$$f(x) = \frac{2x^8 + 6x^4 + 4x^2 + 3}{x^8 + 3x^4 + 2x^2 + 1} = 2 + \frac{1}{x^8 + 3x^4 + 2x^2 + 1}$$

$$\text{Let } g(x) = x^8 + 3x^4 + 2x^2 + 1$$

$$\Rightarrow R_g = [1, \infty)$$

$$\text{Let } h(x) = \frac{1}{g(x)}$$

$$\Rightarrow R_h = (0, 1]$$

$$\text{Now, } f(x) = 2 + \frac{1}{x^8 + 3x^4 + 2x^2 + 1}$$

$$\Rightarrow R_f = (2, 3]$$

34. Let $A = \{x_1, x_2, x_3, x_4, x_5\}$, $B = \{y_1, y_2, y_3, y_4\}$. Function f is defined from A to B . Such that $f(x_1) = y_1$ and $f(x_2) = y_2$ then, number of onto functions from A to B is

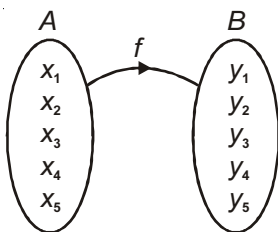
(1) 12

(2) 6

(3) 18

(4) 27

Sol. Answer (3)



$$\therefore f(x_1) = y_1, f(x_2) = y_2$$

Case-I :

When x_3, x_4, x_5 are not related to y_1 and y_2 .

The number of onto functions

$$= \frac{3!}{1! 2!} \times 2!$$

$$= 6$$

Case-II :

When x_3, x_4, x_5 are related to any one of y_1 and y_2 .

The number of onto functions

$$= \frac{3!}{3!} \times 3! \times 2!$$

$$= 12$$

Thus, total number of onto functions

$$= 6 + 12$$

$$= 18$$

